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Mgra functions

Synopsis

The functions prefixed with Mgra make up the Graphics module. The Graphics module provides graphics that can be used to perform annotations.

Functions

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MgraAlloc

Synopsis

Allocate a 2D graphics context.

Syntax

```
AIL_ID MgraAlloc(
    AIL_ID SysId,           //in
    AIL_ID *ContextGraIdPtr //out
)
```

Description

This function allocates a 2D graphics context on the specified system. A 2D graphics context is required to use Aurora Imaging Library graphics to perform annotations. Whenever an annotation is done, 2D graphics context settings are inherited by the graphics. Different contexts can coexist; use their identifier to specify which to use or change.

To modify or inquire 2D graphics context settings, use MgraControl() and MgraInquire().

Note that, upon allocation of an application, a default 2D graphics context is automatically allocated. Rather than using **MgraAlloc**() to allocate a 2D graphics context, you can use this default 2D graphics context, by specifying M_DEFAULT wherever a 2D graphics context identifier is required. Note that there is a different default 2D graphics context for each thread.

When the 2D graphics context is no longer required, release it using MgraFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 2D graphics context's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the 2D graphics context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextGraIdPtr

Specifies the address of the variable in which to write the 2D graphics context identifier or specifies the data type that the function should use to return the 2D graphics context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 2D graphics context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 2D graphics context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_GRA_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 2D graphics context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated 2D graphics context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 2D graphics context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_GRA_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraAllocList

Synopsis

Allocate a 2D graphics list.

Syntax

```
AIL_ID MgraAllocList(
    AIL_ID SysId,          //in
    AIL_INT64 ListGraType, //in
    AIL_ID *ListGraIdPtr   //out
)
```

Description

This function allocates a 2D graphics list, which holds graphics in vector (mathematical) form. Different 2D graphics lists can coexist; use their identifier to specify which to use or change.

The 2D graphics list can be used to annotate the display non-destructively, using MdispControl() with M_ASSOCIATED_GRAPHIC_LIST_ID. Since graphics in a 2D graphics list are vector-based, you can zoom the display without the pixelation (loss of clarity) of the graphics. Also, if you add, modify, or delete graphics, the display can be immediately updated to reflect the changes. To set whether the display is automatically updated, use MdispControl() with M_UPDATE_GRAPHIC_LIST.

The 2D graphics list can also be used to draw destructively in an image, using MgraDraw(). Once drawn, the graphics are raster-based and are part of the image buffer; they cannot be modified. Also, if this annotated image is displayed, zooming it will cause pixelation (loss of clarity) of the graphics and changes in line thickness. Using MgraDraw() is similar to using a graphics function (for example, MgraArc()) to draw destructively in an image.

Graphics can be added to the list using any of the Mgra...() graphics functions, such as MgraArc(). To remove all graphics from the list, use MgraClear(). You can also use MgraControlList() with M_DELETE to remove graphics (one or all) from the list.

When you add graphics to the list, they inherit the current settings of the specified 2D graphics context (as described in MgraAlloc()); however, there is no link between the 2D graphics list and the 2D graphics context. Graphics already in the list are not affected by changes made to the context. To modify or inquire 2D graphics list settings, use MgraControlList() and MgraInquireList().

When the 2D graphics list is no longer required, release it using MgraFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 2D graphics list's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the 2D graphics list.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b

System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ListGraType

Reserved for future expansion and must be set to **M_DEFAULT**.

ListGraIdPtr

Specifies the address of the variable in which to write the 2D graphics list identifier or specifies the data type that the function should use to return the 2D graphics list identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 2D graphics list ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 2D graphics list ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_GRA_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 2D graphics list (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated 2D graphics list. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 2D graphics list identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_GRA_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraArc

Synopsis

Draw an arc in an image without rotation or fill color, or add it to a 2D graphics list.

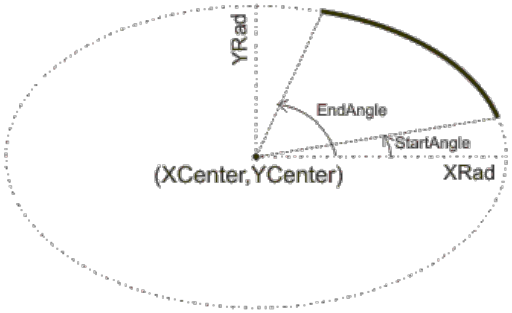
Syntax

```
void MgraArc(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XCenter,            //in
    AIL_DOUBLE YCenter,            //in
    AIL_DOUBLE XRad,               //in
    AIL_DOUBLE YRad,               //in
    AIL_DOUBLE StartAngle,         //in
    AIL_DOUBLE EndAngle            //in
)
```

Description

This function draws an arc destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of an arc to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

The arc is based on an ellipse centered at XCenter and YCenter, with radii XRad and YRad. The arc is the part of the ellipse between the start angle (StartAngle) and the end angle (EndAngle), both measured with respect to the X-axis. The arc inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (MgraControl() with M_COLOR). If part of the arc falls outside of the specified area (image or display), that part is clipped off. The following image shows how the parameters are used to define an arc:



An arc's center position, radii, and angle values are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup. Note that if you set your input coordinate system to M_WORLD and you pass **MgraArc**() an uncalibrated image or the drawing target of the 2D graphics list is uncalibrated, the function will generate an error.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

To create a filled arc, use MgraArcFill(). To create an arc that can be optionally filled and rotated, use MgraArcAngle().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraArc()** internally calls the *AIL_DOUBLE* version of this function (MgraArcDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraArcInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context or 2D graphics list																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the arc or the identifier of a valid 2D graphics list in which to add the arc. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XCenter INQ

Specifies the X-coordinate of the center of the ellipse from which the arc is defined, in the input coordinate system.

YCenter INQ

Specifies the Y-coordinate of the center of the ellipse from which the arc is defined, in the input coordinate system.

XRad INQ

Specifies the radius of the ellipse from which the arc is defined, along the X-axis of the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the radius along the X-axis																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 0.0	Specifies the radius along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b

YRad INQ

Specifies the radius of the ellipse from which the arc is defined, along the Y-axis of the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j –

radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the radius along the Y-axis																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 0.0	Specifies the radius along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

StartAngle INQ

Specifies the angle at which to start drawing the arc, relative to the positive X-axis of the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the starting angle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
-360.0 <= *Value* <= 360.0	Specifies the starting angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

EndAngle INQ

Specifies the angle at which to end drawing the arc, relative to the positive X-axis of the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the ending angle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
-360.0 <= *Value* <= 360.0	Specifies the ending angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraArcDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XCenter, AIL_DOUBLE YCenter,
AIL_DOUBLE XRad, AIL_DOUBLE YRad, AIL_DOUBLE StartAngle, AIL_DOUBLE EndAngle)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XCenter

See XCenter of the main function for a description.

YCenter

See YCenter of the main function for a description.

XRad

See XRad of the main function for a description.

YRad

See YRad of the main function for a description.

StartAngle

See StartAngle of the main function for a description.

EndAngle

See EndAngle of the main function for a description.

```
void MgraArcInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XCenter, AIL_INT64 YCenter,
AIL_INT64 XRad, AIL_INT64 YRad, AIL_DOUBLE StartAngle, AIL_DOUBLE EndAngle)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XCenter

See XCenter of the main function for a description.

YCenter

See YCenter of the main function for a description.

XRad

See XRad of the main function for a description.

YRad

See YRad of the main function for a description.

StartAngle

See StartAngle of the main function for a description.

EndAngle

See EndAngle of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraArcAngle

Synopsis

Draw an arc or sector in an image, with optional rotation and fill color, or add it to a 2D graphics list.

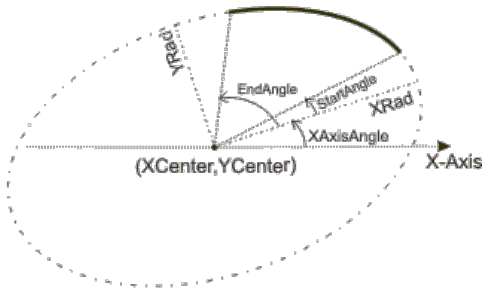
Syntax

```
void MgraArcAngle(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XCenter,            //in
    AIL_DOUBLE YCenter,            //in
    AIL_DOUBLE XRad,               //in
    AIL_DOUBLE YRad,               //in
    AIL_DOUBLE StartAngle,         //in
    AIL_DOUBLE EndAngle,           //in
    AIL_DOUBLE XAxisAngle,         //in
    AIL_INT64 ControlFlag          //in
)
```

Description

This function draws an arc or sector destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of an arc or sector to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

The arc or sector graphic is based on an ellipse centered at XCenter and YCenter, with radii XRad and YRad. The arc is the curve of the ellipse between the start angle (StartAngle) and the end angle (EndAngle), with an orientation along the X-axis (XAxisAngle). The arc or sector inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (MgraControl() with M_COLOR). If part of the arc or sector falls outside of the specified area (image or display), that part is clipped off. The following image shows how the parameters are used to define the arc:



The center position, radii, and angle values are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup. Note that if you set your input coordinate system to M_WORLD and you pass **MgraArcAngle**() an uncalibrated image or the drawing target of the 2D graphics list is uncalibrated, the function will generate an error.

By default, only the arc is drawn in the image (M_CONTOUR). With the ControlFlag parameter, you can change this behavior to draw a sector of the ellipse (M_SECTOR) instead, with lines extending from the center of the ellipse to the start and end points of the arc. However, you can choose to draw it filled with the specified foreground color by specifying the M_FILLED combination constant.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

To create a filled arc (sector) that is not rotated, you can also use MgraArcFill(). To create an arc that is not rotated or filled, you can also use MgraArc().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraArcAngle()** internally calls the *AIL_DOUBLE* version of this function (MgraArcAngleDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraArcAngleInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the arc or sector, or the identifier of a valid 2D graphics list in which to add the arc or sector. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XCenter INQ

Specifies the X-coordinate of the center of the ellipse from which the arc or sector is defined, in the input coordinate system.

YCenter INQ

Specifies the Y-coordinate of the center of the ellipse from which the arc or sector is defined, in the input coordinate system.

XRad INQ

Specifies the radius of the ellipse from which the arc or sector is defined, along the X-axis in the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the radius along the X-axis																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 0.0	Specifies the radius along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b

YRad INQ

Specifies the radius of the ellipse from which the arc or sector is defined, along the Y-axis in the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the radius along the Y-axis																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 0.0	Specifies the radius along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

StartAngle INQ

Specifies the angle at which to start drawing the arc or sector, relative to the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the starting angle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
-360.0 <= *Value* <= 360.0	Specifies the starting angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

EndAngle INQ

Specifies the angle at which to end drawing the arc or sector, relative to the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the ending angle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
-360.0 <= *Value* <= 360.0	Specifies the ending angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

XAxisAngle INQ

Specifies the amount to rotate the ellipse used to define the arc or sector, in degrees, relative to the input coordinate system.

ControlFlag INQ

Specifies whether to draw an arc or sector. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To specify whether to draw an arc or sector

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_CONTOUR.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTOUR +	Specifies that the arc (the curve between the specified start and end angles) is drawn without lines extending from the center of the ellipse to the start and end points of the arc.	c		e	f	g	a		i		j	k	l	m	n	b
M_SECTOR +	Specifies that a sector is drawn with lines extending from the center of the ellipse to the start and end points of the arc, unless the specified start and end angles form a closed curve.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for any of the possible values of the ControlFlag parameter.

You can add the following value to the above-mentioned values to set whether the shape is filled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether the shape is filled																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILLED	Specifies that the shape is filled with specified foreground color (MgraControl() or MgraControlList() with M_COLOR), resulting in a filled sector.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraArcAngleDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XCenter, AIL_DOUBLE YCenter, AIL_DOUBLE XRad, AIL_DOUBLE YRad, AIL_DOUBLE StartAngle, AIL_DOUBLE EndAngle, AIL_DOUBLE XAxisAngle, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XCenter

See XCenter of the main function for a description.

YCenter

See YCenter of the main function for a description.

XRad

See XRad of the main function for a description.

YRad

See YRad of the main function for a description.

StartAngle

See StartAngle of the main function for a description.

EndAngle

See EndAngle of the main function for a description.

XAxisAngle

See XAxisAngle of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraArcAngleInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XCenter, AIL_INT64 YCenter, AIL_INT64 XRad, AIL_INT64 YRad, AIL_DOUBLE StartAngle, AIL_DOUBLE EndAngle, AIL_DOUBLE XAxisAngle, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XCenter

See XCenter of the main function for a description.

YCenter

See YCenter of the main function for a description.

XRad

See XRad of the main function for a description.

YRad

See YRad of the main function for a description.

StartAngle

See StartAngle of the main function for a description.

EndAngle

See EndAngle of the main function for a description.

XAxisAngle

See XAxisAngle of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraArcFill

Synopsis

Draw a filled arc (sector) in an image without rotation, or add it to a 2D graphics list.

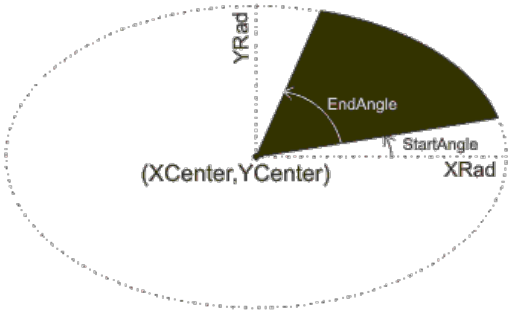
Syntax

```
void MgraArcFill(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XCenter,            //in
    AIL_DOUBLE YCenter,            //in
    AIL_DOUBLE XRad,               //in
    AIL_DOUBLE YRad,               //in
    AIL_DOUBLE StartAngle,         //in
    AIL_DOUBLE EndAngle            //in
)
```

Description

This function draws a filled arc (sector) destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of a filled arc to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling. Note that the arc is filled with the specified foreground color (MgraControl() or MgraControlList() with M_COLOR).

The filled arc is based on an ellipse centered at XCenter and YCenter, with radii XRad and YRad. The arc is the part of the ellipse between the start angle (StartAngle) and the end angle (EndAngle), both measured with respect to the X-axis. The arc inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (MgraControl() with M_COLOR). If part of the filled arc falls outside of the specified area (image or display), that part is clipped off. The following image shows how the parameters are used to define the arc:



A filled arc's center position, radii, and angle values are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup. Note that if you set your input coordinate system to M_WORLD and you pass **MgraArcFill**() an uncalibrated image or the drawing target of the 2D graphics list is uncalibrated, the function will generate an error.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

To create an arc that is not filled, use MgraArc(). To create an arc that can be optionally filled and rotated, use MgraArcAngle().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list

> (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraArcFill()** internally calls the *AIL_DOUBLE* version of this function (MgraArcFillDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraArcFillInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the arc or the identifier of a valid 2D graphics list in which to add the arc. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XCenter INQ

Specifies the X-coordinate of the center of the ellipse from which the arc is defined, in the input coordinate system.

YCenter INQ

Specifies the Y-coordinate of the center of the ellipse from which the arc is defined, in the input coordinate system.

XRad INQ

Specifies the radius of the ellipse from which the arc is defined, along the X-axis of the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the radius along the X-axis																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 0.0	Specifies the radius along the X-axis.	c		e	f	g	a		i		j	k	l	m	n	b

YRad INQ

Specifies the radius of the ellipse from which the arc is defined, along the Y-axis of the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the radius along the Y-axis																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 0.0	Specifies the radius along the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

StartAngle INQ

Specifies the angle at which to start drawing the arc, relative to the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the starting angle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
-360.0 <= *Value* <= 360.0	Specifies the starting angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

EndAngle INQ

Specifies the angle at which to end drawing the arc, relative to the input coordinate system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the ending angle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
-360.0 <= *Value* <= 360.0	Specifies the ending angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraArcFillDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XCenter, AIL_DOUBLE YCenter, AIL_DOUBLE XRad, AIL_DOUBLE YRad, AIL_DOUBLE StartAngle, AIL_DOUBLE EndAngle)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XCenter

See XCenter of the main function for a description.

YCenter

See YCenter of the main function for a description.

XRad

See XRad of the main function for a description.

YRad

See YRad of the main function for a description.

StartAngle

See StartAngle of the main function for a description.

EndAngle

See EndAngle of the main function for a description.

```
void MgraArcFillInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XCenter, AIL_INT64 YCenter,
AIL_INT64 XRad, AIL_INT64 YRad, AIL_DOUBLE StartAngle, AIL_DOUBLE EndAngle)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XCenter

See XCenter of the main function for a description.

YCenter

See YCenter of the main function for a description.

XRad

See XRad of the main function for a description.

YRad

See YRad of the main function for a description.

StartAngle

See StartAngle of the main function for a description.

EndAngle

See EndAngle of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraClear

Synopsis

Clear an image buffer to a specified foreground color, or remove graphics from a 2D graphics list.

Syntax

```
void MgraClear(
    AIL_ID ContextGraId,          //in
    AIL_ID DstImageBufOrListGraId //out
)
```

Description

This function clears the entire specified image buffer to the foreground color of the specified 2D graphics context, or removes all the graphics from the specified 2D graphics list.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

> If you specify a 2D graphics list (DstImageBufOrListGraId), this parameter (ContextGraId) is ignored.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer to clear or the identifier of a valid 2D graphics list to empty. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

Compilation information

Header	Include AIL.h.

Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraControl

Synopsis

Control a specified 2D graphics context setting.

Syntax

```
void MgraControl(
    AIL_ID ContextGraId,    //out
    AIL_INT64 ControlType,  //in
    AIL_DOUBLE ControlValue //in
)
```

Description

This function allows you to control the specified 2D graphics context setting. Most of the control type settings can be inquired using MgraInquire(). To control a 2D graphics list, use MgraControlList().

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context to control. This parameter must be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlType

Specifies the 2D graphics context setting to control.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the value to assign to the 2D graphics context setting.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For general 2D graphics context settings
- For settings that apply to graphics when interactive mode is enabled
- For settings that apply only to text

The following ControlType and corresponding ControlValue parameter settings are used to change general 2D graphics context settings. Unless otherwise specified, these affect all types of graphics.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

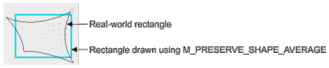
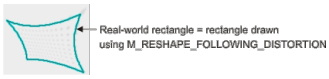
For general 2D graphics context settings																	
ControlType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlValue																	
M_COLOR +	Sets the foreground color. INQ																
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies an RGB value when using the 2D graphics context to draw in an 8-bit, 3-band buffer. The red, green, and blue values must be values between 0 and 255, inclusive. When drawing in a 16-bit or 32-bit color buffer, the bands of the RGB value are cast to the type of the destination buffer's bands. To specify a value for a specific band of a 16-bit or 32-bit color buffer, use **MgraControl**() with M_COLOR combined with the appropriate constant (M_RED, M_GREEN, or M_BLUE).	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters																
	Red Specifies the red band, as a value between 0 and 255.																
	Green Specifies the green band, as a value between 0 and 255.																
	Blue Specifies the blue band, as a value between 0 and 255.																
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. The buffer can be a 1-band or multi-band buffer. The specified value is cast to the buffer type and depth. The default value is -1. For example, this value is equivalent to 0xFF for an 8-bit buffer. Note that a grayscale value can be any integer or floating-point number. If the given value exceeds the range of the possible values that can be stored in each band of the destination buffer, the least significant bits of the value are used.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_DIRECTION	Sets how to draw (render) an arrow pointing in the direction with which a graphic was defined. This only affects graphics that are not filled, and that have such a direction as part of their definition; for example, arc, ellipse, line, rectangle, and ring. Aurora Imaging Library ignores this setting for other graphics. This setting does not affect drawings created using a processing or analysis module draw function (M...Draw()). For more information about how the directions are drawn for different graphics, see the Drawing graphics in a calibrated image subsection of the Drawing graphics section of Chapter 26: Generating graphics. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_NONE.	c		e	f	g	a		i		j	k	l	m	n	b
M_NONE	Specifies that no direction is drawn.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRIMARY_DIRECTION	Specifies to draw an arrow showing the primary direction of the graphic. For example, if you draw the primary direction for a rectangle defined with MgraRectAngle(), the arrow will point in the angular direction set with the Angle parameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRIMARY_DIRECTION + M_SECONDARY_DIRECTION	Specifies to draw an arrow showing both the primary and secondary direction of the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
M_SECONDARY_DIRECTION	Specifies to draw an arrow showing the secondary direction of the graphic. For example, a rectangle's secondary direction is equal to its *PrimaryDirection* - 90°.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_OFFSET_X	Sets the offset to subtract from the source X-coordinates before rendering the graphic. This is useful to take results obtained relative to an offset and draw them relative to the top-left corner of the destination image of a draw operation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b

Value	Specifies the X-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_OFFSET_Y	Sets the offset to subtract from the source Y-coordinates before rendering the graphic. This is useful to take results obtained relative to an offset and draw them relative to the top-left corner of the destination image of a draw operation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_X	Sets the scale factor in the X-direction. This value is useful to zoom drawings of results. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scale factor in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_Y	Sets the scale factor in the Y-direction. This value is useful to zoom drawings of results. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scale factor in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIXTURE	Sets which camera calibration information to use when rendering (drawing or displaying) a graphic that has been defined in world units (M_INPUT_UNITS set to M_WORLD). Aurora Imaging Library can interpret positioning and dimensioning information of the graphic solely using the camera calibration information associated with the destination image buffer (or the image buffer selected to the display). Alternatively, you can have Aurora Imaging Library use the explicitly specified camera calibration information (M_GRAPHIC_SOURCE_CALIBRATION). If only the destination of the graphic is associated with camera calibration information, it will be used to render the graphic. However, if both are available, you should use M_FIXTURE to identify with respect to which of their relative coordinate systems has positioning and dimensioning information been specified. If with respect to that of the destination camera calibration information (M_USE_DESTINATION_FIRST), the source camera calibration information is not used. If with respect to that of the source camera calibration information (M_USE_SOURCE_FIRST), positions and dimensions are transformed from this relative coordinate system to the absolute coordinate system; then, since it is assumed that there is only one absolute coordinate system, these positions and dimensions are transformed from the absolute coordinate system to the pixel coordinate system, using the destination camera calibration information. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_USE_DESTINATION_FIRST	Specifies that the camera calibration information of the destination is used when rendering the graphic. In this case, the graphic follows the relative coordinate system of the destination. This is most often used when working with ROIs. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

M_USE_SOURCE_FIRST	Specifies that the camera calibration information of the graphic, set with M_GRAPHIC_SOURCE_CALIBRATION, is used when rendering the graphic. In this case, the graphic is positioned independent of the destination's relative coordinate system; the destination camera calibration information is only used, if available, for the world to pixel mapping.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_CONVERSION_MODE	Sets how the shape of a graphic, defined in world units, is converted to pixels (rendered). Drawings created using a processing or analysis module draw function (M...Draw()) ignore this control type; instead the graphics are drawn using settings defined in their respective modules. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_PRESERVE_SHAPE_AVERAGE	Specifies to render the graphic so that its shape is preserved, even if it means not respecting the camera calibration information exactly. For example, a rectangle in world units will be mapped to a rectangle in pixel units, even if the accurate conversion would not yield a rectangle. This setting applies only to dots (MgraDot() and MgraDots()) and to rectangles (MgraRect() and MgraRectAngle()). 	c		e	f	g	a		i		j	k	l	m	n	b
M_RESHAPE_FOLLOWING_DISTORTION	Specifies that all points along the contour of the graphic will be converted using the camera calibration information, following any non-linear distortion. This mode is slower, but more accurate. 	c		e	f	g	a		i		j	k	l	m	n	b

M_RESHAPE_FROM_POINTS	Specifies that only a few key points or features will be converted using the camera calibration information; from these converted points, the rest of the graphic will be rendered respecting the shape of the graphic. For example, when converting a line segment from world to pixel units, the two end points are mapped to pixel units, and then a straight line is drawn connecting these two points, ignoring any non-linear distortion in-between. This is the fastest mode, and is accurate as long as there is only linear distortion. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SOURCE_CALIBRATION	Sets the camera calibration information to use to interpret positioning and dimensioning information of a graphic defined in world units (M_INPUT_UNITS set to M_WORLD), if you have set M_FIXTURE to M_USE_SOURCE_FIRST. If the destination also has camera calibration information, positioning and dimensioning information are transformed from the relative coordinate system of the source camera calibration information to its absolute coordinate system. Then, since it is assumed that there is only one absolute coordinate system, the dimensions and positions are transformed from the absolute coordinate system to the pixel coordinate system, using the destination camera calibration information. Note that this camera calibration information is also used if you have set M_FIXTURE to M_USE_DESTINATION_FIRST, but the graphic is drawn in an image buffer that has no camera calibration information. INQ	c		e	f	g	a		i		j	k	l	m	n	b

M_NULL	Specifies that no source camera calibration information is available to interpret positioning and dimensioning information of a graphic defined in world units, except when the graphic is drawn using the draw function (M...Draw()) of a processing or analysis module; in this case, the camera calibration information of the context or result will be used. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
Aurora Imaging Library Identifier	Specifies the identifier of a camera calibration context, image buffer or processing or analysis module result buffer, whose camera calibration information to use; this object is only associated with the 2D graphics context, not copied. If, for example, the relative coordinate system changes after using M_GRAPHIC_SOURCE_CALIBRATION, it will be reflected in the camera calibration information that is used.	c		e	f	g	a		i		j	k	l	m	n	b
M_INPUT_UNITS	Sets the units with which to interpret the graphic's position and dimensional information. This essentially sets the input coordinate system to use. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISPLAY	Specifies to interpret the values in pixel units that, unlike M_PIXEL, are not altered when the display is panned or zoomed. This can be useful if, for example, you always want to display the camera's current frame rate at the top-left corner. M_DISPLAY must only be specified when drawing in a 2D graphics list and the 2D graphics list is used to annotate the display non-destructively, using MdispControl() with M_ASSOCIATED_GRAPHIC_LIST_ID.	c		e	f	g	a		i		j	k	l	m	n	b

M_PIXEL	Specifies to interpret the values in pixel units, with respect to the pixel coordinate system. If you are drawing in a 2D graphics list and the 2D graphics list is used to annotate the display non-destructively (M_ASSOCIATED_GRAPHIC_LIST_ID), the graphic will be zoomed and panned according to the display's zoom and pan settings. This is not the case when setting the units to M_DISPLAY, which also uses pixel units. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WORLD	Specifies to interpret the values in world units, with respect to the relative coordinate system. If you are drawing in a 2D graphics list and the 2D graphics list is used to annotate the display non-destructively (M_ASSOCIATED_GRAPHIC_LIST_ID), the graphic will be zoomed and panned according to the display's zoom and pan settings. When using M_WORLD, a graphic's coordinates are interpreted as real-world values and transformed to pixel values before being drawn or associated with the display non-destructively. Note that using the camera calibration information associated with the image (M_WORLD), ensures that whenever the relative coordinate system moves (for example, with McalRelativeOrigin(), McalSetCoordinateSystem(), or McalFixture()), the non-destructive annotations move accordingly. In this case, the position of the graphics is set according to the position of the relative coordinate system. If world units are specified, drawing a graphic or calling MgraFill() generates an error if the operation is performed on an uncalibrated image and M_GRAPHIC_SOURCE_CALIBRATION is set to M_NULL.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_THICKNESS	Sets the thickness of the lines and the diameter of dots, depending on the drawing operation. Note that for this setting, the thickness of the line is not affected by panning or zooming, and is always interpreted in pixel units. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 1	Specifies the thickness of the lines and diameter of the dots, in pixels. The value must be an odd integer. The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to change the 2D graphics context settings that affect the interactivity of a graphic, drawn in a 2D graphics list associated with a display, when interactive mode is enabled (MdispControl() with M_GRAPHIC_LIST_INTERACTIVE set to M_ENABLE).

Note, you can still change these settings even if interactive mode is disabled, but they will have no effect until interactive mode is enabled. Adding a graphic that does not support one of the following settings to a 2D graphics list will automatically assign the setting of the graphic to M_DISABLE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For settings that apply to graphics when interactive mode is enabled																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_EDITABLE		Sets whether a graphic can be edited via user interaction in an interactive display. If a graphic is not visible or if it is not selectable, it is not editable and this setting is ignored. This setting applies to all types of graphics except dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). INQ															
	M_DISABLE	Specifies that the graphic cannot be edited via user interaction. Note that the graphic can still be modified using MgraControlList().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be edited via user interaction. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESIZABLE		Sets whether a graphic can be resized via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not resizable and this setting is ignored. This setting applies to all types of graphics except a dot (MgraDot()), dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be resized via user interaction and the resize handles will not be displayed if a graphic is selected. Note that the graphic can still be resized using MgraControlList().	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the graphic can be resized via user interaction by clicking and dragging one of the resize handles. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATABLE		Sets whether a graphic can be rotated via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not rotatable and this setting is ignored. This setting applies to all types of graphics except a dot (MgraDot()), dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be rotated via user interaction and the rotate handle will not be displayed if a graphic is selected. Note that the graphic can still be rotated using MgraControlList().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be rotated via user interaction by clicking and dragging the rotation handle. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTABLE		Sets whether a graphic can be selected via user interaction in an interactive display. If a graphic is not visible, it is not selectable and this setting is ignored. This setting applies to all graphics except dots (MgraDots()) and sets of finite length lines (MgraLines() with M_LINE_LIST). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be selected via user interaction. Note that the selection can still be changed using MgraControlList(), but it will not be apparent on the display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be selected by clicking on it. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

M_SPECIFIC_FEATURES_EDITABLE		Sets whether a graphic can be modified via user interaction in an interactive display using handles that are specific to its graphic type. If a graphic is not visible, not selectable, or not editable, it cannot be modified using its type-specific handles and this setting is ignored. This setting applies to all types of graphics except dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be modified via user interaction and its specific feature handles will not be displayed if a graphic is selected. Note that the graphic can still be modified using MgraControlList().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be modified via user interaction by clicking and dragging one of the specific feature handles. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSLATABLE		Sets whether a graphic can be moved (translated) via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not movable and this setting is ignored. This setting applies to all types of graphics except dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be moved via user interaction. Note that the graphic can still be moved using MgraControlList().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be moved via user interaction by clicking and dragging the graphic, its selection box, or its center handle. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_VISIBLE		Sets whether a graphic is rendered on the display. Note that a graphic that is not visible cannot be affected by any user interaction in an interactive display. This setting applies to all types of graphics. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the graphic is not rendered.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that the graphic is rendered. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to change 2D graphics context settings that apply only to text (MgraText()).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For settings that apply only to text																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_BACKCOLOR +		Sets the background color. INQ															
	M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies an RGB value when using the 2D graphics context to draw in an 8-bit, 3-band buffer. The red, green, and blue values must be values between 0 and 255, inclusive. When drawing in a 16-bit or 32-bit color buffer, the bands of the RGB value are cast to the type of the destination buffer's bands. To specify a value for a specific band of a 16-bit or 32-bit color buffer, use **MgraControl**() with M_BACKCOLOR combined with the appropriate constant (M_RED, M_GREEN, or M_BLUE).	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Red Specifies the red band, as a value between 0 and 255.															
		Green Specifies the green band, as a value between 0 and 255.															
		Blue Specifies the blue band, as a value between 0 and 255.															
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies a grayscale value. The buffer can be a 1-band or multi-band buffer. The specified value is cast to the buffer type and depth. The default value is 0. Note that a grayscale value can be any integer or floating-point number. If the given value exceeds the range of the possible values that can be stored in each band of the destination buffer, the least significant bits of the value are used.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_MODE	Sets whether to fill the text's background. INQ	c		e	f	g	a		i		j	k	l	m	n	b

M_OPAQUE	Specifies that the background will be filled with the current background color before drawing text. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSPARENT	Specifies not to change the background before drawing text. This creates a transparent background for printed characters.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_AUTO_SELECT	Sets whether Aurora Imaging Library should search for a suitable font to draw text if the currently selected font (MgraFont()) is a TrueType font that does not support the character code. Aurora Imaging Library will first attempt to make its selection from already used fonts, and then from system fonts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that Aurora Imaging Library will not search for a suitable font. An error is returned if the character code cannot be drawn using the current TrueType font.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that Aurora Imaging Library will search for a suitable font. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE	Sets the size to draw text when using a TrueType font. When using a bitmap font, use M_FONT_X_SCALE and M_FONT_Y_SCALE to set the font size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the text's font size, in points. The default value is 12-point font.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_X_SCALE	Sets the font's horizontal scaling factor. This setting only applies to bitmap fonts. For TrueType fonts, use M_FONT_SIZE to set the font size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor by which to multiply the width of the font's characters. Note that using a font with a scale of 1.0 accelerates text drawing. The default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_Y_SCALE	Sets the font's vertical scaling factor. This setting only applies to bitmap fonts. For TrueType fonts, use M_FONT_SIZE to set the font size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor by which to multiply the height of the font's characters. Note that using a font with a scale of 1.0 accelerates text drawing. The default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b

M_TEXT_ALIGN_HORIZONTAL	Sets the horizontal alignment of text. The alignment is relative to the point at which the text starts, as specified with MgraText(). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that text is horizontally centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that text is left-aligned. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that text is right-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_VERTICAL	Sets the vertical alignment of text. The alignment is relative to the point at which the text starts, as specified with MgraText(). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that text is bottom-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that text is vertically centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP	Specifies that text is top-aligned. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_BORDER	Sets borders around the text. Note that the possible settings can be combined. For example, to draw a box around the text, specify M_TOP + M_BOTTOM + M_LEFT + M_RIGHT. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that a line is drawn underneath the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that a line is drawn to the left of the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_NONE	Specifies that no border is drawn around the text. This setting cannot be combined with any other setting. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that a line is drawn to the right of the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP	Specifies that a line is drawn above the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_DIRECTION	Sets the direction to draw text when using a TrueType font. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_LEFT_TO_RIGHT	Specifies that text will be drawn from left to right. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIGHT_TO_LEFT	Specifies that text will be drawn from right to left.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_COLOR; M_BACKCOLOR.

You can add one of the following values to the above-mentioned values to specify the color band.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the color band (for 16- or 32-bit color buffers)																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BLUE	Specifies the blue color band.	c		e	f	g	a		i		j	k	l	m	n	b
M_GREEN	Specifies the green color band.	c		e	f	g	a		i		j	k	l	m	n	b
M_RED	Specifies the red color band.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraControlList

Synopsis

Control the graphics contained within the 2D graphics list.

Syntax

```
void MgraControlList(
    AIL_ID GraListId,       //out
    AIL_INT LabelOrIndex,   //in
    AIL_INT SubIndex,       //in
    AIL_INT64 ControlType,  //in
    AIL_DOUBLE ControlValue //in
)
```

Description

This function allows you to control the settings of graphics contained within the 2D graphics list. Most of the control type settings can be inquired using MgraInquireList().

When graphics are added to the list, they inherit the current settings of the 2D graphics context. However, subsequent changes to the 2D graphics context (for example, with MgraControl()) do not affect graphics already in the list. The settings of graphics in the list can only be modified using **MgraControlList**().

Using this function, you can also control a sub-element of a graphic in a 2D graphics list. This is useful when a single point of a graphic requires repositioning. For example, you can reposition (M_POSITION_X and M_POSITION_Y) a vertex point of a polygon (MgraLines() with M_POLYGON) by specifying the polygon graphic (LabelOrIndex), and one of its points (SubIndex).

All geometric type changes that you make to the graphics (such as, rotation, scaling, and translation) are applied according to the current input units of each graphic in the list, as specified with M_INPUT_UNITS, on a per graphic basis. For example, if you change the position of two graphics in the list, a dot set in pixel units and a rectangle set in world units, the dot's position will change in pixel units and the rectangle's position will change in world units. However, changing M_INPUT_UNITS to a different value will not convert a graphic's position.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (GraListId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraControlList()** internally calls the *AIL_DOUBLE* version of this function (MgraControlListDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraControlListInt64()).

Parameters

GraListId

Specifies the identifier of the 2D graphics list to control. The 2D graphics list must have been previously allocated on the required system using MgraAllocList().

LabelOrIndex

Specifies the graphic (one or all) to control. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics list or graphic therein

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GRAPHIC_INDEX(AIL_INT Index)	Specifies the index of an existing graphic on which to apply the control setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Specifies the index. The index is from 0 (inclusive) to the total number of graphics in the list minus 1. You can get the number graphics in the list using MgraInquireList() with M_NUMBER_OF_GRAPHICS. If the control setting is not supported on the graphic specified, an error is generated. For example, a dot (MgraDot()) cannot be scaled (M_APPLY_SCALE) or rotated (M_ROTATE).															
M_GRAPHIC_LABEL(AIL_INT Label)	Specifies the label of an existing graphic on which to apply the control setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Label Specifies the label. The label must be greater than 0. You can retrieve the label of the last graphic added to the list using MgraInquireList() with M_LAST_LABEL. If the control setting is not supported on the graphic specified, an error is generated. For example, a dot (MgraDot()) cannot be scaled (M_APPLY_SCALE) or rotated (M_ROTATE).															
M_ALL	Applies the specified control setting to all the graphics contained within the 2D graphics list. The graphics that do not support the specified control type setting remain unchanged.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL_SELECTED	Applies the specified control setting to all the graphics contained within the 2D graphics list that are currently selected (that is, all graphics with M_GRAPHIC_SELECTED set to M_TRUE). The graphics that are not selected remain unchanged.	c		e	f	g	a		i		j	k	l	m	n	b
M_LIST	Applies the specified control setting to the 2D graphics list itself.	c		e	f	g	a		i		j	k	l	m	n	b

SubIndex

Specifies the index of the sub-element of the graphic on which to apply the control setting. If this information is not required or supported, set this parameter to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of a graphic's sub-element																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies to apply the control setting to the graphic itself (instead of just a sub-element).	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= M_NUMBER_OF_SUB_ELEMENTS	Specifies the index of the sub-element of the graphic on which to apply the control setting. The following table lists all the graphics for which you can control their individual sub-elements and outlines how sub-indices are assigned for each of these.	c		e	f	g	a		i		j	k	l	m	n	b

Graphics types	Index of sub-element values
Dots (created using MgraDots()).	The indices, starting from 0, are assigned to each dot in order of creation.
A single line (created using MgraLine() or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_LINE).	0 is assigned to the specified start point, and 1 is assigned to the specified end point of the line.

Sets of finite length lines (created using MgraLines() with M_LINE_LIST).	The indices, starting from 0, are assigned to the specified start point and end point of each line, in order of creation. For example, the start point of the first line has an index of 0 and the end point has an index of 1, and the start point of the second line has an index of 2 and the end point has an index of 3.

Sets of infinite lines (created using MgraLines() with M_INFINITE_LINES).	The indices, starting from 0, are assigned to the first and second specified point of each line, in order of creation. For example, the first specified point of the first line has an index of 0 and its second specified point has an index of 1, and the first specified point of the second line has an index of 2 and its second specified point has an index of 3.
A polygon (created using MgraLines() with M_POLYGON or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_POLYGON).	The indices, starting from 0, are assigned to each polygon vertex in order of creation.

	A polyline (created using MgraLines() with M_POLYLINE or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_POLYLINE).	The indices, starting from 0, are assigned to each polyline point in order of creation.																

ControlType

Specifies the 2D graphics list setting to control.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the value to assign to the 2D graphics list setting.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For changing general settings of the 2D graphics list itself
- For changing the general settings inherited from the 2D graphics context
- For changing the interactive properties of graphics in the list
- For changing the settings of text
- For changing the position and dimensional information of a graphic
- For performing geometric operations on graphics
- For deleting graphics (one or all) from the list
- For specifying general keyboard key settings to interactively modify graphics
- For specifying explicit keyboard keys to interactively modify graphics

The following ControlType and corresponding ControlValue parameter settings are used to change general settings of a 2D graphics list itself. In this case, set the LabelOrIndex parameter to M_LIST and the SubIndex parameter to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For changing general settings of the 2D graphics list itself																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_ANGLE_SNAPPING_VALUE		Sets the angular value to use as a multiple when rotating the graphic. This value only applies when using M_ANGLE_SNAPPING. INQ															
	M_DEFAULT	Specifies the default value; the default value is 15.0°.	c		e	f	g	a		i		j	k	l	m	n	b

	1.0 <= *Value* <= 180.0	Specifies the value to use as a multiple, in degrees. The value must be a divisor of 180.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_EASY_SELECTION		Sets whether Aurora Imaging Library allows the easy selection of graphics on the display. When easy selection is enabled, you can select closed graphics (such as circles) by clicking anywhere inside them, even if they are not filled. To select non-filled graphics when easy selection is disabled, you must click on their contour. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that Aurora Imaging Library does not allow for the easy selection of graphics on the display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that Aurora Imaging Library allows for the easy selection of graphics on the display.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_ANNOTATIONS_COLOR +		Sets the color of the selection box and handles in interactive mode. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies an RGB value when drawing in an 8-bit, 3-band buffer. The red, green, and blue values must be between 0 and 255, inclusive. When drawing in a 16-bit or 32-bit color buffer, the bands of the RGB value are cast to the type of the destination buffer's bands. To specify a value for a specific band of a 16-bit or 32-bit color buffer, use **MgraControlList**() with M_INTERACTIVE_ANNOTATIONS_COLOR combined with the appropriate constant (M_RED, M_GREEN, or M_BLUE).	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Red Specifies the red band, as a value between 0 and 255.															
		Green Specifies the green band, as a value between 0 and 255.															
		Blue Specifies the blue band, as a value between 0 and 255.															
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b

	Value	Specifies a grayscale value. The buffer can be a 1-band or multi-band buffer. The specified value is cast to the buffer type and depth. Note that a grayscale value can be any integer or floating-point number. If the given value exceeds the range of the possible values that can be stored in each band of the destination buffer, the least significant bits of the value are used.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE		Sets the interactive behavior when resizing the graphic using its selection box. To resize the graphic, click and move one of its selection box's eight resize handles. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_FIXED_CORNER.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER +	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER +	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic. If you click on a selection box handle that is not a corner, the opposite side (which includes its corners) does not move when resizing.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_ALT		Sets the interactive behavior when resizing the graphic using its selection box and pressing the Alt key. To resize the graphic, click and move (while pressing Alt) one of its selection box's eight resize handles. If multiple key values (M_MODE_RESIZE_ALT, M_MODE_RESIZE_CTRL, or M_MODE_RESIZE_SHIFT) are set to one of M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO, and multiple keys (Alt, Ctrl, or Shift) are pressed when resizing, M_FIXED_ASPECT_RATIO takes precedence over M_SQUARE_ASPECT_RATIO, and M_SQUARE_ASPECT_RATIO takes precedence over M_NO_CONSTRAINT. If multiple key values (M_MODE_RESIZE_ALT, M_MODE_RESIZE_CTRL, or M_MODE_RESIZE_SHIFT) are set to one of M_FIXED_CENTER or M_FIXED_CORNER, and multiple keys (Alt, Ctrl, or Shift) are pressed when resizing, M_FIXED_CENTER takes precedence over M_FIXED_CORNER. If the graphic's aspect ratio is set to 1 using **MgraControlList**() with M_CONSTRAIN_ASPECT_RATIO, it will remain 1 (M_MODE_RESIZE_ALT will not change it). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SQUARE_ASPECT_RATIO.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that pressing the Alt key does not affect how to resize the graphic. In this case, Aurora Imaging Library uses M_MODE_RESIZE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant, when resizing the graphic while pressing the Alt key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic while pressing the Alt key. This overrides the value set with M_MODE_RESIZE. The aspect ratio combination value that M_MODE_RESIZE uses (M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic while pressing the Alt key. If you click on a selection box handle that is not a corner, the opposite side (which includes its corners) does not move when resizing. This overrides the value set with M_MODE_RESIZE. The aspect ratio combination value that M_MODE_RESIZE uses (M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change, when resizing it while pressing the Alt key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square), when resizing it while pressing the Alt key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODE_RESIZE_CTRL		Sets the interactive behavior when resizing the graphic using its selection box and pressing the Ctrl key. To resize the graphic, click and move (while pressing Ctrl) one of its selection box's eight resize handles. If multiple key values (M_MODE_RESIZE_ALT, M_MODE_RESIZE_CTRL, or M_MODE_RESIZE_SHIFT) are set to one of M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO, and multiple keys (Alt, Ctrl, or Shift) are pressed when resizing, M_FIXED_ASPECT_RATIO takes precedence over M_SQUARE_ASPECT_RATIO, and M_SQUARE_ASPECT_RATIO takes precedence over M_NO_CONSTRAINT. If multiple key values (M_MODE_RESIZE_ALT, M_MODE_RESIZE_CTRL, or M_MODE_RESIZE_SHIFT) are set to one of M_FIXED_CENTER or M_FIXED_CORNER, and multiple keys (Alt, Ctrl, or Shift) are pressed when resizing, M_FIXED_CENTER takes precedence over M_FIXED_CORNER. If the graphic's aspect ratio is set to 1 using **MgraControlList**() with M_CONSTRAIN_ASPECT_RATIO, it will remain 1 (M_MODE_RESIZE_CTRL will not change it). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_FIXED_CENTER.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to resize the graphic. In this case, Aurora Imaging Library uses M_MODE_RESIZE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant, when resizing the graphic while pressing the Ctrl key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic while pressing the Ctrl key. This overrides the value set with M_MODE_RESIZE. The aspect ratio combination value that M_MODE_RESIZE uses (M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b

	M_FIXED_CORNER	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic while pressing the Ctrl key. If you click on a selection box handle that is not a corner, the opposite side (which includes its corners) does not move when resizing. This overrides the value set with M_MODE_RESIZE. The aspect ratio combination value that M_MODE_RESIZE uses (M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change, when resizing it while pressing the Ctrl key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square), when resizing it while pressing the Ctrl key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION		Sets how to establish the graphic's secondary dimension when defining the graphic interactively on the display. This only applies when creating the graphic using MgraInteractive() with M_GRAPHIC_TYPE_RECT set to M_ORIENTED_RECT. In this case, the graphic's secondary dimension refers to the rectangle's height. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ONE_SIDED.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on one side of the width line. The side depends on whether you move your mouse above or below the width line.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on both sides of the width line. Moving your mouse above or below the width line adjusts the height equally on each side.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODE_RESIZE_SECONDARY_DIMENSION_ALT		Sets how to establish the graphic's secondary dimension when defining the graphic interactively on the display and pressing the Alt key. This only applies when creating the graphic using MgraInteractive() with M_GRAPHIC_TYPE_RECT set to M_ORIENTED_RECT. In this case, the graphic's secondary dimension refers to the rectangle's height. Unless you disable M_MODE_RESIZE_SECONDARY_DIMENSION_ALT, its value overrides the value set with M_MODE_RESIZE_SECONDARY_DIMENSION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Alt key does not affect how to establish the graphic's secondary dimension when defining it interactively on the display. In this case, Aurora Imaging Library uses M_MODE_RESIZE_SECONDARY_DIMENSION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part, while pressing the Alt key. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on one side of the width line. The side depends on whether you move your mouse (while pressing Alt) above or below the width line.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part, while pressing the Alt key. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on both sides of the width line. Moving your mouse (while pressing Alt) above or below the width line adjusts the height equally on each side.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION_CTRL		Sets how to establish the graphic's secondary dimension when defining the graphic interactively on the display and pressing the Ctrl key. This only applies when creating the graphic using MgraInteractive() with M_GRAPHIC_TYPE_RECT set to M_ORIENTED_RECT. In this case, the graphic's secondary dimension refers to the rectangle's height. Unless you disable M_MODE_RESIZE_SECONDARY_DIMENSION_CTRL, its value overrides the value set with M_MODE_RESIZE_SECONDARY_DIMENSION. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_SYMMETRIC.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to establish the graphic's secondary dimension when defining it interactively on the display. In this case, Aurora Imaging Library uses M_MODE_RESIZE_SECONDARY_DIMENSION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part, while pressing the Ctrl key. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on one side of the width line. The side depends on whether you move your mouse (while pressing Ctrl) above or below the width line.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part, while pressing the Ctrl key. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on both sides of the width line. Moving your mouse (while pressing Ctrl) above or below the width line adjusts the height equally on each side.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION_SHIFT		Sets how to establish the graphic's secondary dimension when defining the graphic interactively on the display and pressing the Shift key. This only applies when creating the graphic using MgraInteractive() with M_GRAPHIC_TYPE_RECT set to M_ORIENTED_RECT. In this case, the graphic's secondary dimension refers to the rectangle's height. Unless you disable M_MODE_RESIZE_SECONDARY_DIMENSION_SHIFT, its value overrides the value set with M_MODE_RESIZE_SECONDARY_DIMENSION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to establish the graphic's secondary dimension when defining it interactively on the display. In this case, Aurora Imaging Library uses M_MODE_RESIZE_SECONDARY_DIMENSION.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part, while pressing the Shift key. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on one side of the width line. The side depends on whether you move your mouse (while pressing Shift) above or below the width line.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part, while pressing the Shift key. For example, when interactively defining an oriented rectangle graphic, the first two clicks on the display define a line that represents the rectangle's width, and the third click defines its height on both sides of the width line. Moving your mouse (while pressing Shift) above or below the width line adjusts the height equally on each side.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SHIFT		Sets the interactive behavior when resizing the graphic using its selection box and pressing the Shift key. To resize the graphic, click and move (while pressing Shift) one of the selection box's eight resize handles. If multiple key values (M_MODE_RESIZE_ALT, M_MODE_RESIZE_CTRL, or M_MODE_RESIZE_SHIFT) are set to one of M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO, and multiple keys (Alt, Ctrl, or Shift) are pressed when resizing, M_FIXED_ASPECT_RATIO takes precedence over M_SQUARE_ASPECT_RATIO, and M_SQUARE_ASPECT_RATIO takes precedence over M_NO_CONSTRAINT. If multiple key values (M_MODE_RESIZE_ALT, M_MODE_RESIZE_CTRL, or M_MODE_RESIZE_SHIFT) are set to one of M_FIXED_CENTER or M_FIXED_CORNER, and multiple keys (Alt, Ctrl, or Shift) are pressed when resizing, M_FIXED_CENTER takes precedence over M_FIXED_CORNER. If the graphic's aspect ratio is set to 1 using **MgraControlList**() with M_CONSTRAIN_ASPECT_RATIO, it will remain 1 (M_MODE_RESIZE_SHIFT will not change it). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_FIXED_ASPECT_RATIO.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to resize the graphic. In this case, Aurora Imaging Library uses M_MODE_RESIZE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant, when resizing the graphic while pressing the Shift key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic while pressing the Shift key. This overrides the value set with M_MODE_RESIZE. The aspect ratio combination value that M_MODE_RESIZE uses (M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic while pressing the Shift key. If you click on a selection box handle that is not a corner, the opposite side (which includes its corners) does not move when resizing. This overrides the value set with M_MODE_RESIZE. The aspect ratio combination value that M_MODE_RESIZE uses (M_FIXED_ASPECT_RATIO, M_NO_CONSTRAINT, or M_SQUARE_ASPECT_RATIO) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change, when resizing it while pressing the Shift key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square), when resizing it while pressing the Shift key. This overrides the aspect ratio combination value set for M_MODE_RESIZE. The value that M_MODE_RESIZE uses (M_FIXED_CENTER or M_FIXED_CORNER) is not overridden.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE		Sets the interactive behavior when rotating the graphic using its selection box. To rotate the graphic, click and move its selection box's rotate handle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NO_CONSTRAINT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to rotate the graphic freely.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODE_ROTATE_ALT	Sets the interactive behavior when rotating the graphic using its selection box and pressing the Alt key. To rotate the graphic, click and move (while pressing Alt) its selection box's rotate handle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ANGLE_SNAPPING.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE, when rotating it while pressing the Alt key. This overrides the value set with M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that pressing the Alt key does not affect how to rotate the graphic. In this case, Aurora Imaging Library uses M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_CONSTRAINT	Specifies to rotate the graphic freely, when rotating it while pressing the Alt key. This overrides the value set with M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE_CTRL	Sets the interactive behavior when rotating the graphic using its selection box and pressing the Ctrl key. To rotate the graphic, click and move (while pressing Ctrl) its selection box's rotate handle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE, when rotating it while pressing the Ctrl key. This overrides the value set with M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that pressing the Ctrl key does not affect how to rotate the graphic. In this case, Aurora Imaging Library uses M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_CONSTRAINT	Specifies to rotate the graphic freely, when rotating it while pressing the Ctrl key. This overrides the value set with M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE_SHIFT	Sets the interactive behavior when rotating the graphic using its selection box and pressing the Shift key. To rotate the graphic, click and move (while pressing Shift) its selection box's rotate handle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE, when rotating it while pressing the Shift key. This overrides the value set with M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to rotate the graphic. In this case, Aurora Imaging Library uses M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to rotate the graphic freely, when rotating it while pressing the Shift key. This overrides the value set with M_MODE_ROTATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE		Sets the interactive behavior when moving (translating) the graphic using its selection box. To move the graphic, hover over its selection box border until the mouse arrow pointer changes to the move pointer, and then click and drag the graphic to a different location on the display. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NO_CONSTRAINT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE_ALT		Sets the interactive behavior when moving (translating) the graphic using its selection box and pressing the Alt key. To move the graphic, hover over its selection box border until the mouse arrow pointer changes to the move pointer, and then click and drag the graphic (while pressing Alt) to a different location on the display. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes, when moving it while pressing the Alt key. This overrides the value set with M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Alt key does not affect how to move the graphic. In this case, Aurora Imaging Library uses M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely, when moving it while pressing the Alt key. This overrides the value set with M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODE_TRANSLATE_CTRL		Sets the interactive behavior when moving (translating) the graphic using its selection box and pressing the Ctrl key. To move the graphic, hover over its selection box border until the mouse arrow pointer changes to the move pointer, and then click and drag the graphic (while pressing Ctrl) to a different location on the display. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes, when moving it while pressing the Ctrl key. This overrides the value set with M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to move the graphic. In this case, Aurora Imaging Library uses M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely, when moving it while pressing the Ctrl key. This overrides the value set with M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE_SHIFT		Sets the interactive behavior when moving (translating) the graphic using its selection box and pressing the Shift key. To move the graphic, hover over its selection box border until the mouse arrow pointer changes to the move pointer, and then click and drag the graphic (while pressing Shift) to a different location on the display. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_AXIS_ALIGNED.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes, when moving it while pressing the Shift key. This overrides the value set with M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to move the graphic. In this case, Aurora Imaging Library uses M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely, when moving it while pressing the Shift key. This overrides the value set with M_MODE_TRANSLATE.	c		e	f	g	a		i		j	k	l	m	n	b

M_MULTIPLE_SELECTION	Sets whether to permit interactive multiple selection using the Ctrl key. When this setting is enabled, a user can select multiple graphics by holding down the Ctrl key and clicking on the graphics that need to be selected. Note that, even if this is disabled, multiple selection is still possible programmatically using M_GRAPHIC_SELECTED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that interactive multiple selection is not permitted.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that interactive multiple selection is permitted. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MULTIPLE_SELECTION_KEY	Sets the keyboard key that you can press to select multiple graphics with your mouse on the display. Clicking on a graphic that you have already selected, while pressing the specified key, deselects that graphic. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_KEY_CTRL.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ALT	Specifies that you can press the Alt key to select (or deselect) multiple graphics.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CTRL	Specifies that you can press the Ctrl key to select (or deselect) multiple graphics.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_SHIFT	Specifies that you can press the Shift key to select (or deselect) multiple graphics.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTED_COLOR +	Sets the color of the selected graphics in interactive mode. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies an RGB value when drawing in an 8-bit, 3-band buffer. The red, green, and blue values must be between 0 and 255, inclusive. When drawing in a 16-bit or 32-bit color buffer, the bands of the RGB value are cast to the type of the destination buffer's bands. To specify a value for a specific band of 16-bit or 32-bit color buffer, use **MgraControlList**() with M_SELECTED_COLOR combined with the appropriate constant (M_RED, M_GREEN, or M_BLUE).	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														

			Red Specifies the red band, as a value between 0 and 255.														
			Green Specifies the green band, as a value between 0 and 255.														
			Blue Specifies the blue band, as a value between 0 and 255.														
	M_COLOR_BLACK		Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE		Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY		Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN		Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE		Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN		Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN		Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA		Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED		Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW		Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY		Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN		Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE		Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY		Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN		Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE		Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. The buffer can be a 1-band or multi-band buffer. The specified value is cast to the buffer type and depth. Note that a grayscale value can be any integer or floating-point number. If the given value exceeds the range of the possible values that can be stored in each band of the destination buffer, the least significant bits of the value are used.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTION_RADIUS		Sets the size of the selection-radius in interactive mode. Selection-radius refers to how close a mouse click needs to be to a graphic to select it. For example if a graphic is at a position that is 4 units radially away from where you clicked, the graphic will appear selected if the selection-radius is set to a value greater than or equal to 4 units. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the size of the selection-radius, in display units (these are pixel units that remain unaltered when you pan or zoom the display). The default value is 8.0.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for [when ControlType=M_MODE_RESIZE: M_FIXED_CENTER; M_FIXED_CORNER].

You can add one of the following values to the above-mentioned values to specify the aspect ratio when resizing the graphic.

The following values will not modify the graphic's aspect ratio if it is set to 1 using **MgraControlList**() with M_CONSTRAIN_ASPECT_RATIO.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the aspect ratio when resizing the graphic																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant when resizing it.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change when resizing it. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square) when resizing it.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to change the general settings inherited from the 2D graphics context when a graphic is initially added to the list. Unless otherwise specified, these affect all types of graphics.

In this case, the LabelOrIndex parameter can be set to one or all graphics and the SubIndex parameter must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

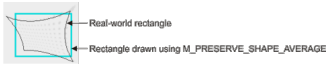
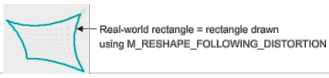
For changing the general settings inherited from the 2D graphics context																	
ControlType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlValue																
M_COLOR +	Sets the foreground color of the graphic. INQ															
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies an RGB value when drawing in an 8-bit, 3-band buffer. The red, green, and blue values must be between 0 and 255, inclusive. When drawing in a 16-bit or 32-bit color buffer, the bands of the RGB value are cast to the type of the destination buffer's bands. To specify a value for a specific band of a 16-bit or 32-bit color buffer, use **MgraControlList**() with M_COLOR combined with the appropriate constant (M_RED, M_GREEN, or M_BLUE).	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b

Value	Specifies a grayscale value. The buffer can be a 1-band or multi-band buffer. The specified value is cast to the buffer type and depth. The default value is -1. For example, this value is equivalent to 0xFF for an 8-bit buffer. Note that a grayscale value can be any integer or floating-point number. If the given value exceeds the range of the possible values that can be stored in each band of the destination buffer, the least significant bits of the value are used.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_DIRECTION	Sets how to draw (render) an arrow pointing in the direction with which a graphic was defined. This only affects graphics that are not filled, and that have such a direction as part of their definition; for example, arc, ellipse, line, rectangle, and ring. Aurora Imaging Library ignores this setting for other graphics. This setting does not affect drawings created using a processing or analysis module draw function (M...Draw()). For more information about how the directions are drawn for different graphics, see the Drawing graphics in a calibrated image subsection of the Drawing graphics section of Chapter 26: Generating graphics. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_NONE	Specifies that no direction is drawn.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRIMARY_DIRECTION	Specifies to draw an arrow showing the primary direction of the graphic. For example, if you draw the primary direction for a rectangle defined with MgraRectAngle(), the arrow will point in the angular direction set with the Angle parameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_PRIMARY_DIRECTION + M_SECONDARY_DIRECTION	Specifies to draw an arrow showing both the primary and secondary direction of the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
M_SECONDARY_DIRECTION	Specifies to draw an arrow showing the secondary direction of the graphic. For example, a rectangle's secondary direction is equal to its *PrimaryDirection* - 90°.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_OFFSET_X	Sets the offset to subtract from the source X-coordinates before rendering the graphic. This is useful to take results obtained relative to an offset and draw them relative to the top-left corner of the destination image of a draw operation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_OFFSET_Y	Sets the offset to subtract from the source Y-coordinates before rendering the graphic. This is useful to take results obtained relative to an offset and draw them relative to the top-left corner of the destination image of a draw operation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_X	Sets the scaling factor in the X-direction. This value is useful to zoom in on drawings of results. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scaling factor in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_Y	Sets the scaling factor in the Y-direction. This value is useful to zoom in on drawings of results. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scaling factor in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIXTURE	Sets the camera calibration information to use when rendering (drawing or displaying) a graphic that has been defined in world units (M_INPUT_UNITS set to M_WORLD). Aurora Imaging Library can interpret positioning and dimensioning information of the graphic solely using the camera calibration information associated with the destination image buffer (or the image buffer selected to the display). Alternatively, you can have Aurora Imaging Library use the explicitly specified camera calibration information (M_GRAPHIC_SOURCE_CALIBRATION). If only the destination or the graphic is associated with camera calibration information, it will be used to render the graphic. However, if both are available, you should use M_FIXTURE to identify with respect to which of their relative coordinate systems has positioning and dimensioning information been specified. If with respect to that of the destination camera calibration information (M_USE_DESTINATION_FIRST), the source camera calibration information is not used. If with respect to that of the source camera calibration information (M_USE_SOURCE_FIRST), positions and dimensions are transformed from this relative coordinate system to the absolute coordinate system; then, since it is assumed that there is only one absolute coordinate system, these positions and dimensions are transformed from the absolute coordinate system to the pixel coordinate system, using the destination camera calibration information. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_USE_DESTINATION_FIRST	Specifies that the camera calibration information of the destination is used when rendering the graphic. In this case, the graphic follows the relative coordinate system of the destination. This is most often used when working with ROIs.	c		e	f	g	a		i		j	k	l	m	n	b

M_USE_SOURCE_FIRST	Specifies that the camera calibration information of the graphic, set with M_GRAPHIC_SOURCE_CALIBRATION, is used when rendering the graphic. In this case, the graphic is positioned independent of the destination's relative coordinate system; the destination camera calibration information is only used, if available, for the world to pixel mapping.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_CONVERSION_MODE	Sets how the shape of a graphic, defined in world units, is converted to pixels (rendered). Drawings created using a processing or analysis module draw function (M...Draw()) ignore this control type; instead the graphics are drawn using settings defined in their respective modules. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_PRESERVE_SHAPE_AVERAGE	Specifies to render the graphic so that its shape is preserved even if it means not respecting the camera calibration information exactly. For example, a rectangle in world units will be mapped to a rectangle in pixel units, even if the accurate conversion would not yield a rectangle. This setting applies only to dots (MgraDot() and MgraDots()) and to rectangles (MgraRect() and MgraRectAngle()). 	c		e	f	g	a		i		j	k	l	m	n	b
M_RESHAPE_FOLLOWING_DISTORTION	Specifies that all points along the contour of the graphic will be converted using the camera calibration information, following any non-linear distortion. This mode is slower, but more accurate. 	c		e	f	g	a		i		j	k	l	m	n	b

M_RESHAPE_FROM_POINTS	Specifies that only a few key points or features will be converted using the camera calibration information; from these points, the rest of the graphic will be rendered respecting the shape of the graphic. For example, when converting a line segment from world to pixel units, the two end points are mapped to pixel units, and then a straight line is drawn connecting these two points, ignoring any non-linear distortion in-between. This is the fastest mode, and is accurate as long as there is only linear distortion.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SOURCE_CALIBRATION	Sets the camera calibration information to use to interpret positioning and dimensioning information of a graphic defined in world units (M_INPUT_UNITS set to M_WORLD), if you have set M_FIXTURE to M_USE_SOURCE_FIRST. If the destination also has camera calibration information, positioning and dimensioning information are transformed from the relative coordinate system of the source camera calibration information to its absolute coordinate system. Then, since it is assumed that there is only one absolute coordinate system, the dimensions and positions are transformed from the absolute coordinate system to the pixel coordinate system, using the destination camera calibration information. Note that this camera calibration information is also used if you have set M_FIXTURE to M_USE_DESTINATION_FIRST, but the graphic is drawn in an image buffer that has no camera calibration information. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that no source camera calibration information is available to interpret positioning and dimensioning information of a graphic defined in world units.	c		e	f	g	a		i		j	k	l	m	n	b

	Aurora Imaging Library identifier	Specifies the identifier of a camera calibration context, image buffer or processing or analysis module result buffer, whose camera calibration information to use. Note that the camera calibration information of this object is copied to an internal camera calibration context; the specified object is not associated with the 2D graphics list. Since the information is being copied, MgraInquireList() will not return the same identifier as the one passed to **MgraControlList**(); instead it will return the identifier of the copy. This identifier is automatically generated by Aurora Imaging Library. If, for example, the relative coordinate system changes after using M_GRAPHIC_SOURCE_CALIBRATION, it will have no effect on the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
M_INPUT_UNITS		Sets the unit with which to interpret the graphic's positional and dimensional information, for graphics already added to the 2D graphics list. This essentially sets the input coordinate system to use. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISPLAY	Specifies to interpret the values in pixel units that, unlike M_PIXEL, are not altered when the display is panned or zoomed. This can be useful if, for example, you always want to display the camera's current frame rate at the top-left corner. M_DISPLAY must only be specified when the 2D graphics list is used to annotate the display non-destructively, using MdispControl() with M_ASSOCIATED_GRAPHIC_LIST_ID.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies to interpret the values in pixel units, with respect to the pixel coordinate system. If the 2D graphics list is used to annotate the display non-destructively (M_ASSOCIATED_GRAPHIC_LIST_ID), the graphics will be zoomed and panned according to the display's zoom and pan settings. This is not the case when setting the units to M_DISPLAY, which also uses pixel units.	c		e	f	g	a		i		j	k	l	m	n	b

M_WORLD	Specifies to interpret the values in world units, with respect to the relative coordinate system. If the 2D graphics list is used to annotate the display non-destructively (M_ASSOCIATED_GRAPHIC_LIST_ID), the graphic will be zoomed and panned according to the display's zoom and pan settings. When using M_WORLD, a graphic's coordinates are interpreted as real-world values and transformed to pixel values before being drawn or associated with the display non-destructively. Note that using the camera calibration information associated with the image (M_WORLD), ensures that whenever the relative coordinate system moves (for example, with McalRelativeOrigin(), McalSetCoordinateSystem(), or McalFixture()), the non-destructive annotations move accordingly. In this case, the position of the graphics is set according to the position of the relative coordinate system. If world units are specified, calling MgraDraw() generates an error if the operation is performed on an uncalibrated image and M_GRAPHIC_SOURCE_CALIBRATION is set to M_NULL. When the 2D graphics list is associated with the display, nothing will be drawn.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_THICKNESS	Sets the thickness of the lines and the diameter of dots, depending on the drawing operation. Note that for this setting, the thickness of the line is not affected by panning or zooming, and is always interpreted in pixel units. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the thickness of the lines and diameter of the dots, in pixels. The value must be an odd integer. The default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to change settings inherited from the 2D graphics context that affect the interactivity of a graphic, drawn in a 2D graphics list associated with a display, when interactive mode is enabled (MdispControl() with M_GRAPHIC_LIST_INTERACTIVE set to M_ENABLE).

You can still change these settings even if interactive mode is disabled, but they will have no effect until interactive mode is enabled. If you attempt to change one of these settings to M_ENABLE for a graphics type that cannot be assigned this value, an error is generated.

In this case, the LabelOrIndex parameter can be set to one or all graphics and the SubIndex parameter must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For changing the interactive properties of graphics in the list																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_EDITABLE		Sets whether a graphic can be edited, via user interaction in an interactive display. If a graphic is not visible or if it is not selectable, it is not editable and this setting is ignored. This setting applies to all types of graphics except dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). Enabling this setting when specifying an unsupported graphic type will generate an error. INQ															
	M_DISABLE	Specifies that the graphic cannot be edited via user interaction. Note that the graphic can still be modified using **MgraControlList**().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be edited via user interaction.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SELECTED		Sets whether a graphic is selected. In an interactive display, a selected graphic will be surrounded by a selection box with handles that allow user interaction. This setting applies to all types of graphics. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the graphic is not selected.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the graphic is selected.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESIZABLE		Sets whether a graphic can be resized, via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not resizable and the setting is ignored. This setting applies to all types of graphics except a dot (MgraDot()), dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). Enabling this setting when specifying an unsupported graphic type will generate an error. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the graphic cannot be resized via user interaction and the resize handles will not be displayed if a graphic is selected. Note that the graphic can still be resized using **MgraControlList**().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be resized via user interaction by clicking and dragging one of the resize handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATABLE		Sets whether a graphic can be rotated, via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not rotatable and the setting is ignored. This setting applies to all types of graphics except a dot (MgraDot()), dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). Enabling this setting when specifying an unsupported graphic type will generate an error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be rotated via user interaction and the rotate handle will not be displayed if a graphic is selected. Note that the graphic can still be rotated using **MgraControlList**().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be rotated via user interaction by clicking and dragging the rotation handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTABLE		Sets whether a graphic can be selected, via user interaction in an interactive display. If a graphic is not visible, it is not selectable and this setting is ignored. This setting applies to all graphics except dots (MgraDots()) and sets of finite length lines (MgraLines() with M_LINE_LIST). Enabling this setting when specifying an unsupported graphic type will generate an error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be selected via user interaction. Note that the graphic can still be selected using **MgraControlList**().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be selected by clicking on it.	c		e	f	g	a		i		j	k	l	m	n	b

M_SPECIFIC_FEATURES_EDITABLE		Sets whether a graphic can be modified using handles that are specific to its graphic type via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it cannot be modified using type-specific handles and the setting is ignored. This setting applies to all types of graphics except dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). Enabling this setting when specifying an unsupported graphic type will generate an error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be modified via user interaction and its specific feature handles will not be displayed if a graphic is selected. Note that the graphic can still be modified using **MgraControlList**().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be modified via user interaction by clicking and dragging one of the type-specific handles.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSLATABLE		Sets whether a graphic can be translated (moved), via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not movable and the setting is ignored. This setting applies to all types of graphics except dots (MgraDots()), text (MgraText()), sets of finite length lines (MgraLines() with M_LINE_LIST), and drawings created using a processing or analysis module draw function (M...Draw()). Enabling this setting when specifying an unsupported graphic type will generate an error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be moved via user interaction. Note that the graphic can still be moved using **MgraControlList**().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be moved via user interaction by clicking and dragging the graphic, its selection box, or its center handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_VISIBLE		Sets whether a graphic is rendered on the display. Note that a graphic that is not visible will not be affected by any user interaction in an interactive display. This setting applies to all types of graphics. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the graphic is not rendered.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the graphic is rendered.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to change the settings inherited from the 2D graphics context when text (MgraText()) is added to the list.

In this case, the LabelOrIndex parameter can be set to the label or index of text in the 2D graphics list and the SubIndex parameter must be set to M_DEFAULT. Attempting to change one of the following settings on graphics other than text will generate an error.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For changing the settings of text																	
ControlType ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BACKCOLOR +	Sets the background color of the text. INQ															
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies an RGB value when drawing in an 8-bit, 3-band buffer. The red, green, and blue values must be between 0 and 255, inclusive. When drawing in a 16-bit or 32-bit color buffer, the bands of the RGB value are cast to the type of the destination buffer's bands. To specify a value for a specific band of a 16-bit or 32-bit color buffer, use **MgraControlList**() with M_BACKCOLOR combined with the appropriate constant (M_RED, M_GREEN, or M_BLUE).	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. The buffer can be a 1-band or multi-band buffer. The specified value is cast to the buffer type and depth. Note that a grayscale value can be any integer or floating-point number. If the given value exceeds the range of the possible values that can be stored in each band of the destination buffer, the least significant bits of the value are used.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_MODE		Sets whether to fill the text's background. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPAQUE	Specifies that the background will be filled with the current background color before drawing text.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRANSPARENT	Specifies not to change the background before drawing text. This creates a transparent background for printed characters.	c		e	f	g	a		i		j	k	l	m	n	b

M_FONT	Sets the font of the characters in the text. Note that the non-bitmap fonts will use TrueType and Unicode features to draw text. This allows you to draw text using different sizes and TrueType fonts installed on your computer. This also allows you to draw any Unicode text (depending on the font). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT	Same as M_FONT_DEFAULT_SMALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_LARGE	Specifies a large bitmap font, where each character is drawn in a 16x32 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_MEDIUM	Specifies a medium bitmap font, where each character is drawn in a 12x24 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_SMALL	Specifies a small bitmap font, where each character is drawn in a 8x16 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_TTF	Specifies the default TrueType font of your operating system.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FontFamily:Weight:Slant*") 1	Specifies the font according to the following format, *Family*: *Weight*: *Slant*. *Family* must be set to the name of the font's family, such as Arial, Times New Roman, and Wingdings. *Weight* can be set to one of the following: Thin, ExtraLight, UltraLight, Light, Book, Normal, Regular, Medium, SemiBold, DemiBold, Bold, ExtraBold, UltraBold, Heavy, Black, ExtraBlack, or UltraBlack. *Slant* can be set to one of the following: Italic, Oblique, or Roman. You can omit *Weight* and *Slant*; also, you can provide the *Weight* and *Slant* in any order. > Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*FontFamily:Weight:Slant*")).	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FontFile*") 1	Specifies a full path to a TrueType file name. > Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*FontFile*")).	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_AUTO_SELECT	Sets whether Aurora Imaging Library should search for a suitable font to draw the text, if the currently selected font (M_FONT) is a TrueType font that does not support the character code. Aurora Imaging Library will first attempt to make its selection from already used fonts, and then from system fonts. INQ	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies that Aurora Imaging Library will not search for a suitable font. An error is returned if the character code cannot be drawn using the current TrueType font.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that Aurora Imaging Library will search for a suitable font.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE	Sets the font size, when using a TrueType font. When using a bitmap font, use M_FONT_X_SCALE and M_FONT_Y_SCALE to set the font size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the text's font size, in points.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_X_SCALE	Sets the font's horizontal scaling factor. This setting only applies to bitmap fonts. For TrueType fonts, use M_FONT_SIZE to set the font size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor with which to multiply the width of the font characters. Note that using a font with a scale of 1.0 accelerates text drawing.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_Y_SCALE	Sets the font's vertical scaling factor. This setting only applies to bitmap fonts. For TrueType fonts, use M_FONT_SIZE to set the font size. INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor with which to multiply the height of the font characters. Note that using a font with a scale of 1.0 accelerates text drawing.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_HORIZONTAL	Sets the horizontal alignment of the text. The alignment is relative to the point at which the text starts, as specified using MgraText() or using **MgraControlList**() with M_POSITION_X. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that the text is horizontally centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that the text is left-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that the text is right-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_VERTICAL	Sets the vertical alignment of the text. The alignment is relative to the point at which the text starts, as specified using MgraText() or using **MgraControlList**() with M_POSITION_Y. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that the text is bottom-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that the text is vertically centered.	c		e	f	g	a		i		j	k	l	m	n	b

M_TOP	Specifies that the text is top-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_BORDER	Sets borders around the text. Note that the possible settings can be combined. For example, to draw a box around the text, use M_TOP + M_BOTTOM + M_LEFT + M_RIGHT. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that a line is drawn underneath the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that a line is drawn to the left of the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_NONE	Specifies that no border is drawn around the text. This setting cannot be combined with any other setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that a line is drawn to the right of the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP	Specifies that a line is drawn above the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_DIRECTION	Sets the direction to draw the text, when using a TrueType font. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT_TO_RIGHT	Specifies that text will be drawn from left to right.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT_TO_LEFT	Specifies that text will be drawn from right to left.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Combination values for M_INTERACTIVE_ANNOTATIONS_COLOR; M_SELECTED_COLOR; M_COLOR; M_BACKCOLOR.

You can add one of the following values to the above-mentioned values to specify the color band.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the color band (for 16- or 32-bit color buffers)																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BLUE	Specifies the blue color band.	c		e	f	g	a		i		j	k	l	m	n	b
M_GREEN	Specifies the green color band.	c		e	f	g	a		i		j	k	l	m	n	b
M_RED	Specifies the red color band.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to change a graphic's characteristics that have been set using the Mgra...() function with which the graphic was added to the list. In this case, the LabelOrIndex parameter can be set to one or all graphics. Unless otherwise specified, the SubIndex parameter should be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For changing the position and dimensional information of a graphic

ControlType / ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ANGLE	Sets the angle of the graphic. This setting applies to all types of graphics except dot (MgraDot()), text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()). INQ															
-360.0 <= *Value* <= 360.0	Specifies the angle, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_END	Sets the angle at which to stop drawing the arc. This setting only applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the end angle, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_START	Sets the angle at which to start drawing the arc. This setting only applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b

Value	Specifies the start angle, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
M_ARC_STYLE	Sets whether to draw an arc or a sector. This setting only applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTOUR	Specifies that the arc (the curve between the specified start and end angles) is drawn without lines extending from the center of the ellipse to the start and end points of the arc.	c		e	f	g	a		i		j	k	l	m	n	b
M_SECTOR	Specifies that a sector is drawn with lines extending from the center of the ellipse to the start and end points of the arc, unless the specified start and end angles form a closed curve.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONSTRAIN_ASPECT_RATIO	Sets whether to force the width and height of a graphic to be equal. This affects both interactive and programmatic manipulations. This setting applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()) and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the width and height of a graphic are not forced to be equal.	c		e	f	g	a		i		j	k	l	m	n	b
1.0	Specifies that the width and height of a graphic are forced to be equal. This is considered a 1:1 (square) aspect ratio. Changing this setting on a graphic that does not meet this condition will change the dimensions of the graphic accordingly. Any attempts to change one of the dimensions, either programmatically or interactively, will scale the other so that the condition (1:1) is met.	c		e	f	g	a		i		j	k	l	m	n	b

M_FILLED	Sets whether to fill the graphic. This setting only applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), polygons (MgraLines() with M_POLYGON), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the graphic is not filled.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the graphic is filled.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_TYPE +	Sets how to interpret M_POSITION_X and M_POSITION_Y for a rectangle. This setting only applies to rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_AND_DIMENSION	Specifies to interpret M_POSITION_X and M_POSITION_Y as the rectangle's center.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_AND_DIMENSION	Specifies to interpret M_POSITION_X and M_POSITION_Y as the rectangle's top-left corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X	Sets the X-position of a graphic, or of one of its sub-elements (position-points). If SubIndex is set to M_DEFAULT, this setting affects the entire graphic. If a sub-element is specified, a single point, within the graphic, is affected by this setting. This setting applies to all types of graphics except drawings created using a processing or analysis module draw function (M...Draw()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-position. You can specify any positive or negative value.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y	Sets the Y-position of a graphic, or of one of its sub-elements (position-points). If SubIndex is set to M_DEFAULT, this setting affects the entire graphic. If a sub-element is specified, a single point, within the graphic, is affected by this setting. This setting applies to all types of graphics except drawings created using a processing or analysis module draw function (M...Draw()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-position. You can specify any positive or negative value.	c		e	f	g	a		i		j	k	l	m	n	b

M_RADIUS_X	Sets the radius of the arc in the X-direction. This setting only applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_Y	Sets the radius of the arc in the Y-direction. This setting only applies to arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE_HEIGHT	Sets the height of the rectangle. This setting only applies to rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the height.	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE_WIDTH	Sets the width of the rectangle. This setting only applies to rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). INQ	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the width.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to perform geometric operations on graphics. Using one of the following operations on a graphic will change its current position and dimensional settings (the settings listed in the For changing the position and dimensional information of a graphic table). Once an operation has occurred, the value specified cannot be inquired.

In this case, the LabelOrIndex parameter can be set to one or all graphics. Unless otherwise specified, the SubIndex parameter should be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing geometric operations on graphics																
ControlType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlValue																

M_APPLY_SCALE	Performs a uniform scaling operation on the graphic. A uniform scaling operation enlarges or shrinks graphics by a scale factor in both X- and Y-directions, relative to the point specified by M_POSITION_X and M_POSITION_Y; the point specified by M_POSITION_X and M_POSITION_Y remains the same after a uniform scaling operation. For example, if you specify a scale factor of one half to dots (MgraDots()), each dot will have its distance from M_POSITION_X and M_POSITION_Y reduced by one half. This setting applies to all types of graphics except dot (MgraDot()) and drawings created using a processing or analysis module draw function (M...Draw()).														
Value > 0.0	Specifies the scale factor.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESIZE_HEIGHT	Increases or decreases the graphic's height. This does not affect the graphic's X- and Y-position (M_POSITION_X and M_POSITION_Y remains the same). Aurora Imaging Library typically resizes the height on both sides of the graphic; therefore, the height is actually modified by twice the specified resize value. This does not occur for rectangle graphics whose position (M_POSITION_TYPE) is interpreted as its top-left corner (M_CORNER_AND_DIMENSION). In that case, Aurora Imaging Library only applies the resize value to the opposite side. This setting applies to all types of graphics except dot (MgraDot()), text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()).	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the value by which to resize the height.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESIZE_WIDTH	Increases or decreases the graphic's width. This does not affect the graphic's X- and Y-position (M_POSITION_X and M_POSITION_Y remains the same). Aurora Imaging Library typically resizes the width on both sides of the graphic; therefore, the width is actually modified by twice the specified resize value. This does not occur for rectangle graphics whose position (M_POSITION_TYPE) is interpreted as its top-left corner (M_CORNER_AND_DIMENSION). In that case, Aurora Imaging Library only applies the resize value to the opposite side. This setting applies to all types of graphics except dot (MgraDot()), text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()).	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the value by which to resize the width.	c		e	f	g	a		i		j	k	l	m	n	b

M_ROTATE	Performs a rotation operation on the graphic, or one of its sub-elements. Graphics are rotated around the point specified with M_POSITION_X and M_POSITION_Y. This setting applies to all types of graphics except dot (MgraDot()), text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()).	c		e	f	g	a		i		j	k	l	m	n	b
-360.0 <= *Value* <= 360.0	Specifies the angle of rotation, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. The value specified will be added to the graphic's current angle (M_ANGLE). An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSLATE_X	Performs a horizontal translation of a graphic, or of one of its sub-elements. If SubIndex is set to M_DEFAULT, this setting affects the entire graphic. If a sub-element is specified, a single point, within the graphic, is affected by this setting. This setting applies to all types of graphics except drawings created using a processing or analysis module draw function (M...Draw()).	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the horizontal displacement. You can specify any positive or negative value. The value specified will be added to the graphic's current X-position (M_POSITION_X).	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSLATE_Y	Performs a vertical translation of a graphic, or of one of its sub-elements. If SubIndex is set to M_DEFAULT, this setting affects the entire graphic. If a sub-element is specified, a single point, within the graphic, is affected by this setting. This setting applies to all types of graphics except drawings created using a processing or analysis module draw function (M...Draw()).	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the vertical displacement. You can specify any positive or negative value. The value specified will be added to the graphic's current Y-position (M_POSITION_Y).	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_POSITION_TYPE.

You can add the following value to the above-mentioned value to specify that a rectangle should stay at the same

location in the image regardless of how Aurora Imaging Library interprets its reference position.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying that a rectangle should stay at the same location regardless of how Aurora Imaging Library interprets its position																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_SAME_LOCATION	Specifies that, when you set M_POSITION_TYPE, Aurora Imaging Library also adjusts the rectangle's actual position in the image so it remains at the same location.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to delete graphics from the 2D graphics list. The LabelOrIndex parameter can be set to one or all graphics and the SubIndex parameter must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For deleting graphics (one or all) from the list																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_DELETE		Deletes the graphic (one or all), as specified by the LabelOrIndex parameter setting. When a graphic is deleted, the indices of the 2D graphics list are reassigned; that is, index values greater than that of the removed graphic are reduced by one.															
	M_DEFAULT	Implements the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to specify general settings for using keyboard keys to interactively modify graphics. In this case, set the LabelOrIndex parameter to M_LIST and the SubIndex parameter to M_DEFAULT. These settings apply to the 2D graphics list itself.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying general keyboard key settings to interactively modify graphics																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_ACTION_KEYS		Sets whether the 2D graphics list allows you to interactively modify its graphics by pressing keys on the keyboard (keyboard events). INQ																
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the 2D graphics list ignores keyboard events.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the 2D graphics list allows keyboard events to interactively modify its graphics. In this case, you must still enable the 2D graphics list's interactivity (MdispControl() with M_GRAPHIC_LIST_INTERACTIVE).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_MODIFIER_SPEED		Set the additional keyboard key that you can press to select an alternate speed to rotate, resize, or translate interactively selected graphics with the keyboard. This affects M_ACTION_KEY_ROTATE_..., M_ACTION_KEY_RESIZE_..., and M_ACTION_KEY_TRANSLATE_... settings. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_CTRL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ALT	Specifies to use the Alt key to select an alternate speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL	Specifies to use the Ctrl key to select an alternate speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SHIFT	Specifies to use the Shift key to select an alternate speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_RESIZE_INCREMENT		Sets the value with which to increase or decrease the width or height of the interactively selected graphic for a single key press. This affects the M_ACTION_KEY_RESIZE_... settings. Note that if the graphic is in world units, the increment is converted to world units using the calibration's average pixel size in the resizing direction. To keep the graphic's position fixed, Aurora Imaging Library typically applies this increment value to both sides of the graphic. Therefore, the graphic's width or height is actually modified by twice the specified value. This does not occur for rectangle graphics whose position (M_POSITION_TYPE) is interpreted as its top-left corner (M_CORNER_AND_DIMENSION). In that case, Aurora Imaging Library only applies the increment to the opposite side. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_RESIZE_INCREMENT_ALTERNATE	Sets an alternate value with which to increase or decrease the width or height of the interactively selected graphic for a single key press. M_ACTION_RESIZE_INCREMENT_ALTERNATE behaves like M_ACTION_RESIZE_INCREMENT. Aurora Imaging Library uses M_ACTION_RESIZE_INCREMENT_ALTERNATE instead of M_ACTION_RESIZE_INCREMENT when you press the key specified for M_ACTION_MODIFIER_SPEED while you resize the graphic using the keyboard. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_ROTATE_INCREMENT	Sets the angle with which to rotate the interactively selected graphic for a single key press. This affects the M_ACTION_KEY_ROTATE_... settings. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0°.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_ROTATE_INCREMENT_ALTERNATE	Sets an alternate angle with which to rotate the interactively selected graphic for a single key press. M_ACTION_ROTATE_INCREMENT_ALTERNATE behaves like M_ACTION_ROTATE_INCREMENT. Aurora Imaging Library uses M_ACTION_ROTATE_INCREMENT_ALTERNATE instead of M_ACTION_ROTATE_INCREMENT when you press the key specified for M_ACTION_MODIFIER_SPEED while you rotate the graphic using the keyboard. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 15.0°.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_TRANSLATE_AXES	Set the axes along which Aurora Imaging Library translates a graphic that is in world units, when the translation is done using the keyboard. This affects M_ACTION_TRANSLATE_INCREMENT and M_ACTION_TRANSLATE_INCREMENT_ALTERNATE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PIXEL.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ACCORDING_TO_CALIBRATION	Specifies to convert the translation increment to world units using the calibration's average pixel size in the translation direction, then apply it along the world axes.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies to apply the translation increment to the graphic position, along the image axes, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_TRANSLATE_INCREMENT		Sets the value with which to translate (move) the interactively selected graphics for a single key press. This affects the M_ACTION_KEY_TRANSLATE_... settings. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display). Note that if the graphic is in world units, the behavior depends on M_ACTION_TRANSLATE_AXES.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_TRANSLATE_INCREMENT_ALTERNATE		Sets an alternate value with which to translate (move) the interactively selected graphics for a single key press. M_ACTION_TRANSLATE_INCREMENT_ALTERNATE behaves like M_ACTION_TRANSLATE_INCREMENT. Aurora Imaging Library uses M_ACTION_TRANSLATE_INCREMENT_ALTERNATE instead of M_ACTION_TRANSLATE_INCREMENT when you press the key specified for M_ACTION_MODIFIER_SPEED while you move the graphic using the keyboard. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 10.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display). Note that if the graphic is in world units, the increment depends on M_ACTION_TRANSLATE_AXES.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to specify explicit keyboard keys to interactively modify graphics. In this case, set the LabelOrIndex parameter to M_LIST and the SubIndex parameter to M_DEFAULT. These settings apply to the 2D graphics list itself.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying explicit keyboard keys to interactively modify graphics																	
ControlType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)	

	ControlValue															
M_ACTION_KEY_CANCEL		Sets the keyboard keys that you can press to cancel the interactive graphic modification you are currently performing. Aurora Imaging Library recommends using the Esc key. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. If you press the specified cancel key while you are interactively translating (moving), resizing, or rotating the graphic, Aurora Imaging Library returns the graphic to its initial state (before you started interacting with it). If you press the specified cancel key while interactively creating the graphic, Aurora Imaging Library removes from the display any part of the graphic you started to define, and also removes the graphic from the 2D graphics list (the graphic is typically added to the 2D graphics list before it is entirely defined). If you press the specified cancel key when there is no interaction to cancel, Aurora Imaging Library deselects all selected graphics. M_ACTION_KEY_CANCEL only has an effect if you enable M_ACTION_KEYS. INQ														
	M_DEFAULT	Same as M_NONE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_DELETE		Sets the keyboard keys that you can press to delete the interactively selected graphics. Aurora Imaging Library recommends using the Delete key. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_DELETE only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NONE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_HEIGHT_DOWN		Sets the keyboard keys that you can press to reduce the height of the interactively selected graphics. Aurora Imaging Library recommends using the Down Arrow key and the Shift key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_RESIZE_HEIGHT_DOWN only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_KEY_ARROW_DOWN + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_HEIGHT_UP		Sets the keyboard keys that you can press to increase the height of the interactively selected graphics. Aurora Imaging Library recommends using the Up Arrow key and the Shift key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_RESIZE_HEIGHT_UP only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_UP + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_WIDTH_DOWN		Sets the keyboard keys that you can press to reduce the width of the interactively selected graphics. Aurora Imaging Library recommends using the Left Arrow key and the Shift key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_RESIZE_WIDTH_DOWN only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_WIDTH_UP		Sets the keyboard keys that you can press to increase the width of the interactively selected graphics. Aurora Imaging Library recommends using the Right Arrow key and the Shift key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_RESIZE_WIDTH_UP only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ROTATE_CLOCKWISE		Sets the keyboard keys that you can press to rotate the interactively selected graphics clockwise. Aurora Imaging Library recommends using the Right Arrow key and the Alt key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_ROTATE_CLOCKWISE only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ROTATE_COUNTER_CLOCKWISE		Sets the keyboard keys that you can press to rotate the interactively selected graphics counter-clockwise. Aurora Imaging Library recommends using the Left Arrow key and the Alt key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_ROTATE_COUNTER_CLOCKWISE only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_DOWN		Sets the keyboard keys that you can press to translate (move) the interactively selected graphics down. Aurora Imaging Library recommends specifying the Down Arrow key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_TRANSLATE_DOWN only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_KEY_ARROW_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_LEFT		Sets the keyboard keys that you can press to translate (move) the interactively selected graphics left. Aurora Imaging Library recommends specifying the Left Arrow key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_TRANSLATE_LEFT only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_RIGHT		Sets the keyboard keys that you can press to translate (move) the interactively selected graphics right. Aurora Imaging Library recommends specifying the Right Arrow key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_TRANSLATE_RIGHT only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_UP		Sets the keyboard keys that you can press to translate (move) the interactively selected graphics up. Aurora Imaging Library recommends specifying the Up Arrow key, which is the default. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. M_ACTION_KEY_TRANSLATE_UP only has an effect if you enable M_ACTION_KEYS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_UP.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the ControlValue parameter listed in For specifying explicit keyboard keys to interactively modify graphics.

You can add one or more of the following values to the above-mentioned values to specify an additional keyboard key.

For example, to specify that you must press the Right Arrow key while pressing the Alt key to rotate the interactively selected graphic clockwise, set the ControlType parameter to M_ACTION_KEY_ROTATE_CLOCKWISE, and set the ControlValue parameter to M_KEY_ARROW_RIGHT + M_KEY_ALT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying additional keyboard keys																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_KEY_ALT	Specifies the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CTRL	Specifies the Ctrl key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_SHIFT	Specifies the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraControlListDouble (AIL_ID GraListId, AIL_INT LabelOrIndex, AIL_INT SubIndex, AIL_INT64 ControlType,
AIL_DOUBLE ControlValue)
```

Details

Parameters

GraListId

See GraListId of the main function for a description.

LabelOrIndex

See LabelOrIndex of the main function for a description.

SubIndex

See SubIndex of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void MgraControlListInt64 (AIL_ID GraListId, AIL_INT LabelOrIndex, AIL_INT SubIndex, AIL_INT64 ControlType,
AIL_INT64 ControlValue)
```

Details

Parameters

GraListId

See GraListId of the main function for a description.

LabelOrIndex

See LabelOrIndex of the main function for a description.

SubIndex

See SubIndex of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraCopy

Synopsis

Copy one or more graphics from one 2D graphics list to another.

Syntax

```
void MgraCopy(
    AIL_ID SrcListGraId,                    //in
    AIL_ID DstListGraId,                    //out
    AIL_INT64 Operation,                    //in
    AIL_INT InsertLocation,                 //in
    AIL_INT NumGraphics,                    //in
    const AIL_INT *SrcIndexOrLabelArrayPtr, //in
    AIL_INT *DstLabelArrayPtr,              //out
    AIL_INT64 ControlFlag                   //in
)
```

Description

This function copies or moves one or more graphics from a source 2D graphics list to a destination 2D graphics list. When moved, the graphics are erased from the source 2D graphics list. You can specify the location in the destination 2D graphics list at which to insert the copied graphics, which offers some control on the drawing order.

Note that if you want to replace all existing elements of the destination 2D graphics list, you must clear it first using MgraClear().

This function supports in-place processing: the source and destination 2D graphics list identifiers can refer to the same 2D graphics list. In the case of an in-place move operation (M_MOVE), the graphic referred to by the InsertLocation parameter must not be one of the graphics being moved (SrcIndexOrLabelArrayPtr).

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

SrcListGraId

Specifies the identifier of a valid 2D graphics list from which to copy or move the graphics. You must have allocated the 2D graphics list using MgraAllocList().

DstListGraId

Specifies the identifier of a valid 2D graphics list in which to copy or move the graphics. You must have allocated the 2D graphics list using MgraAllocList(). In-place processing is supported; DstListGraId can refer to the same 2D graphics list as SrcListGraId.

Operation

Specifies the type of copy operation to perform. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of copy operation to perform																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Same as M_COPY + M_LABEL_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b
M_COPY +	Specifies graphics from the source 2D graphics list are copied to the destination 2D graphics list. The graphics are not deleted from the source 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOVE +	Specifies graphics from the source 2D graphics list are moved to the destination 2D graphics list. The graphics are deleted from the source 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For specifying the type of copy operation to perform.

You can add one of the following values to the above-mentioned values to specify whether indices or labels are given to the SrcIndexOrLabelArrayPtr parameter.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to use indices or labels																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INDEX_VALUE	Specifies index values are given to the SrcIndexOrLabelArrayPtr parameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE	Specifies labels are given to the SrcIndexOrLabelArrayPtr parameter. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

InsertLocation

Specifies the location inside the destination 2D graphics list at which to insert the graphics. Note that the indices or labels passed to this parameter must refer to valid graphics. Otherwise, an error is generated.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the insert location in the destination 2D graphics list																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_END_OF_LIST.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_INDEX(AIL_INT Index)	Specifies that the graphics will be inserted before the graphic with the specified index in the destination 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

	Index Specifies the index. The index is from 0 (inclusive) to the total number of graphics in the list minus 1. If the given index value is not in that range, an error is generated. You can get the number graphics in the list using MgraInquireList() with M_NUMBER_OF_GRAPHICS. You can get the index corresponding to the label of a graphic using MgraInquireList() with M_INDEX_VALUE.														
M_GRAPHIC_LABEL(AIL_INT Label)	Specifies that the graphics will be inserted before the graphic with the specified label in the destination 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	Label Specifies the label. The label must be greater than 0. If the label does not exist in the destination 2D graphics list, an error is generated. You can get the label of the last graphic added to the list using MgraInquireList() with M_LAST_LABEL. You can get the label corresponding to the index of a graphic using MgraInquireList() with M_LABEL_VALUE.														
M_END_OF_LIST	Specifies that the graphics are appended to the end of the destination 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b

NumGraphics

Specifies the number of graphics to copy or move.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the SrcIndexOrLabelArrayPtr parameter.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the number of graphics to copy or move																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ALL	Specifies that all the graphics from the source 2D graphics list must be copied or moved.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of graphics from the source 2D graphics list to be copied or moved.	c		e	f	g	a		i		j	k	l	m	n	b

SrcIndexOrLabelArrayPtr
Accepts the address of one of the following:

- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumGraphics)
- M_NULL

Specifies the address of the array containing the indices or labels of the graphics to copy from the source 2D

graphics list. If the M_INDEX_VALUE combination constant is used with the Operation parameter, Aurora Imaging Library expects the array to contain graphics indices. Otherwise, Aurora Imaging Library expects it to contain graphics labels.

The array must contain at least NumGraphics indices or labels. If NumGraphics is set to M_ALL, you must set this parameter to **M_NULL**.

Note that the indices or labels passed to this parameter must refer to valid graphics. Otherwise, an error is generated.

DstLabelArrayPtr
Accepts the address of one of the following:

- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumGraphics)
- M_NULL

Specifies the address of the array in which to write the labels that have been automatically given to the copied or moved graphics in the destination 2D graphics list. The labels are written in the same order as the graphics are specified in the SrcIndexOrLabelArrayPtr parameter. If NumGraphics is set to M_ALL, the labels are written in index order.

The array must be at least as large as the number of graphics being moved or copied (NumGraphics). If NumGraphics is set to M_ALL, determine the required size of the array using MgraInquireList() with M_NUMBER_OF_GRAPHICS.

If you do not need the labels of the copied or moved graphics, this parameter can be set to **M_NULL**.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraDot

Synopsis

Draw a dot in an image or add a dot to a 2D graphics list.

Syntax

```
void MgraDot(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XPos,               //in
    AIL_DOUBLE YPos                //in
)
```

Description

This function draws a dot destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of a dot to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

The dot is based on a point positioned at XPos and YPos. The dot inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If the dot falls outside of the specified area, it is not created.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

The dot's coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraDot**() an uncalibrated image, the function will generate an error.

To create one or more dots, use MgraDots().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraDot()** internally calls the *AIL_DOUBLE* version of this function (MgraDotDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraDotInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the dot or the identifier of a valid 2D graphics list in which to add the dot. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XPos INQ

Specifies the X-coordinate of the dot in the input coordinate system.

YPos INQ

Specifies the Y-coordinate of the dot in the input coordinate system.

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraDotDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XPos, AIL_DOUBLE YPos)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XPos

See XPos of the main function for a description.

YPos

See YPos of the main function for a description.

```
void MgraDotInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XPos, AIL_INT64 YPos)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XPos

See XPos of the main function for a description.

YPos

See YPos of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraDots

Synopsis

Draw one or more dots in an image, or add them to a 2D graphics list.

Syntax

```
void MgraDots(
    AIL_ID ContextGraId,            //in
    AIL_ID DstImageBufOrListGraId,  //out
    AIL_INT NumberOfDots,           //in
    const AIL_DOUBLE *PosXArrayPtr, //in
    const AIL_DOUBLE *PosYArrayPtr, //in
    AIL_INT64 ControlFlag           //in
)
```

Description

This function draws one or more dots destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of the dots to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

Dots are based on the points positioned at PosXArrayPtr and PosYArrayPtr. The dots inherit all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the dots fall outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

The coordinates of the dots are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraDots**() an uncalibrated image, the function will generate an error.

To create a single dot without the option of creating multiple dots, use MgraDot().

Note that prior to Aurora Imaging Library 9.0, this function only supported integer positions; the PosXArrayPtr and PosYArrayPtr parameters only accepted arrays of type *long*. As of Aurora Imaging Library 9.0, this function includes support for positions with floating-point precision. To implement this change and maintain backwards compatibility when using a C++ compiler (*.cpp), **MgraDots**() is available as an inline function which automatically calls MgraDotsDouble(), MgraDotsInt32(), and, MgraDotsInt64() depending on whether PosXArrayPtr and PosYArrayPtr receive arrays of type *AIL_DOUBLE*, *AIL_INT32*, or *AIL_INT64*, respectively. To maintain backwards compatibility when using a C compiler (*.c), **MgraDots**() maps to MgraDotsInt32() when working on a 32-bit system, or MgraDotsInt64() when working on a 64-bit system; you must explicitly call MgraDotsDouble() to pass PosXArrayPtr and PosYArrayPtr arrays of type *AIL_DOUBLE*. If you are an advanced user and want to retrieve a pointer to **MgraDots**(), you must use the Double, Int64, or Int32 version of this function, since **MgraDots**() is actually a macro or an overloaded function.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that presented above is an example prototype of this function. You can call MgraDots() with the same prototype as any of the type-specific functions below, without using the function's type-specific name; that is, use MgraDots() and not, for example, MgraDotsInt32(). The data is automatically cast into the appropriate type. This is true except when using a C compiler. When using a C compiler, you must always call this function with its

> type-specific name (you cannot use the standard name of this function).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the identifier of the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies the identifier of the 2D graphics context, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the dots or the identifier of a valid 2D graphics list in which to add the dots. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

NumberOfDots

Specifies the number of dots to draw.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the PosXArrayPtr or PosYArrayPtr parameter.

PosXArrayPtr INQ
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumberOfDots)

Specifies the address of the array containing the X-coordinate(s) of the dots to be drawn in the input coordinate system.

PosYArrayPtr INQ
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumberOfDots)

Specifies the address of the array containing the Y-coordinate(s) of the dots to be drawn in the input coordinate system.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Type-specific versions of the function when using a C compiler

```
void MgraDotsDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumberOfDots, const AIL_DOUBLE
*PosXArrayPtr, const AIL_DOUBLE *PosYArrayPtr, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumberOfDots

See NumberOfDots of the main function for a description.

PosXArrayPtr

See PosXArrayPtr of the main function for a description.

PosYArrayPtr

See PosYArrayPtr of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraDotsInt32 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumberOfDots, const AIL_INT32 *PosXArrayPtr, const AIL_INT32 *PosYArrayPtr, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumberOfDots

See NumberOfDots of the main function for a description.

PosXArrayPtr

See PosXArrayPtr of the main function for a description.

PosYArrayPtr

See PosYArrayPtr of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraDotsInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumberOfDots, const AIL_INT64 *PosXArrayPtr, const AIL_INT64 *PosYArrayPtr, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumberOfDots

See NumberOfDots of the main function for a description.

PosXArrayPtr

See PosXArrayPtr of the main function for a description.

PosYArrayPtr

See PosYArrayPtr of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraDraw

Synopsis

Draw graphics, contained in a 2D graphics list, in an image.

Syntax

```
void MgraDraw(
    AIL_ID GraListId,     //in
    AIL_ID DestImageId,   //out
    AIL_INT64 ControlFlag //in
)
```

Description

This function draws graphics, contained in the specified 2D graphics list, destructively (raster-based) in the specified image.

A graphic's position and dimension values are interpreted with respect to the input coordinate system, specified using MgraControl() with M_INPUT_UNITS. Note that you can also use MgraControlList() with M_INPUT_UNITS to modify the input units of graphics that have already been added to the 2D graphics list.

If you attempt to draw the graphics contained in the 2D graphics list using world units, but without passing **MgraDraw**() a calibrated image, the function will generate an error.

If parts of the graphics fall outside of the specified image buffer area, those parts are clipped off.

If you are drawing graphics in a 2D graphics list with interactive mode enabled (MdispControl() with M_GRAPHIC_LIST_INTERACTIVE set to M_ENABLE), selected graphics (graphics with M_GRAPHIC_SELECTED set to M_TRUE) are typically surrounded by annotations that indicate the graphic is selected, such as the selection box (represented by a dotted line) with handles that allow user interaction. By default, this function will draw the annotations that indicate the graphic is selected, unless otherwise specified.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (GraListId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

GraListId

Specifies the identifier of the 2D graphics list containing the graphics to destructively draw in the image. The 2D graphics list must have been previously allocated on the required system using MgraAllocList().

DestImageId

Specifies the identifier of the destination image buffer in which to draw. If the 2D graphics list contains graphics set in world units, a camera calibration context must be associated with the destination image buffer; otherwise, an error is generated.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

Note that you can also annotate an image non-destructively by drawing into its display's overlay buffer.

ControlFlag

Specifies whether to also draw the annotations that indicate whether a graphic is selected, such as the selection box and handles. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether to draw annotations that indicate a graphic is selected																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that annotations that indicate whether a graphic is selected will be drawn destructively in the image buffer, along with the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_INTERACTIVE_ANNOTATION	Specifies that annotations that indicate whether a graphic is selected will not be drawn in the image buffer, even if there are selected graphics in the 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraFill

Synopsis

Perform a boundary-type seed fill.

Syntax

```
void MgraFill(
    AIL_ID ContextGraId,  //in
    AIL_ID DstImageBufId, //out
    AIL_DOUBLE XStart,    //in
    AIL_DOUBLE YStart     //in
)
```

Description

This function performs a boundary-type seed fill. It fills in an area of the specified image with the foreground color of the specified 2D graphics context, starting from the seed position (XStart, YStart). Filling occurs on adjacent pixels (vertically and horizontally to original seed pixel) that have the same value as the original seed pixel.

If **MgraFill**() is used on a multi-band buffer, each band will be processed separately. This means that each band of the adjacent pixels will be compared with the corresponding band of the seed pixel. This can produce strange results if, for example, you try to fill the inside of a red circle with blue. The blue will spread to the whole image since the red circle does not exist in the blue band.

A seed's coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraFill**() an uncalibrated image, the function will generate an error.

Note that **MgraFill**() cannot be performed on graphics contained within a 2D graphics list.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraFill()** internally calls the *AIL_DOUBLE* version of this function (MgraFillDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraFillInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufId

Specifies the identifier of a valid image buffer in which to perform the filling operation. You must have allocated the image buffer using MbufAlloc...().

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XStart

Specifies the X-coordinate of the seed position in the input coordinate system. If the specified point is not within an enclosed area, filling occurs until the boundaries of the buffer are encountered.

YStart

Specifies the Y-coordinate of the seed position in the input coordinate system. If the specified point is not within an enclosed area, filling occurs until the boundaries of the buffer are encountered.

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraFillDouble (AIL_ID ContextGraId, AIL_ID DstImageBufId, AIL_DOUBLE XStart, AIL_DOUBLE YStart)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufId

See DstImageBufId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

```
void MgraFillInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufId, AIL_INT64 XStart, AIL_INT64 YStart)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufId

See DstImageBufId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraFont

Synopsis

Associate a text font with a 2D graphics context.

Syntax

```
void MgraFont(
    AIL_ID ContextGraId, //out
    AIL_INT64 FontName   //in
)
```

Description

This function associates a character font with the specified 2D graphics context for use with subsequent MgraText() function calls.

To change the font for graphics already added to a 2D graphics list, use MgraControlList().

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context with which to associate the character font. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

FontName INQ

Specifies the font with which to write text. This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the font																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FONT_DEFAULT	Same as M_FONT_DEFAULT_SMALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_LARGE	Specifies a large bitmap font, where each character is drawn in a 16x32 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b

M_FONT_DEFAULT_MEDIUM	Specifies a medium bitmap font, where each character is drawn in a 12x24 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_SMALL	Specifies a small bitmap font, where each character is drawn in a 8x16 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_TTF	Specifies the default TrueType font of your operating system. > Note, if using a C compiler, you must pass the value as AIL_FONT_NAME(M_FONT_DEFAULT_TTF).	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FontFamily:Weight:Slant*") 1	Specifies the font according to the following format, *Family*: *Weight*: *Slant*. *Family* must be set to the name of the font's family, such as Arial, Times New Roman, and Wingdings. *Weight* can be set to one of the following: Book, Thin, ExtraLight, UltraLight, Light, Normal, Regular, Medium, SemiBold, DemiBold, Bold, ExtraBold, UltraBold, Heavy, Black, ExtraBlack, or UltraBlack. *Slant* can be set to one of the following: Italic, Oblique, or Roman. You can omit *Weight* and *Slant*; also, you can provide the *Weight* and *Slant* in any order. > Note, if using a C compiler, you must pass the value as AIL_FONT_NAME(AIL_TEXT("*FontFamily:Weight:Slant*")).	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FontFile*") 1	Specifies a full path to a TrueType file name. > Note, if using a C compiler, you must pass the value as AIL_FONT_NAME(AIL_TEXT("*FontFile*")).	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraFontScale

Synopsis

Set the font scale of a 2D graphics context.

Syntax

```
void MgraFontScale(
    AIL_ID ContextGraId,   //out
    AIL_DOUBLE XFontScale, //in
    AIL_DOUBLE YFontScale  //in
)
```

Description

This function sets both the vertical and horizontal scaling factor settings of the specified 2D graphics context for use with subsequent MgraText() function calls. These settings only apply to bitmap fonts. When using TrueType fonts, use MgraControl() with M_FONT_SIZE to set the font size.

To change the font scale for graphics already added to a 2D graphics list, use MgraControlList().

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context for which to set the font scale. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

XFontScale INQ

Specifies the X-scale-factor by which to multiply the width of the font characters. This parameter can be independently set to any positive floating-point value. The default scale factor is 1.0.

> Note, using a font with a scale of 1.0 accelerates text drawing.

YFontScale INQ

Specifies the Y-scale-factor by which to multiply the height of the font characters. This parameter can be independently set to any positive floating-point value. The default scale factor is 1.0.

> Note, using a font with a scale of 1.0 accelerates text drawing.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraFree

Synopsis

Free a 2D graphics context or a 2D graphics list.

Syntax

```
void MgraFree(
    AIL_ID ObjectId //in
)
```

Description

This function deallocates a 2D graphics context or a 2D graphics list previously allocated with MgraAlloc() or MgraAllocList().

If you free a 2D graphics list that is annotating the display non-destructively (specified using MdispControl() with M_ASSOCIATED_GRAPHIC_LIST_ID), the 2D graphics list is automatically disassociated from the display.

All 2D graphics contexts or 2D graphics lists allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ObjectId

Specifies the identifier of the 2D graphics context or 2D graphics list to deallocate. If **M_DEFAULT** is specified, an error will occur.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraGetHookInfo

Synopsis

Get information about a 2D graphics list event.

Syntax

```
AIL_INT MgraGetHookInfo(
    AIL_ID EventId,     //in
    AIL_INT64 InfoType, //in
    void *UserVarPtr    //out
)
```

Description

This function allows you to get information about the event that caused the hook-handler function to be called. **MgraGetHookInfo**() should only be called within the scope of a 2D graphics list hook-handler function (see MgraHookFunction()).

Parameters

EventId

Specifies the 2D graphics list event identifier received by the hook-handler function (see MgraHookFunction()). The event will be of the type passed to the HookType parameter of the hook-handler function.

See the Parameter associations section for possible values that can be specified.

InfoType

Specifies the type of information to get.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The table below lists possible values for the **EventId** and **InfoType** parameters and possible values returned to the **UserVarPtr** parameter.

- For getting information about a 2D graphics list event

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For getting information about a 2D graphics list event

EventId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InfoType																
UserVarPtr - Possible values returned																
An M_GRAPHIC_LIST_MODIFIED event ID +	Specifies that the event that called the hook-handler function was due to the 2D graphics list being modified, either by an Aurora Imaging Library function or by interactive manipulations.															
M_GRAPHIC_LIST_ID +	Retrieves the Aurora Imaging Library identifier of the 2D graphics list that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
Graphics list identifier	Retrieves the Aurora Imaging Library identifier of the 2D graphics list containing the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
An M_GRAPHIC_MODIFIED event ID +	Specifies that the event that called the hook-handler function was due to a graphic in the 2D graphics list being modified, either by an Aurora Imaging Library function or by interactive manipulations. If a graphic's selection (MgraControlList() with M_GRAPHIC_SELECTED) is changed either programmatically or interactively, this event will not be triggered; the M_GRAPHIC_SELECTION_MODIFIED event will be triggered instead.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_CONTROL_TYPE +	Inquires the type of modification made to the graphics in the list that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE	Specifies that the angle of the graphic was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_END	Specifies that the angle at which to stop drawing the arc was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_ANGLE_SNAPPING_VALUE	Specifies that the angular value used as a multiple when rotating the graphic was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ANGLE_START	Specifies that the angle at which to start drawing the arc was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_APPLY_SCALE	Specifies that a uniform scaling operation was performed on the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ARC_STYLE	Specifies that the option of whether to draw an arc or a sector was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_BACKCOLOR	Specifies that the background color of the text was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_BACKGROUND_MODE	Specifies that the option of whether to fill the text's background was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR	Specifies that the foreground color of the graphic was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_CONSTRAIN_ASPECT_RATIO	Specifies that the option of whether to force the width and height of a graphic to be equal was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_DELETE	Specifies that one of the following has been called on the 2D graphics list, resulting in some graphics being removed from the 2D graphics list: MgraControlList() with M_DELETE, MgraClear(), or when MgraCopy() with M_MOVE deletes graphics from a source 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DRAW_DIRECTION	Specifies that the option of how an arrow, pointing in the direction with which a graphic was defined is rendered (drawn), was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DRAW_OFFSET_X	Specifies that the offset to subtract from the source X-coordinates before rendering the graphic was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_DRAW_OFFSET_Y	Specifies that the offset to subtract from the source Y-coordinates before rendering the graphic was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_DRAW_ZOOM_X	Specifies that the scaling factor in the X-direction was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_DRAW_ZOOM_Y	Specifies that the scaling factor in the Y-direction was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_EASY_SELECTION	Specifies that the option of whether Aurora Imaging Library allows the easy selection of graphics on the display was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_EDITABLE	Specifies that the option of whether a graphic can be edited, via user interaction in an interactive display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FILLED	Specifies that the option of whether to fill the graphic was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FIXTURE	Specifies that the camera calibration information used when rendering (drawing or displaying) a graphic that has been defined in world units (M_INPUT_UNITS set to M_WORLD) was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FONT	Specifies that the font of the characters in the text was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FONT_AUTO_SELECT	Specifies that the option of whether Aurora Imaging Library should search for a suitable font to draw the text, if the currently selected font (M_FONT) is a TrueType font that does not support the character code, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FONT_SIZE	Specifies that the font size, when using a TrueType font, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FONT_X_SCALE	Specifies that the font's horizontal scaling factor was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_FONT_Y_SCALE	Specifies that the font's vertical scaling factor was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_CONVERSION_MODE	Specifies that how the shape of a graphic, defined in world units, is converted to pixels (rendered), was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_CREATE	Specifies that a new graphic has been added to the 2D graphics list, either using an Aurora Imaging Library function or through interactive manipulations.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_INTERACTIVE	Specifies that interactive manipulations, such as translating, rotating, and resizing, have changed the properties of a graphic in the 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_SOURCE_CALIBRATION	Specifies that the identifier of the camera calibration information, used to interpret positioning and dimensioning information of a graphic defined in world units, was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INPUT_UNITS	Specifies that the unit with which to interpret the graphic's positional and dimensional information, for graphics already added to the 2D graphics list, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_INTERACTIVE_ANNOTATIONS_COLOR	Specifies that the color of the selection box and handles in interactive mode was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LINE_THICKNESS	Specifies that the thickness of the lines and the diameter of dots, depending on the drawing operation was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE	Specifies that the interactive behavior, when resizing the graphic using its selection box, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE_ALT	Specifies that the interactive behavior, when resizing the graphic using its selection box and pressing the Alt key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_MODE_RESIZE_CTRL	Specifies that the interactive behavior, when resizing the graphic using its selection box and pressing the Ctrl key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE_SECONDARY_DIMENSION	Specifies that the option of whether to establish the graphic's secondary dimension, when defining the graphic interactively on the display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE_SECONDARY_DIMENSION_ALT	Specifies that the option of whether to establish the graphic's secondary dimension, when defining the graphic interactively on the display and pressing the Alt key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE_SECONDARY_DIMENSION_CTRL	Specifies that the option of whether to establish the graphic's secondary dimension, when defining the graphic interactively on the display and pressing the Ctrl key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE_SECONDARY_DIMENSION_SHIFT	Specifies that the option of whether to establish the graphic's secondary dimension, when defining the graphic interactively on the display and pressing the Shift key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_RESIZE_SHIFT	Specifies that the interactive behavior, when resizing the graphic using its selection box and pressing the Shift key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_ROTATE	Specifies that the interactive behavior, when rotating the graphic using its selection box, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_ROTATE_ALT	Specifies that the interactive behavior, when rotating the graphic using its selection box and pressing the Alt key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_ROTATE_CTRL	Specifies that the interactive behavior, when rotating the graphic using its selection box and pressing the Ctrl key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_MODE_ROTATE_SHIFT	Specifies that the interactive behavior, when rotating the graphic using its selection box and pressing the Shift key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_TRANSLATE	Specifies that the interactive behavior, when moving (translating) the graphic using its selection box, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_TRANSLATE_ALT	Specifies that the interactive behavior, when moving (translating) the graphic using its selection box and pressing the Alt key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_TRANSLATE_CTRL	Specifies that the interactive behavior, when moving (translating) the graphic using its selection box and pressing the Ctrl key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MODE_TRANSLATE_SHIFT	Specifies that the interactive behavior, when moving (translating) the graphic using its selection box and pressing the Shift key, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MOVE	Specifies that a call to MgraCopy(), with an in-place move (M_MOVE) operation, changed the order of the graphics inside the 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MULTIPLE_SELECTION	Specifies that the option of whether to permit interactive multiple selection using the Ctrl key was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MULTIPLE_SELECTION_KEY	Specifies that the keyboard key that you can press to select multiple graphics with your mouse on the display was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_POSITION_TYPE +	Specifies that how M_POSITION_X and M_POSITION_Y are interpreted for a rectangle was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_POSITION_X	Specifies that the X-position of a graphic, or of one of its sub-elements (position-points), was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_POSITION_Y	Specifies that the Y-position of a graphic, or of one of its sub-elements (position-points), was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_RADIUS_X	Specifies that the radius of the arc in the X-direction was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_RADIUS_Y	Specifies that the radius of the arc in the Y-direction was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_RECTANGLE_HEIGHT	Specifies that the height of the rectangle was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_RECTANGLE_WIDTH	Specifies that the width of the rectangle was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_RESIZABLE	Specifies that the option of whether a graphic can be resized, via user interaction in an interactive display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_RESIZE_HEIGHT	Specifies that the graphic's height was increased or decreased.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RESIZE_WIDTH	Specifies that the graphic's width was increased or decreased.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ROTATABLE	Specifies that the option of whether a graphic can be rotated, via user interaction in an interactive display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ROTATE	Specifies that a rotation operation was performed on the graphic, or one of its sub-elements.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SELECTABLE	Specifies that the option of whether a graphic can be selected, via user interaction in an interactive display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SELECTED_COLOR	Specifies that the color of the selected graphics in interactive mode was modified.	c		e	f	g	a		i		j	k	l	m	n	b

		M_SELECTION_RADIUS	Specifies that the size of the selection-radius in interactive mode was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SPECIFIC_FEATURES_EDITABLE	Specifies that the option of whether a graphic can be modified, using handles that are specific to its graphic type via user interaction in an interactive display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TEXT_ALIGN_HORIZONTAL	Specifies that the horizontal alignment of the text was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TEXT_ALIGN_VERTICAL	Specifies that the vertical alignment of the text was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TEXT_BORDER	Specifies that the border drawn around the text was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TEXT_DIRECTION	Specifies that the direction to draw the text, when using a TrueType font, was modified.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRANSLATABLE	Specifies that the option of whether a graphic can be translated (moved), via user interaction in an interactive display, was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRANSLATE_X	Specifies that a horizontal translation of a graphic, or of one of its sub-elements, was performed. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRANSLATE_Y	Specifies that a vertical translation of a graphic, or of one of its sub-elements, was performed. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_VISIBLE	Specifies that the option of whether a graphic is rendered on the display was modified. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LABEL_VALUE +		Retrieves the label of the modified graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_ALL	Specifies that all graphics in the 2D graphics list were modified using MgraClear() or MgraControlList() with M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ALL_SELECTED	Specifies that all selected graphics in the 2D graphics list were modified using MgraControlList() with M_ALL_SELECTED.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LIST	Specifies that a 2D graphics list's setting was modified using MgraControlList() with M_LIST.	c		e	f	g	a		i		j	k	l	m	n	b
		M_MULTIPLE_LABELS	Specifies that more than one graphic was modified. This can happen when calling MgraCopy() and with interactive manipulations (for example, translating many graphics).	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the label of the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_ID +		Retrieves the Aurora Imaging Library identifier of the 2D graphics list containing the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
		Graphics list identifier	Retrieves the Aurora Imaging Library identifier of the 2D graphics list containing the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_SUB_INDEX +		Retrieves the index of the sub-element of the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies that the event affected the whole graphic.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the index of the sub-element of the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
An M_GRAPHIC_SELECTION_MODIFIED event ID +			Specifies that the event that called the hook-handler function was due to a graphic being selected or deselected.	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_LABEL_VALUE +	Retrieves the label of the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_MULTIPLE_LABELS	Specifies that more than one graphic was selected. This can happen when calling MgraControlList() with M_GRAPHIC_SELECTED and with interactive manipulations (for example, Ctrl-clicking many graphics).	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_LABEL	Specifies a deselection triggered the event and no graphic was selected.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the label of the graphic that was selected.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LABEL_VALUE_DESELECTED +	Retrieves the label of the graphic that was deselected when the event was triggered.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_MULTIPLE_LABELS	Specifies that more than one graphic was deselected.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_LABEL	Specifies that no graphic was deselected.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the label of the graphic that was deselected.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_ID +	Retrieves the Aurora Imaging Library identifier of the 2D graphics list containing the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
Graphics list identifier	Retrieves the Aurora Imaging Library identifier of the 2D graphics list containing the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b

An M_INTERACTIVE_GRAPHIC_STATE_MODIFIED event ID +	Specifies that the event that called the hook-handler function was due to the 2D graphics list's interactive state being modified, either by an Aurora Imaging Library function or by interactive manipulations.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LABEL_VALUE +	Retrieves the label of the graphic that triggered the event. For example, it can return the label of the graphic being dragged when the state of interactivity (M_INTERACTIVE_GRAPHIC_STATE) is M_STATE_GRAPHIC_DRAGGED.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_LABEL	Specifies that an Aurora Imaging Library function triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the label of the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_ID +	Retrieves the Aurora Imaging Library identifier of the 2D graphics list that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
Graphics list identifier	Retrieves the Aurora Imaging Library identifier of the 2D graphics list containing the graphic that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_GRAPHIC_PREVIOUS_STATE +	Inquires the state of interactivity of the 2D graphics list before the event occurred. This always returns a different value than M_INTERACTIVE_GRAPHIC_STATE.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_STATE_BEING_CREATED	Specifies that a new graphic was being created.	c		e	f	g	a		i		j	k	l	m	n	b
M_STATE_GRAPHIC_DRAGGED	Specifies that a graphic was being dragged using the mouse.	c		e	f	g	a		i		j	k	l	m	n	b
M_STATE_GRAPHIC_HOVERED	Specifies that the cursor was hovering over a graphic.	c		e	f	g	a		i		j	k	l	m	n	b

	M_STATE_HANDLE_DRAGGED	Specifies that a graphic's handle was being dragged using the mouse.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_HANDLE_HOVERED	Specifies that the cursor was hovering over a graphic's handle.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_IDLE	Specifies that the cursor was not hovering over anything and that no graphic was queued for creation.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_WAITING_FOR_CREATION	Specifies that a graphic was queued for creation.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_GRAPHIC_STATE +		Inquires the current state of interactivity of the 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_BEING_CREATED	Specifies that a new graphic is being created.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_GRAPHIC_DRAGGED	Specifies that a graphic is being dragged using the mouse.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_GRAPHIC_HOVERED	Specifies that the cursor is currently hovering over a graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_HANDLE_DRAGGED	Specifies that a graphic's handle is being dragged using the mouse.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_HANDLE_HOVERED	Specifies that the cursor is currently hovering over a graphic's handle.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_IDLE	Specifies that the cursor is not hovering over anything and that no graphic is queued for creation.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_WAITING_FOR_CREATION	Specifies that a graphic is queued for creation.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for [when EventId=An M_GRAPHIC_MODIFIED event ID, and InfoType=M_GRAPHIC_CONTROL_TYPE, : M_POSITION_TYPE].

The following value might be combined with the above-mentioned value to specify that a rectangle's location in the image will not change, regardless of how Aurora Imaging Library interprets its reference position.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Returns whether a rectangle's location in the image will not change																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_SAME_LOCATION	Specifies that, when you set M_POSITION_TYPE, Aurora Imaging Library also adjusts the rectangle's actual position in the image so it remains at the same location.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For getting information about a 2D graphics list event.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																
EventId	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InfoType																
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested information to an *AIL_ID*. Note that M_TYPE_AIL_ID should only be used with M_GRAPHIC_LIST_ID.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is **M_NULL** if successful. If the operation fails, a non-null (!**M_NULL**) value is returned.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraHookFunction

Synopsis

Hook a function to a 2D graphics list event.

Syntax

```
void MgraHookFunction(
    AIL_ID GraListId,                        //in
    AIL_INT HookType,                        //in
    AIL_GRA_HOOK_FUNCTION_PTR HookHandlerPtr, //in
    void *UserDataPtr                        //in-out
)
```

Description

This function allows you to attach or detach a user-defined function to a specified 2D graphics list event (for example, a modification to the 2D graphics list's contents). Once a hook-handler function is defined and hooked to an event, it is automatically called when the event occurs.

You can hook more than one function to an event by making separate calls to **MgraHookFunction**() for each function that you want to hook. Aurora Imaging Library automatically chains and keeps an internal list of all these hooked functions. When a function is hooked, this new function is added to the end of the list. When the event happens, all user-defined functions in the list will be executed in the same order that they were hooked to the event. You can also remove any function from the list; in this case, Aurora Imaging Library preserves the order of the remaining functions in the list.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (GraListId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

GraListId

Specifies the identifier of the 2D graphics list to which to hook a function. The 2D graphics list must have been previously allocated on the required system using MgraAllocList().

HookType

Specifies the 2D graphics list event to which to hook the function. This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics list event																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GRAPHIC_LIST_MODIFIED +	Calls the hook-handler function each time the specified 2D graphics list is modified in any way, either by an Aurora Imaging Library function or by interactive manipulations. Any of the other events will also trigger this event, and it will always be the last event to be triggered.	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_MODIFIED +	Calls the hook-handler function each time the specified 2D graphics list's graphics are modified by an Aurora Imaging Library function or by interactive manipulations. This includes the creation of a new graphic (for example, using MgraLine() or by interactive manipulations), its deletion (using MgraControlList() with M_DELETE or MgraClear()) and any modification to a graphic using MgraControlList(), except for changing the selection (MgraControlList() with M_GRAPHIC_SELECTED). Note that this event can be triggered even if no modification occurred. For example, calling MgraControlList() with M_POSITION_X and passing the current value returned by MgraInquireList() with M_POSITION_X will not modify the graphic, but will still trigger the event.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SELECTION_MODIFIED +	Calls the hook-handler function each time a graphic is selected or deselected, either by an Aurora Imaging Library function or by interactive manipulations.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_GRAPHIC_STATE_MODIFIED +	Calls the hook-handler function each time the specified 2D graphics list's interactive state is modified by an Aurora Imaging Library function or by interactive manipulations.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For specifying the 2D graphics list event.

You can add the following value to the above-mentioned values to set whether to detach the hook-handler function.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For detaching the hook-handler function																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_UNHOOK	Unhooks the specified function if hooked to the specified event. When you use M_UNHOOK, you must provide the same values for all the parameters as when you originally hooked the function.	c		e	f	g	a		i		j	k	l	m	n	b

HookHandlerPtr

Specifies the address of the function that should be called when the specified event occurs. The hook-handler function must be declared as follows:

AIL_INT MFTYPE HookHandler(

```
    AIL_INT HookType,
    AIL_ID EventId,
    void *UserDataPtr
)
```

Parameters

HookType

Type of 2D graphics list event that generated the call.

EventId

Event identifier to pass to MgraGetHookInfo() when inquiring about the hooked event.

UserDataPtr

User data pointer that was passed to the UserDataPtr parameter of **MgraHookFunction**().

Upon successful completion, the hook-handler function should return **M_NULL**. Note MFTYPE and AIL_GRA_HOOK_FUNCTION_PTR are reserved Aurora Imaging Library predefined types for functions and data pointers.

UserDataPtr

Specifies the address of the user data that you want to make available to the hook-handler function. This address is passed to the hook-handler function, through its UserDataPtr parameter, when the specified event occurs. Set this parameter to **M_NULL** if not used.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraInquire

Synopsis

Inquire information about a 2D graphics context.

Syntax

```
AIL_INT MgraInquire(
    AIL_ID ContextGraId,   //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

Description

This function inquires information about a specified 2D graphics context. To inquire about a 2D graphics list, use MgraInquireList().

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context about which to inquire information. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

InquireType

Specifies the 2D graphics context setting to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **MgraInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For inquiring about general 2D graphics context settings
- For inquiring about a setting related to interactivity
- For inquiring about a setting specific to text

To inquire general 2D graphics context settings, the InquireType parameter should be set to one of the following values. Unless otherwise specified, these affect all types of graphics.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about general 2D graphics context settings																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_COLOR +	Inquires the foreground color. SET															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_DIRECTION +		Inquires how to draw (render) an arrow pointing in the direction with which a graphic was defined. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that no direction is drawn.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PRIMARY_DIRECTION	Specifies to draw an arrow showing the primary direction of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_PRIMARY_DIRECTION + M_SECONDARY_DIRECTION	Specifies to draw an arrow showing both the primary and secondary direction of the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SECONDARY_DIRECTION	Specifies to draw an arrow showing the secondary direction of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_OFFSET_X +	Inquires the offset subtracted from the source X-coordinates. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_OFFSET_Y +	Inquires the offset subtracted from the source Y-coordinates. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_X +	Inquires the scale factor in the X-direction. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scale factor in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_Y +	Inquires the scale factor in the Y-direction. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scale factor in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIXTURE +	Inquires the camera calibration information used when rendering (drawing or displaying) a graphic defined in world units, when both source camera calibration information and destination camera calibration information are available to interpret positioning and dimensioning information. If with respect to that of the source camera calibration information (M_USE_SOURCE_FIRST), positions and dimensions are transformed from this relative coordinate system to the absolute coordinate system; then, since it is assumed that there is only one absolute coordinate system, these positions and dimensions are transformed from the absolute coordinate system to the pixel coordinate system, using the destination camera calibration information. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_USE_DESTINATION_FIRST	Specifies that the camera calibration information of the destination is used when rendering the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_USE_SOURCE_FIRST	Specifies that the camera calibration information of the graphic, set with M_GRAPHIC_SOURCE_CALIBRATION, is used when rendering the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_CONVERSION_MODE +	Inquires how the shape of a graphic, defined in world units, is converted to pixels (rendered). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_PRESERVE_SHAPE_AVERAGE	Specifies to render the graphic so that its shape is preserved, even if it means not respecting the camera calibration information exactly. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RESHAPE_FOLLOWING_DISTORTION	Specifies that all points along the contour of the graphic will be converted using the camera calibration information, following any non-linear distortion. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_RESHAPE_FROM_POINTS	Specifies that only a few key points or features will be converted using the camera calibration information; from these converted points, the rest of the graphic will be rendered respecting the shape of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SOURCE_CALIBRATION +		Inquires the identifier of the camera calibration information used to interpret positioning and dimensioning information of a graphic defined in world units. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that no source camera calibration information is available to interpret positioning and dimensioning information of a graphic defined in world units, except when the graphic is drawn using the draw function (M...Draw()) of a processing or analysis module; in this case, the camera calibration information of the context or result will be used.	c		e	f	g	a		i		j	k	l	m	n	b
	Aurora Imaging Library Identifier	Specifies the identifier of a camera calibration context, image buffer or processing or analysis module result buffer, whose camera calibration information to use; this object is only associated with the 2D graphics context, not copied. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INPUT_UNITS +		Inquires the units with which to interpret the graphic's positional and dimensional information. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISPLAY	Specifies to interpret the values in pixel units that, unlike M_PIXEL, are not altered when the display is panned or zoomed. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies to interpret the values in pixel units, with respect to the pixel coordinate system. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies to interpret the values in world units, with respect to the relative coordinate system. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_LINE_THICKNESS +	Inquires the thickness of the lines and diameter of the dots. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the thickness of the lines and diameter of the dots, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_OWNER_SYSTEM +	Inquires the Aurora Imaging Library identifier (*AIL_ID*) of the system on which the 2D graphics context has been allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
M_OWNER_SYSTEM_TYPE +	Inquires the type of system on which the 2D graphics context was allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_CLARITY_UHD_TYPE	Specifies an Aurora Imaging Library Clarity UHD system.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_GENTL_TYPE	Specifies an Aurora Imaging Library GenTL system.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_GEVIQ_TYPE	Specifies an Aurora Imaging Library GevIQ system.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_GIGE_VISION_TYPE	Specifies an Aurora Imaging Library GigE Vision system.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_HOST_TYPE	Specifies the Host.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_IRIS_GTX_TYPE	Specifies an Aurora Imaging Library Iris GTX system.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYSTEM_RADIENTEVCL_TYPE	Specifies an Aurora Imaging Library Radient eV-CL system.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SYSTEM_RAPIXOCL_TYPE	Specifies an Aurora Imaging Library Rapixo Pro CL system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_RAPIXOCOF_TYPE	Specifies an Aurora Imaging Library Rapixo CoF system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_RAPIXOCXP_TYPE	Specifies an Aurora Imaging Library Rapixo CXP system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_USB3_VISION_TYPE	Specifies an Aurora Imaging Library USB3 Vision system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_V4L2_TYPE	Specifies an Aurora Imaging Library Video4Linux2 system.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire 2D graphics context settings that affect the interactivity of a graphic, drawn in a 2D graphics list associated with a display when interactive mode is enabled (MdispControl() with M_GRAPHIC_LIST_INTERACTIVE set to M_ENABLE), the InquireType parameter should be set to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a setting related to interactivity																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_EDITABLE +		Inquires whether a graphic can be edited via user interaction in an interactive display. If a graphic is not visible or if it is not selectable, it is not editable and this setting is ignored. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be edited via user interaction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be edited via user interaction.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESIZABLE +		Inquires whether a graphic can be resized via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not resizable and this setting is ignored. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be resized via user interaction and the resize handles will not be displayed if a graphic is selected. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the graphic can be resized via user interaction by clicking and dragging one of the resize handles.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATABLE +		Inquires whether a graphic can be rotated via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not rotatable and this setting is ignored. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be rotated via user interaction and the rotate handle will not be displayed if a graphic is selected. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be rotated via user interaction by clicking and dragging the rotation handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTABLE +		Inquires whether a graphic in a 2D graphics list can be selected via user interaction in an interactive display. If a graphic is not visible, it is not selectable and this setting is ignored. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be selected via user interaction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be selected by clicking on it.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPECIFIC_FEATURES_EDITABLE +		Inquires whether a graphic can be modified via user interaction in an interactive display using handles that are specific to its graphic type. If a graphic is not visible, not selectable, or not editable, it cannot be modified using its type-specific handles and this setting is ignored. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be modified via user interaction and its specific feature handles will not be displayed if a graphic is selected. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be modified via user interaction by clicking and dragging one of the specific feature handles.	c		e	f	g	a		i		j	k	l	m	n	b

M_TRANSLATABLE +	Inquires whether a graphic can be moved (translated) via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not movable and this setting is ignored. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the graphic cannot be moved via user interaction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the graphic can be moved via user interaction by clicking and dragging the graphic, its selection box, or its center handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_VISIBLE +	Inquires whether a graphic is rendered on the display. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the graphic is not rendered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the graphic is rendered.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire 2D graphics context settings that apply only to text (MgraText()), the InquireType parameter should be set to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a setting specific to text																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_BACKCOLOR +	Inquires the background color. SET															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_MODE +		Inquires whether the text's background is filled. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPAQUE	Specifies that the background will be filled with the current background color before drawing text.	c		e	f	g	a		i		j	k	l	m	n	b

M_TRANSPARENT	Specifies not to change the background before drawing text. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT +	Inquires the character font. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_LARGE	Specifies a large bitmap font, where each character is drawn in a 16x32 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_MEDIUM	Specifies a medium bitmap font, where each character is drawn in a 12x24 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_SMALL	Specifies a small bitmap font, where each character is drawn in a 8x16 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_TTF	Specifies a TrueType font. This value is returned when you have specified a TrueType font using MgraFont().	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_AUTO_SELECT +	Inquires whether Aurora Imaging Library will search for a suitable font to draw text if the currently selected font is a TrueType font that does not support the character code. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that Aurora Imaging Library will not search for a suitable font. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that Aurora Imaging Library will search for a suitable font.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE +	Inquires the size text is drawn when using a TrueType font. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the text's font size, in points.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_X_SCALE +	Inquires the font's horizontal scaling factor. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor by which to multiply the width of the font's characters. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_Y_SCALE +	Inquires the font's vertical scaling factor. SET SET	c		e	f	g	a		i		j	k	l	m	n	b

UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor by which to multiply the height of the font's characters. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_HORIZONTAL +	Inquires the horizontal alignment of text. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that text is horizontally centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that text is left-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that text is right-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_VERTICAL +	Inquires the vertical alignment of text. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that text is bottom-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that text is vertically centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP	Specifies that text is top-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_BORDER +	Inquires how borders are drawn around the text. Note that a combination of the values below can be returned. Bitwise operators must be used to verify the presence of a specific border. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that a line is drawn underneath the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that a line is drawn to the left of the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_NONE	Specifies that no border is drawn around the text. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that a line is drawn to the right of the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP	Specifies that a line is drawn above the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_DIRECTION +	Inquires the direction text is drawn when using a TrueType font. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_LEFT_TO_RIGHT	Specifies that text will be drawn from left to right.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIGHT_TO_LEFT	Specifies that text will be drawn from right to left.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_COLOR; M_BACKCOLOR.

You can add one of the following values to the above-mentioned values to get the color used in the 2D graphics context for a 16-bit or 32-bit color buffer.

You must inquire each color band (R,G, and B) separately.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the color value used (for 16- or 32-bit color buffers)																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)	
M_BLUE	Inquires the blue color band.	c		e	f	g	a		i		j	k	l	m	n	b	
M_GREEN	Inquires the green color band.	c		e	f	g	a		i		j	k	l	m	n	b	
M_RED	Inquires the red color band.	c		e	f	g	a		i		j	k	l	m	n	b	

Combination values for
Tables:

- For inquiring about general 2D graphics context settings
- For inquiring about a setting related to interactivity
- For inquiring about a setting specific to text

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)	
UserVarPtr - Possible values returned																	
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.																
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b	
M_TYPE_AIL_ID	Casts the requested information to an *AIL_ID*. Note that M_TYPE_AIL_ID should only be used with M_OWNER_SYSTEM and M_GRAPHIC_SOURCE_CALIBRATION.	c		e	f	g	a		i		j	k	l	m	n	b	
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b	

M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT*. If the requested information does not fit into an *AIL_INT*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraInquireList

Synopsis

Inquire information about a 2D graphics list or a graphic contained within the 2D graphics list.

Syntax

```
AIL_INT MgraInquireList(
    AIL_ID GraListId,      //in
    AIL_INT LabelOrIndex,  //in
    AIL_INT SubIndex,      //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

Description

This function inquires information about a specified 2D graphics list, or a graphic contained within the 2D graphics list.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (GraListId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

GraListId

Specifies the identifier of the 2D graphics list about which to inquire information. The 2D graphics list must have been previously allocated on the required system using MgraAllocList().

LabelOrIndex

Specifies the graphic about which to inquire, or specifies to inquire about the 2D graphics list itself. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the graphic or the 2D graphics list																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GRAPHIC_INDEX(AIL_INT Index)	Specifies the index of an existing graphic about which to inquire.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Specifies the index. The index is from 0 (inclusive) to the total number of graphics in the list minus 1. You can get the number graphics in the list with M_NUMBER_OF_GRAPHICS. If the inquired value is not supported on the graphic specified, an error is generated.															
M_GRAPHIC_LABEL(AIL_INT Label)	Specifies the label of an existing graphic about which to inquire.	c		e	f	g	a		i		j	k	l	m	n	b

	Parameters														
	Label Specifies the label. The label must be greater than 0. You can get the label of the last graphic added to the list with M_LAST_LABEL. If the inquired value is not supported on the graphic specified, an error is generated.														
M_LIST	Inquires information about the 2D graphics list itself.	c		e	f	g	a		i		j	k	l	m	n	b

SubIndex

Specifies the index of the sub-element of the graphic about which to inquire. If this information is not required or supported, set this parameter to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of a graphic's sub-element																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies to inquire about the graphic itself (instead of just a sub-element).	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= M_NUMBER_OF_SUB_ELEMENTS	Specifies the index of the sub-element of the graphic about which to inquire. The following table lists all the graphics for which you can inquire information about their individual sub-elements and outlines how sub-indices are assigned for each of these. Graphics types: Index of sub-element values Dots (created using MgraDots()): The indices, starting from 0, are assigned to each dot in order of creation.	c		e	f	g	a		i		j	k	l	m	n	b

A single line (created using MgraLine() or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_LINE).	0 is assigned to the specified start point, and 1 is assigned to the specified end point of the line.
Sets of finite length lines (created using MgraLines() with M_LINE_LIST).	The indices, starting from 0, are assigned to the specified start point and end point of each line, in order of creation. For example, the start point of the first line has an index of 0 and the end point has an index of 1, and the start point of the second line has an index of 2 and the end point has an index of 3.

Sets of infinite lines (created using MgraLines() with M_INFINITE_LINES).	The indices, starting from 0, are assigned to the first and second specified point of each line, in order of creation. For example, the first specified point of the first line has an index of 0 and its second specified point has an index of 1, and the first specified point of the second line has an index of 2 and its second specified point has an index of 3.
A polygon (created using MgraLines() with M_POLYGON or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_POLYGON).	The indices, starting from 0, are assigned to each polygon vertex in order of creation.

	A polyline (created using MgraLines() with M_POLYLINE or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_POLYLINE).	The indices, starting from 0, are assigned to each polyline point in order of creation.

InquireType

Specifies the setting to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **MgraInquireList**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For inquiring about a setting of a 2D graphics list itself
- For inquiring about a general graphic setting
- For inquiring about a setting related to interactivity
- For inquiring about a setting specific to text
- For inquiring about a setting regarding the position and dimensional information of a graphic
- For inquiring about general keyboard key settings to interactively modify graphics
- For inquiring about explicit keyboard keys to interactively modify graphics

To inquire about a setting of a 2D graphics list, the InquireType parameter should be set to one of the following values. In this case, you must set the LabelOrIndex parameter to M_LIST and the SubIndex parameter to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a setting of a 2D graphics list itself																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																

M_ANGLE_SNAPPING_VALUE +		Inquires the value to use as a multiple when rotating the graphic. This value only applies when using M_MODE_ROTATE with M_ANGLE_SNAPPING. SET														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	1.0 <= *Value* <= 180.0	Specifies the value to use as a multiple, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EASY_SELECTION +		Inquires whether Aurora Imaging Library allows for the easy selection of graphics on the display. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that Aurora Imaging Library does not allow for the easy selection of graphics on the display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that Aurora Imaging Library allows for the easy selection of graphics on the display.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_ANNOTATIONS_COLOR +		Inquires the color of the selection box and handles when in interactive mode. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_GRAPHIC_STATE +		Inquires the current state of interactivity of the 2D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_STATE_BEING_CREATED	Specifies that a new graphic is being created.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_GRAPHIC_DRAGGED	Specifies that a graphic is being dragged using the mouse.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_GRAPHIC_HOVERED	Specifies that the cursor is currently hovering over a graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_HANDLE_DRAGGED	Specifies that a graphic's handle is being dragged using the mouse.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_HANDLE_HOVERED	Specifies that the cursor is currently hovering over a graphic's handle.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_IDLE	Specifies that the cursor is not hovering over anything and that no graphic is queued for creation.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STATE_WAITING_FOR_CREATION	Specifies that a graphic is queued for creation.	c		e	f	g	a		i		j	k	l	m	n	b

M_LAST_LABEL +		Inquires the label that was automatically assigned to the last graphic added to the list in the current thread. This label value will never change, even if the order of the graphics in the list changes. After adding a graphic to the 2D graphics list, you should use this inquire type to establish the label assigned to it. This label can be used as a means of identifying the graphic independently from its index, which might change if a graphic is deleted from the list. You cannot retrieve the label value of the last added graphic if it has been deleted or moved to another 2D graphics list. If a graphic was added to the list from a different thread, the value returned will be that of the last graphic added to the list from the current thread; it will not return the label value of graphics added from a separate thread. This can be problematic when using MgraInteractive() to add graphics to the list interactively, since the graphics might be added from a separate thread. In this case, use MgraGetHookInfo() with M_GRAPHIC_LABEL_VALUE. See the Creating and modifying graphics interactively section of Chapter 26: Generating graphics for an example of how to inquire the label of a graphic added interactively.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_LABEL	Specifies the last-added graphic is no longer in the list or no graphics have been added to the list.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the label of the last-added graphic in the list.	c		e	f	g	a		i		j	k	l	m	n	b

M_MODE_RESIZE +		Inquires the interactive behavior when resizing the graphic using its selection box. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER +	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER +	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_ALT +		Inquires the interactive behavior when resizing the graphic using its selection box and pressing the Alt key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Alt key does not affect how to resize the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant, when resizing the graphic while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change, when resizing it while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square), when resizing it while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_CTRL +		Inquires the interactive behavior when resizing the graphic using its selection box and pressing the Ctrl key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to resize the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant, when resizing the graphic while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change, when resizing it while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square), when resizing it while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION +		Inquires the interactive behavior when changing the graphic's secondary dimension during its creation. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION_ALT +		Inquires the interactive behavior when changing the graphic's secondary dimension during its creation, and pressing the Alt key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Alt key does not affect how to establish the graphic's secondary dimension when defining it interactively on the display. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part, while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part, while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION_CTRL +		Inquires the interactive behavior when changing the graphic's secondary dimension during its creation, and pressing the Ctrl key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to establish the graphic's secondary dimension when defining it interactively on the display. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part, while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part, while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SECONDARY_DIMENSION_SHIFT +		Inquires the interactive behavior when changing the graphic's secondary dimension during its creation, and pressing the Shift key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to establish the graphic's secondary dimension when defining it interactively on the display. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ONE_SIDED	Specifies to establish the graphic's secondary dimension on one side of a previously established part, while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYMMETRIC	Specifies to establish the graphic's secondary dimension symmetrically on both sides of a previously established part, while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_RESIZE_SHIFT +		Inquires the interactive behavior when resizing the graphic using its selection box and pressing the Shift key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to resize the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant, when resizing the graphic while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CENTER	Specifies that all sides of the selection box move symmetrically, and the center does not move, when resizing the graphic while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIXED_CORNER	Specifies that the center moves, and the opposite corner does not move, when resizing the graphic while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change, when resizing it while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square), when resizing it while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE +		Inquires the interactive behavior when rotating the graphic using its selection box. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to rotate the graphic freely.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE_ALT +		Inquires the interactive behavior when rotating the graphic using its selection box and pressing the Alt key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE, when rotating it while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Alt key does not affect how to rotate the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to rotate the graphic freely, when rotating it while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE_CTRL +		Inquires the interactive behavior when rotating the graphic using its selection box and pressing the Ctrl key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE, when rotating it while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to rotate the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to rotate the graphic freely, when rotating it while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_ROTATE_SHIFT +		Inquires the interactive behavior when rotating the graphic using its selection box and pressing the Shift key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANGLE_SNAPPING	Specifies to rotate the graphic by snapping it to an angle that is a multiple of the value set by M_ANGLE_SNAPPING_VALUE, when rotating it while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that pressing the Shift key does not affect how to rotate the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to rotate the graphic freely, when rotating it while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE +		Inquires the interactive behavior when moving (translating) the graphic. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE_ALT +		Inquires the interactive behavior when moving (translating) the graphic and pressing the Alt key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes, when moving it while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Alt key does not affect how to move the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely, when moving it while pressing the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE_CTRL +		Inquires the interactive behavior when moving (translating) the graphic and pressing the Ctrl key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes, when moving it while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that pressing the Ctrl key does not affect how to move the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely, when moving it while pressing the Ctrl key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODE_TRANSLATE_SHIFT +		Inquires the interactive behavior when moving (translating) the graphic and pressing the Shift key. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AXIS_ALIGNED	Specifies to move the graphic along the X- and Y-axes, when moving it while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that pressing the Shift key does not affect how to move the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CONSTRAINT	Specifies to move the graphic freely, when moving it while pressing the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MULTIPLE_SELECTION +		Inquires whether interactive multiple selection is permitted. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that interactive multiple selection is not permitted.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that interactive multiple selection is permitted.	c		e	f	g	a		i		j	k	l	m	n	b
M_MULTIPLE_SELECTION_KEY +		Inquires the keyboard key that you can press to select multiple graphics with your mouse on the display. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ALT	Specifies that you can press the Alt key to select (or deselect) multiple graphics.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CTRL	Specifies that you can press the Ctrl key to select (or deselect) multiple graphics.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SHIFT	Specifies that you can press the Shift key to select (or deselect) multiple graphics.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_GRAPHICS +		Inquires the number of graphics in the list.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the number of graphics.	c		e	f	g	a		i		j	k	l	m	n	b
M_OWNER_SYSTEM +		Inquires the identifier of the system on which the 2D graphics list was allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
	System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
M_OWNER_SYSTEM_TYPE +		Inquires the type of system on which the 2D graphics list was allocated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_CLARITY_UHD_TYPE	Specifies an Aurora Imaging Library Clarity UHD system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_GENTL_TYPE	Specifies an Aurora Imaging Library GenTL system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_GEVIQ_TYPE	Specifies an Aurora Imaging Library GevIQ system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_GIGE_VISION_TYPE	Specifies an Aurora Imaging Library GigE Vision system.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SYSTEM_HOST_TYPE	Specifies the Host.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_IRIS_GTX_TYPE	Specifies an Aurora Imaging Library Iris GTX system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_RADIENTEVCL_TYPE	Specifies an Aurora Imaging Library Radient eV-CL system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_RAPIXOCL_TYPE	Specifies an Aurora Imaging Library Rapixo Pro CL system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_RAPIXOCOF_TYPE	Specifies an Aurora Imaging Library Rapixo CoF system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_RAPIXOCXP_TYPE	Specifies an Aurora Imaging Library Rapixo CXP system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_USB3_VISION_TYPE	Specifies an Aurora Imaging Library USB3 Vision system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SYSTEM_V4L2_TYPE	Specifies an Aurora Imaging Library Video4Linux2 system.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTED_COLOR +		Inquires the color of the selected graphics in interactive mode. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTION_RADIUS +		Inquires the size of the selection-radius when in interactive mode. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 1.0	Specifies the size of the selection-radius, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for [when InquireType=M_MODE_RESIZE: M_FIXED_CENTER; M_FIXED_CORNER].

One of the following values might be combined with the above-mentioned values to get the aspect ratio used when resizing the graphic.

The following values will not modify the graphic's aspect ratio if it is set to 1 using MgraControlList() with M_CONSTRAIN_ASPECT_RATIO.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the aspect ratio when resizing the graphic																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FIXED_ASPECT_RATIO	Specifies that the aspect ratio of the graphic remains constant when resizing it.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_CONSTRAINT	Specifies that the aspect ratio of the graphic can change when resizing it.	c		e	f	g	a		i		j	k	l	m	n	b
M_SQUARE_ASPECT_RATIO	Specifies that the aspect ratio of the graphic will be 1:1 (square) when resizing it.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about a general graphic setting, the InquireType parameter should be set to one of the following values. You can set the LabelOrIndex parameter to the label or index of a graphic in the list, and the SubIndex parameter to M_DEFAULT. Unless otherwise specified, these affect all types of graphics.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a general graphic setting																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																	
M_COLOR +	Inquires the foreground color of the graphic. SET SET																
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_DIRECTION +	Inquires how to draw (render) an arrow pointing in the direction with which a graphic was defined. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_NONE	Specifies that no direction is drawn. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRIMARY_DIRECTION	Specifies to draw an arrow showing the primary direction of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PRIMARY_DIRECTION + M_SECONDARY_DIRECTION	Specifies to draw an arrow showing both the primary and secondary direction of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SECONDARY_DIRECTION	Specifies to draw an arrow showing the secondary direction of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_OFFSET_X +	Inquires the offset subtracted from the source X-coordinates. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_OFFSET_Y +	Inquires the offset subtracted from the source Y-coordinates. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate offset to subtract, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_X +	Inquires the scale factor in the X-direction. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scaling factor in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ZOOM_Y +	Inquires the scale factor in the Y-direction. SET SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the scaling factor in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIXTURE +		Inquires the camera calibration information used when rendering (drawing or displaying) a graphic defined in world units, when both source camera calibration information and destination camera calibration information are available to interpret positioning and dimensioning information. If with respect to that of the source camera calibration information (M_USE_SOURCE_FIRST), positions and dimensions are transformed from this relative coordinate system to the absolute coordinate system; then, since it is assumed that there is only one absolute coordinate system, these positions and dimensions are transformed from the absolute coordinate system to the pixel coordinate system, using the destination camera calibration information. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_USE_DESTINATION_FIRST	Specifies that the camera calibration information of the destination is used when rendering the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_USE_SOURCE_FIRST	Specifies that the camera calibration information of the graphic, set with M_GRAPHIC_SOURCE_CALIBRATION, is used when rendering the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_CONVERSION_MODE +		Inquires how the shape of a graphic, defined in world units, is converted to pixels (rendered). SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PRESERVE_SHAPE_AVERAGE	Specifies to render the graphic so that its shape is preserved even if it means not respecting the camera calibration information exactly. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_RESHAPE_FOLLOWING_DISTORTION	Specifies that all points along the contour of the graphic will be converted using the camera calibration information, following any non-linear distortion. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RESHAPE_FROM_POINTS	Specifies that only a few key points or features will be converted using the camera calibration information; from these points, the rest of the graphic will be rendered respecting the shape of the graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SOURCE_CALIBRATION +		Inquires the identifier of the camera calibration information used to interpret positioning and dimensioning information of a graphic defined in world units. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that no source camera calibration information is available to interpret positioning and dimensioning information of a graphic defined in world units.	c		e	f	g	a		i		j	k	l	m	n	b
	Aurora Imaging Library Identifier	Specifies the identifier of the internal camera calibration context that will be used.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE +		Inquires the type of the graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_ARC	Specifies an arc (created using MgraArc(), MgraArcFill(), MgraArcAngle(), or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_ARC).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_COLLECTION	Specifies a drawing created using the M...Draw() function of a processing or analysis module, like MmodDraw() or MmeasDraw().	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_DOT	Specifies a dot (created using MgraDot() or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_DOT).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_DOTS	Specifies dots (created using MgraDots()).	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRAPHIC_TYPE_INFINITE_LINES	Specifies infinite lines (created using MgraLines() with M_INFINITE_LINES).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINE	Specifies a line (created using MgraLine() or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_LINE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINES	Specifies lines (created using MgraLines() with M_LINE_LIST).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POLYGON	Specifies a polygon (created using MgraLines() with M_POLYGON or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_POLYGON).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POLYLINE	Specifies a polyline (created using MgraLines() with M_POLYLINE or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_POLYLINE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_RECT	Specifies a rectangle (created using MgraRect(), MgraRectFill(), MgraRectAngle(), or user-defined using MgraInteractive() set to M_GRAPHIC_TYPE_RECT).	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_TEXT	Specifies text (created using MgraText()).	c		e	f	g	a		i		j	k	l	m	n	b
M_INDEX_VALUE +		Inquires the index associated with the graphic. In this case, the LabelOrIndex parameter must be set to the label of the required graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies that the graphic label is invalid. This is returned if the specified label is not associated with a graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the index value associated with the specified graphic.	c		e	f	g	a		i		j	k	l	m	n	b

M_INPUT_UNITS +	Inquires the units with which to interpret the graphic's position and dimensional information, for graphics already added to the 2D graphics list. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISPLAY	Specifies to interpret the values in pixel units that, unlike M_PIXEL, are not altered when the display is panned or zoomed. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL	Specifies to interpret the values in pixel units, with respect to the pixel coordinate system. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WORLD	Specifies to interpret the values in world units, with respect to the relative coordinate system. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE +	Inquires the label that was automatically associated with the graphic when it was added to the list. This label value will never change, even if the order of the graphics in the list changes.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the label value that was automatically associated with the specified graphic.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_THICKNESS +	Inquires the thickness of the lines and diameter of the dots. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the thickness of the lines and diameter of the dots, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_SUB_ELEMENTS +	Inquires the number of sub-elements (position-points) within the specified graphic. Note that in this case the SubIndex parameter must be set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 0	Specifies the number of sub-elements. This value can only be greater than zero for the following graphics: dots, line, lines, polylines, infinite lines, and polygons. For example, a line contains two sub-elements (position points).	c		e	f	g	a		i		j	k	l	m	n	b

To inquire 2D graphics context settings that affect the interactivity of a graphic, drawn in a 2D graphics list associated with a display when interactive mode is enabled (MdispControl() with M_GRAPHIC_LIST_INTERACTIVE set to M_ENABLE), the InquireType parameter should be set to one of the following values. You can set the LabelOrIndex parameter to the label or index of a specific graphic, and the SubIndex parameter must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a setting related to interactivity																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_EDITABLE +		Inquires whether a graphic can be edited via user interaction in an interactive display. If a graphic is not visible or if it is not selectable, it is not editable and this setting is ignored. SET SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be edited via user interaction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be edited via user interaction.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SELECTED +		Inquires whether the graphic is selected. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the graphic is not selected.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the graphic is selected.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESIZABLE +		Inquires whether a graphic can be resized via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not resizeable and this setting is ignored. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the graphic cannot be resized via user interaction and the resize handles will not be displayed if a graphic is selected. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be resized via user interaction by clicking and dragging one of the resize handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATABLE +		Inquires whether a graphic can be rotated via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not rotatable and this setting is ignored. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be rotated via user interaction and the rotate handle will not be displayed if a graphic is selected. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be rotated via user interaction by clicking and dragging the rotation handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTABLE +		Inquires whether a graphic in a 2D graphics list can be selected via user interaction in an interactive display. If a graphic is not visible, it is not selectable and this setting is ignored. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be selected via user interaction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the graphic can be selected by clicking on it.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPECIFIC_FEATURES_EDITABLE +		Inquires whether a graphic can be modified via user interaction in an interactive display using handles that are specific to its graphic type. If a graphic is not visible, not selectable, or not editable, it cannot be modified using its type-specific handles and this setting is ignored. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the graphic cannot be modified via user interaction and its specific feature handles will not be displayed if a graphic is selected. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies that the graphic can be modified via user interaction by clicking and dragging one of the type-specific handles.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSLATABLE +	Inquires whether a graphic can be moved (translated) via user interaction in an interactive display. If a graphic is not visible, not selectable, or not editable, it is not movable and this setting is ignored. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the graphic cannot be moved via user interaction. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the graphic can be moved via user interaction by clicking and dragging the graphic, its selection box, or its center handle.	c		e	f	g	a		i		j	k	l	m	n	b
M_VISIBLE +	Inquires whether a graphic is rendered on the display. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the graphic is not rendered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the graphic is rendered.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about a setting specific to text in the 2D graphics list, the InquireType parameter should be set to one of the following values. You must set the LabelOrIndex parameter to the label or index of text, and the SubIndex parameter to M_DEFAULT. Inquiring one of the following settings for graphics that are not text will generate an error.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a setting specific to text																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_BACKCOLOR +	Inquires the background color of the text. SET SET															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies a grayscale value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_MODE +		Inquires whether the text's background is filled. SET SET	c		e	f	g	a		i		j	k	l	m	n	b

UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_OPAQUE	Specifies that the background will be filled with the current background color before drawing text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSPARENT	Specifies not to change the background before drawing text. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT +	Inquires the font of the characters in the text. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_LARGE	Specifies a large bitmap font, where each character is drawn in a 16x32 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_MEDIUM	Specifies a medium bitmap font, where each character is drawn in a 12x24 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_DEFAULT_SMALL	Specifies a small bitmap font, where each character is drawn in a 8x16 pixel area.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_TTF	Specifies a TrueType font. This value is returned when using MgraControlList() with M_FONT.	c		e	f	g	a		i		j	k	l	m	n	b
"*FontFamily:Weight:Slant*"	Specifies the font according to the following format, *Family*: *Weight*: *Slant*. INFO	c		e	f	g	a		i		j	k	l	m	n	b
"*FontFile*"	Specifies a full path to a TrueType file name.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_AUTO_SELECT +	Inquires whether Aurora Imaging Library will search for a suitable font to draw text if the currently selected font is a TrueType font that does not support the character code. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that Aurora Imaging Library will not search for a suitable font. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that Aurora Imaging Library will search for a suitable font.	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_SIZE +	Inquires the size in which text is drawn for a TrueType font. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the text's font size, in points.	c		e	f	g	a		i		j	k	l	m	n	b

M_FONT_X_SCALE +	Inquires the bitmap font's horizontal scaling factor. SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor with which to multiply the width of the font characters. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FONT_Y_SCALE +	Inquires the bitmap font's vertical scaling factor. SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the factor with which to multiply the height of the font characters. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_HORIZONTAL +	Inquires the horizontal alignment of the text. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that the text is horizontally centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that the text is left-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIGHT	Specifies that the text is right-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_ALIGN_VERTICAL +	Inquires the vertical alignment of the text. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that the text is bottom-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER	Specifies that the text is vertically centered.	c		e	f	g	a		i		j	k	l	m	n	b
M_TOP	Specifies that the text is top-aligned.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_BORDER +	Inquires how borders are drawn around the text. Note that a combination of the values below can be returned. Bitwise operators must be used to verify the presence of a specific border. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOTTOM	Specifies that a line is drawn underneath the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_LEFT	Specifies that a line is drawn to the left of the text.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NONE	Specifies that no border is drawn around the text. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIGHT	Specifies that a line is drawn to the right of the text.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TOP	Specifies that a line is drawn above the text.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEXT_DIRECTION +		Inquires the direction text is drawn when using a TrueType font. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_LEFT_TO_RIGHT	Specifies that text will be drawn from left to right.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIGHT_TO_LEFT	Specifies that text will be drawn from right to left.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_INTERACTIVE_ANNOTATIONS_COLOR; M_SELECTED_COLOR; M_COLOR; M_BACKCOLOR.

You can add one of the following values to the above-mentioned values to get the color used for a 16-bit or 32-bit color buffer.

You must inquire each color band (R,G, and B) separately.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the color value used (for 16- or 32-bit color buffers)																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_BLUE	Inquires the blue color band.	c		e	f	g	a		i		j	k	l	m	n	b
M_GREEN	Inquires the green color band.	c		e	f	g	a		i		j	k	l	m	n	b
M_RED	Inquires the red color band.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the position and dimensional information settings of a graphic (typically set when the graphic is added to the list), the InquireType parameter should be set to one of the following values. You must set the LabelOrIndex parameter to the label or index of a graphic that supports that setting. Unless otherwise specified, the SubIndex parameter should be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a setting regarding the position and dimensional information of a graphic																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																

M_ANGLE +	Inquires the angle of the graphic. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup. You can inquire this value for all graphics except a single dot (MgraDot()), multiple dots (MgraDots()), text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()). Inquiring this value for any of the unsupported types of graphics will generate an error. SET SET SET														
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
-360.0 <= *Value* <= 360.0	Specifies the angle, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_END +	Inquires the angle at which to stop drawing the arc. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the end angle, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_START +	Inquires the angle at which to start drawing the arc. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the start angle, in degrees, relative to the input coordinate system specified using M_INPUT_UNITS. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_ARC_STYLE +	Inquires whether to draw an arc or a sector. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTOUR	Specifies that the arc (the curve between the specified start and end angles) is drawn without lines extending from the center of the ellipse to the start and end points of the arc.	c		e	f	g	a		i		j	k	l	m	n	b
M_SECTOR	Specifies that a sector is drawn with lines extending from the center of the ellipse to the start and end points of the arc, unless the specified start and end angles form a closed curve.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +	Inquires the X-coordinate of a graphic's center. Except for rectangles, M_CENTER_X returns the same value as M_POSITION_X. For rectangles, M_CENTER_X and M_POSITION_X can have different values, depending on how Aurora Imaging Library interprets the rectangle's position (which is established with M_POSITION_TYPE). Use M_CENTER_X to be sure you are inquiring about the center position of any graphic. You can inquire this value for all graphics except text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()). Inquiring this value for any of the unsupported types of graphics will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate of a graphic's center.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y +	Inquires the Y-coordinate of a graphic's center. Except for rectangles, M_CENTER_Y returns the same value as M_POSITION_Y. For rectangles, M_CENTER_Y and M_POSITION_Y can have different values, depending on how Aurora Imaging Library interprets the rectangle's position (which is established with M_POSITION_TYPE). Use M_CENTER_Y to be sure you are inquiring about the center position of any graphic. You can inquire this value for all graphics except text (MgraText()), and drawings created using a processing or analysis module draw function (M...Draw()). Inquiring this value for any of the unsupported types of graphics will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate of a graphic's center.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONSTRAIN_ASPECT_RATIO +		Inquires whether the width and height are forced to be equal. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()) and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an Aurora Imaging Library error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the width and height of a graphic are not forced to be equal.	c		e	f	g	a		i		j	k	l	m	n	b
	1.0	Specifies that the width and height of a graphic are forced to be equal. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_BOTTOM_LEFT_X +		Inquires the X-coordinate of the bottom-left corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-coordinate of the selection box's bottom-left corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_BOTTOM_LEFT_Y +		Inquires the Y-coordinate of the bottom-left corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Value	Specifies the Y-coordinate of the selection box's bottom-left corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_BOTTOM_RIGHT_X +	Inquires the X-coordinate of the bottom-right corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate of the selection box's bottom-right corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_BOTTOM_RIGHT_Y +	Inquires the Y-coordinate of the bottom-right corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate of the selection box's bottom-right corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_TOP_LEFT_X +	Inquires the X-coordinate of the top-left corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate of the selection box's top-left corner.	c		e	f	g	a		i		j	k	l	m	n	b

M_CORNER_TOP_LEFT_Y +	Inquires the Y-coordinate of the top-left corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate of the selection box's top-left corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_TOP_RIGHT_X +	Inquires the X-coordinate of the top-right corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate of the selection box's top-right corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_TOP_RIGHT_Y +	Inquires the Y-coordinate of the top-right corner of the graphic's selection box. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), dots (MgraDots()), lines (MgraLines() with M_LINE_LIST), polygons (MgraLines() with M_POLYGON), polylines (MgraLines() with M_POLYLINE), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-coordinate of the selection box's top-right corner.	c		e	f	g	a		i		j	k	l	m	n	b

M_FILLED +	Inquires whether the graphic is filled. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()), polygons (MgraLines() with M_POLYGON), and rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the graphic is not filled.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the graphic is filled.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_TYPE +	Inquires how to interpret M_POSITION_X and M_POSITION_Y. You can inquire this value for rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_AND_DIMENSION	Specifies to interpret M_POSITION_X and M_POSITION_Y as the rectangle's center.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_AND_DIMENSION	Specifies to interpret M_POSITION_X and M_POSITION_Y as the rectangle's top-left corner.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +	Inquires the specified X-position of a graphic or of one of its sub-elements. This value is set whenever a graphic is defined, either programmatically or interactively. The point returned for the position of a graphic is its center for arcs, sets of dots, a single line, sets of finite-length lines, sets of infinite lines, polygons, and polylines. The point returned for the position of text is dependent upon M_TEXT_ALIGN_HORIZONTAL and M_TEXT_ALIGN_VERTICAL. The point returned for the position of a rectangle is its center or its top-left corner, dependent upon the setting of M_POSITION_TYPE. If SubIndex is set to M_DEFAULT, this setting is inquired for the entire graphic. If a sub-element is specified, this setting is inquired for a single point within the graphic. You can inquire this value for all graphics except drawings created using a processing or analysis module draw function (M...Draw()). Inquiring this value on an unsupported graphic will generate an error. SET SET SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-position. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Inquires the specified Y-position of a graphic or of one of its sub-elements. This value is set whenever a graphic is defined, either programmatically or interactively. The point returned for the position of a graphic is its center for arcs, sets of dots, a single line, sets of finite-length lines, sets of infinite lines, polygons, and polylines. The point returned for the position of text is dependent upon M_TEXT_ALIGN_HORIZONTAL and M_TEXT_ALIGN_VERTICAL. The point returned for the position of a rectangle is its center or its top-left corner, dependent upon the setting of M_POSITION_TYPE. If SubIndex is set to M_DEFAULT, this setting is inquired for the entire graphic. If a sub-element is specified, this setting is inquired for a single point within the graphic. You can inquire this value for all graphics except drawings created using a processing or analysis module draw function (M...Draw()). Inquiring this value on an unsupported graphic will generate an error. SET SET SET SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-position. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_X +		Inquires the radius of the arc, in the X-direction. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the radius.	c		e	f	g	a		i		j	k	l	m	n	b
M_RADIUS_Y +		Inquires the radius of the arc, in the Y-direction. You can inquire this value for arcs (MgraArc(), MgraArcFill(), and MgraArcAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the radius.	c		e	f	g	a		i		j	k	l	m	n	b

M_RECTANGLE_HEIGHT +	Inquires the height of the rectangle. You can inquire this value for rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the height.	c		e	f	g	a		i		j	k	l	m	n	b
M_RECTANGLE_WIDTH +	Inquires the width of a rectangle. You can inquire this value for rectangles (MgraRect(), MgraRectFill(), and MgraRectAngle()). Inquiring this value for any other type of graphic will generate an error. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the width.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about general settings for using keyboard keys to interactively modify graphics, the InquireType parameter should be set to one of the following values. These settings apply to the 2D graphics list itself. In this case, you must set the LabelOrIndex parameter to M_LIST and the SubIndex parameter to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about general keyboard key settings to interactively modify graphics																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_ACTION_KEYS +	Inquires whether the 2D graphics list allows you to interactively modify its graphics by pressing keys on the keyboard (keyboard events). SET															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the 2D graphics list ignores keyboard events.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the 2D graphics list allows keyboard events to interactively modify its graphics. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_MODIFIER_SPEED +		Inquires the combination keyboard key that you can press to select an alternate speed to rotate, resize, or translate interactively selected graphics with the keyboard. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ALT	Specifies to use the Alt key to select an alternate speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL	Specifies to use the Ctrl key to select an alternate speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SHIFT	Specifies to use the Shift key to select an alternate speed.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_RESIZE_INCREMENT +		Inquires the value with which to increase or decrease the width or height of the interactively selected graphic for a single key press. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_RESIZE_INCREMENT_ALTERNATE +		Inquires an alternate value with which to increase or decrease the width or height of the interactively selected graphic for a single key press. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_ROTATE_INCREMENT +	Inquires the angle with which to rotate the interactively selected graphic for a single key press. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_ROTATE_INCREMENT_ALTERNATE +	Inquires an alternate angle with which to rotate the interactively selected graphic for a single key press. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_TRANSLATE_AXES +	Inquires the axes along which Aurora Imaging Library translates a graphic that is in world units, when the translation is done using the keyboard. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_ACCORDING_TO_CALIBRATION	Specifies to convert the translation increment to world units using the calibration's average pixel size in the translation direction, then apply it along the world axes.	c		e	f	g	a		i		j	k	l	m	n	b
M_PIXEL	Specifies to apply the translation increment to the graphic position, along the image axes, in display units (these are pixel units that remain unaltered when you pan or zoom the display).	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_TRANSLATE_INCREMENT +	Inquires the value with which to translate (move) the interactively selected graphics for a single key press. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_TRANSLATE_INCREMENT_ALTERNATE +		Inquires an alternate value with which to translate (move) the interactively selected graphics for a single key press. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the increment, in display units (these are pixel units that remain unaltered when you pan or zoom the display). INFO	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about explicit keyboard keys to interactively modify graphics, the InquireType parameter should be set to one of the following values. These settings apply to the 2D graphics list itself. In this case, you must set the LabelOrIndex parameter to M_LIST and the SubIndex parameter to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about explicit keyboard keys to interactively modify graphics																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_ACTION_KEY_CANCEL +		Inquires the keyboard keys that you can press to cancel the interactive graphic modification you are currently performing. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_DELETE +		Inquires the keyboard keys that you can press to delete the interactively selected graphics. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_HEIGHT_DOWN +		Inquires the keyboard keys that you can press to reduce the height of the interactively selected graphics. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_HEIGHT_UP +		Inquires the keyboard keys that you can press to increase the height of the interactively selected graphics. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_WIDTH_DOWN +		Inquires the keyboard keys that you can press to reduce the width of the interactively selected graphics. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESIZE_WIDTH_UP +		Inquires the keyboard keys that you can press to increase the width of the interactively selected graphics. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ROTATE_CLOCKWISE +		Inquires the keyboard keys that you can press to rotate the interactively selected graphics clockwise. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ROTATE_COUNTER_CLOCKWISE +		Inquires the keyboard keys that you can press to rotate the interactively selected graphics counter-clockwise. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_DOWN +		Inquires the keyboard keys that you can press to translate (move) the interactively selected graphics down. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_LEFT +		Inquires the keyboard keys that you can press to translate (move) the interactively selected graphics left. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_RIGHT +		Inquires the keyboard keys that you can press to translate (move) the interactively selected graphics right. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_UP +		Inquires the keyboard keys that you can press to translate (move) the interactively selected graphics up. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_BACK	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For inquiring about explicit keyboard keys to interactively modify graphics.

You can add one or more of the following values to the above-mentioned values to get additional keyboard keys that were explicitly specified (the values you get back with the UserVarPtr parameter might be combined).

> You do not add the following values to the above mentioned values. One or more of the following values might be combined with the value returned (UserVarPtr) when the InquireType parameter is set to one of the above mentioned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about additional keyboard keys (the values you get back with the UserVarPtr parameter might be combined)																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_KEY_ALT	Specifies the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CTRL	Specifies the Ctrl key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_SHIFT	Specifies the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for

Tables:

- For inquiring about a setting of a 2D graphics list itself
- For inquiring about a general graphic setting
- For inquiring about a setting related to interactivity

- For inquiring about a setting specific to text
- For inquiring about a setting regarding the position and dimensional information of a graphic
- For inquiring about general keyboard key settings to interactively modify graphics
- For inquiring about explicit keyboard keys to interactively modify graphics

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																	
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.																
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested information to an *AIL_ID*. Note that M_TYPE_AIL_ID should only be used with M_OWNER_SYSTEM and M_GRAPHIC_SOURCE_CALIBRATION.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT*. If the requested information does not fit into an *AIL_INT*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraInteractive

Synopsis

Change the state of interactivity of graphics in a 2D graphics list.

Syntax

```
void MgraInteractive(
    AIL_ID ContextGraId,   //out
    AIL_ID ListGraId,      //in
    AIL_INT64 GraphicType, //in
    AIL_INT64 InitFlag,    //in
    AIL_INT64 CreationMode //in
)
```

Description

This function changes the state of interactivity (M_INTERACTIVE_GRAPHIC_STATE) of graphics in a 2D graphics list. It is primarily used to change the state of interactivity of the graphics such that a user can interactively add a graphic of the specified type to the 2D graphics list (M_STATE_WAITING_FOR_CREATION). In addition, the function can end any interactive manipulations occurring in the display and return to an idle state (M_STATE_IDLE).

When in the waiting for creation state (GraphicType set to an M_GRAPHIC_TYPE_... value), the user is expected to define a graphic interactively in the display. Note, however, that **MgraInteractive**() is asynchronous and only changes the state of interactivity; it will not wait for the user to define the graphic. You should have your application wait for the user to finish defining the graphic, before calling further functions; to do so, hook a function to the change in state of interactivity using MgraHookFunction() with M_INTERACTIVE_GRAPHIC_STATE_MODIFIED. When the user begins defining the graphic, the state of interactivity of the 2D graphics list (MgraInquireList() or MgraGetHookInfo() with M_INTERACTIVE_GRAPHIC_STATE) changes to M_STATE_BEING_CREATED. After a user has defined the graphic interactively, the state of interactivity returns to M_STATE_IDLE and you can continue your application from that moment.

This function is especially useful if your application uses toolbars for a graphical user interface. For example, you can use another API to create a toolbar, with one of its buttons calling **MgraInteractive**() with M_GRAPHIC_TYPE_RECT. After clicking on the toolbar button, the user can define a rectangle in the display to specify where to perform processing or analysis in the image.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (ListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

For more information, see Chapter 26: Generating graphics.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies to use the default 2D graphics context of the current Aurora Imaging Library application context. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies a 2D graphics context is not necessary; you should use this setting only when GraphicType is set to M_CANCEL or M_STOP.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ListGraId

Specifies the identifier of a valid 2D graphics list whose state of interactivity must be changed. You must have allocated the 2D graphics list using MgraAllocList().

GraphicType

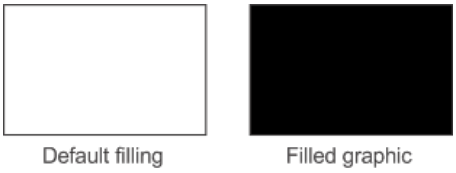
Specifies the type of graphic to define interactively and add to the 2D graphics list, or specifies to cancel or stop the 2D graphics list's state of interactivity.

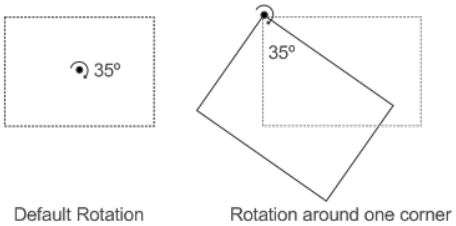
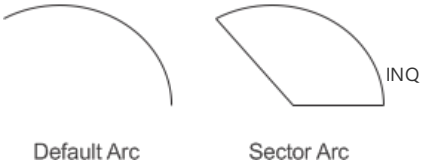
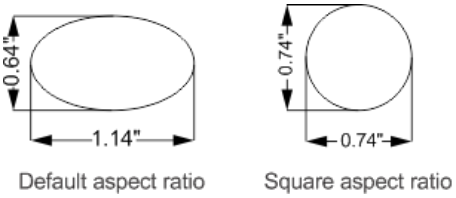
See the Parameter associations section for possible values that can be specified.

InitFlag

Specifies optional behavior when defining the graphic in the display. If this behavior is not required or supported, set this parameter to **M_DEFAULT**. This parameter can be set to one or a combination of the following values. If the value is not supported for the graphic you are defining, an error is generated.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying optional behavior when defining the graphic																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILLED	Specifies that the graphic is filled (MgraControlList() with M_FILLED set to M_TRUE). This setting only applies to arcs (M_GRAPHIC_TYPE_ARC), polygons (M_GRAPHIC_TYPE_POLYGON), and rectangles (M_GRAPHIC_TYPE_RECT).  Default filling Filled graphic INQ	c		e	f	g	a		i		j	k	l	m	n	b

M_ROTATE_AROUND_CORNER	Specifies that the rectangle's position is set to its corner (MgraControlList() with M_POSITION_TYPE set to M_CORNER_AND_DIMENSION). If not specified, the rectangle's position is set to its center by default. This setting only applies to rectangles (M_GRAPHIC_TYPE_RECT).  INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_SECTOR	Specifies that the arc is drawn as a sector (MgraControlList() with M_ARC_STYLE set to M_SECTOR). This setting only applies to arcs (M_GRAPHIC_TYPE_ARC).  	c		e	f	g	a		i		j	k	l	m	n	b
M_SQUARE_ASPECT_RATIO	Specifies that the graphic has a 1:1 (square) aspect ratio (MgraControlList() with M_CONSTRAIN_ASPECT_RATIO set to 1.0). This setting only applies to arcs (M_GRAPHIC_TYPE_ARC) and rectangles (M_GRAPHIC_TYPE_RECT) that are not oriented (M_GRAPHIC_TYPE_RECT with M_ORIENTED_RECT).  INQ	c		e	f	g	a		i		j	k	l	m	n	b

CreationMode

Specifies how to interactively define the graphic.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **GraphicType** and **CreationMode** parameters.

- For defining the graphic interactively
- For ending user interaction

The following GraphicType and corresponding CreationMode parameter settings let you define the graphic interactively. Specifically, the following values change the 2D graphics list's state of interactivity (MgraInquireList() with M_INTERACTIVE_GRAPHIC_STATE) to M_STATE_WAITING_FOR_CREATION, allowing you to add a graphic to the 2D graphics list by defining the graphic interactively in the display.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For defining the graphic interactively

GraphicType / CreationMode	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GRAPHIC_TYPE_ARC	Adds an arc to the 2D graphics list by defining the arc interactively in the display. In addition to defining an arc, you can use this value to define a circle or an ellipse (which are extensions of the arc mechanism).															
M_DEFAULT	Specifies the legacy default behavior; instead of M_DEFAULT, Aurora Imaging Library recommends using M_AXIS_ALIGNED_ELLIPSE. Do not specify M_DEFAULT unless required for backward compatibility. M_DEFAULT ignores 2D graphics list settings (MgraControlList()) related to whether resizing is with respect to the center (M_FIXED_CENTER) or the corner (M_FIXED_CORNER) of the selection box. M_DEFAULT requires you to click on the ellipse center in the display, and drag both ellipse radii with the mouse (this can be considered a legacy version of an axis-aligned ellipse). Aurora Imaging Library adds the ellipse to the 2D graphics list on the first mouse-down event.	c		e	f	g	a		i		j	k	l	m	n	b
M_ARC_THREE_POINTS	Specifies to define a circular arc. This requires three clicks in the display. The first and third clicks determine the arc's start angle and end angle (not necessarily in this order); the second click establishes the position of the arc. Aurora Imaging Library adds the arc to the 2D graphics list on the first mouse-move event after the second mouse-up event. Aurora Imaging Library does not allow the three clicks to form a straight line.	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_ALIGNED_ELLIPSE	Specifies to define an axis-aligned ellipse. This requires you to click on one corner of the ellipse bounding box in the display, and then drag the mouse to the opposite corner. Aurora Imaging Library adds the ellipse to the 2D graphics list on the first mouse-down event. Instead of defining a corner of the ellipse bounding box, the first click can define the ellipse center. This can be the default behavior (MgraControlList() with M_MODE_RESIZE set to M_FIXED_CENTER), or the behavior when resizing while pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key. You can use M_AXIS_ALIGNED_ELLIPSE to define a circle (which is a special type of ellipse), by setting the InitFlag parameter to M_SQUARE_ASPECT_RATIO. You can also call MgraControlList() with M_SQUARE_ASPECT_RATIO as a combination value for M_MODE_RESIZE to specify this as the default behavior, or the behavior when pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key.	c		e	f	g	a		i		j	k	l	m	n	b

M_CIRCLE	Specifies to define a circle. This requires you to click on a point on the circle's circumference in the display, and then drag the mouse to the diametrically opposite point. Aurora Imaging Library adds the circle to the 2D graphics list on the first mouse-down event. Instead of defining a point on the circle's circumference, the first click can define the circle's center. This can be the default behavior (MgraControlList() with M_MODE_RESIZE set to M_FIXED_CENTER), or the behavior when resizing while pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE_DOT	Adds a dot to the 2D graphics list by defining the dot interactively in the display.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to define a dot. This requires you to click on the position in the display at which to place the dot. Aurora Imaging Library adds the dot to the 2D graphics list on the mouse-down event.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE_LINE	Adds a line to the 2D graphics list by defining the line interactively in the display.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to define a line. This requires you to click on the line's start point in the display, and then drag the mouse to the line's end point. Aurora Imaging Library adds the line to the 2D graphics list on the first mouse-down event. Instead of defining the line's start point, the first click can define the line's center. This can be the default behavior (MgraControlList() with M_MODE_RESIZE set to M_FIXED_CENTER, or the behavior when resizing while pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key. When creating the line, its end point can be seen as its angle of rotation. You can use MgraControlList() to specify the default rotation (orientation) behavior (M_MODE_ROTATE), or the rotation (orientation) behavior while pressing the Alt (M_MODE_ROTATE_ALT), Ctrl (M_MODE_ROTATE_CTRL), or Shift (M_MODE_ROTATE_SHIFT) key.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE_POLYGON	Adds a polygon to the 2D graphics list by defining the polygon interactively in the display.	c		e	f	g	a		i		j	k	l	m	n	b

M_DEFAULT	Specifies to define a polygon. This requires you to left-click each polygon vertex in the display, except for the last vertex, which you must define with a right-click. Aurora Imaging Library adds the polygon to the 2D graphics list on the first mouse-down event. By default, creating a polygon while pressing the Alt key snaps the next polygon segment to 15° angles. You can call MgraControlList() with M_MODE_ROTATE... to change this behavior.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE_POLYLINE	Adds a polyline to the 2D graphics list by defining the polyline interactively in the display.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to define a polyline. This requires you to left-click each polyline vertex in the display, except for the last vertex, which you must define with a right-click. Aurora Imaging Library adds the polyline to the 2D graphics list on the first mouse-down event. By default, creating a polyline while pressing the Alt key snaps the next polyline segment to 15° angles. You can call MgraControlList() with M_MODE_ROTATE... to change this behavior.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE_RECT	Adds a rectangle to the 2D graphics list by defining the rectangle interactively in the display. By default, you define the rectangle according to its center. This default behavior is established using MgraControlList() with M_POSITION_TYPE set to M_CENTER_AND_DIMENSION. To define the rectangle according to its top-left corner, set the InitFlag parameter (**MgraInteractive**()) to M_ROTATE_AROUND_CORNER. This is equivalent to setting M_POSITION_TYPE to M_CORNER_AND_DIMENSION.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the legacy default behavior; instead of M_DEFAULT, Aurora Imaging Library recommends using M_AXIS_ALIGNED_RECT. Do not specify M_DEFAULT unless required for backward compatibility. M_DEFAULT ignores 2D graphics list settings (MgraControlList()) related to whether resizing is with respect to the center (M_FIXED_CENTER) or the corner (M_FIXED_CORNER) of the selection box. You also cannot use M_DEFAULT with M_ROTATE_AROUND_CORNER. M_DEFAULT requires you to click on a rectangle corner in the display, and then drag the mouse to the opposite corner (this can be considered a legacy version of an axis-aligned rectangle). Aurora Imaging Library adds the rectangle to the 2D graphics list on the first mouse-down event.	c		e	f	g	a		i		j	k	l	m	n	b

	M_AXIS_ALIGNED_RECT	Specifies to define an axis-aligned rectangle. This requires you to click on a rectangle corner in the display, and then drag the mouse to the opposite corner. Aurora Imaging Library adds the rectangle to the 2D graphics list on the first mouse-down event. Instead of defining a rectangle corner, the first click can define the rectangle center. This can be the default behavior (MgraControlList() with M_MODE_RESIZE set to M_FIXED_CENTER), or the behavior when resizing while pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key. You can use M_AXIS_ALIGNED_RECT to define a square (which is a special type of rectangle), by setting the InitFlag parameter to M_SQUARE_ASPECT_RATIO. You can also call MgraControlList() with M_SQUARE_ASPECT_RATIO as a combination value for M_MODE_RESIZE to specify this as the default behavior, or the behavior when pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key.	c		e	f	g	a		i		j	k	l	m	n	b

			c		e	f	g	a		i		j	k	l	m	n	b
	M_ORIENTED_RECT	Specifies to define an oriented rectangle. This requires three clicks in the display. By default, the first click defines a fixed corner of the rectangle, the second click defines its width (a line extending to the fixed corner), and the third click defines its height on one side of the width line. Aurora Imaging Library adds the rectangle to the 2D graphics list on the first mouse-up event. Instead of defining the rectangle's width using a fixed corner, the first two clicks can define the width symmetrically on both sides of a center point; that is, the first click represents the fixed center of the width line, and the second click extends the width line equally on both sides of the fixed center. This can be the default behavior (MgraControlList() with M_MODE_RESIZE set to M_FIXED_CENTER), or the behavior when pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key. Instead of defining the rectangle's height on one side of the width line, the third click can define the height symmetrically on both sides; that is, moving your mouse above or below the width line (established by the first two clicks) adjusts the height equally on the other side until the third click. This can be the default behavior (MgraControlList() with M_MODE_RESIZE_SECONDARY_DIMENSION), or the behavior when pressing the Alt (M_MODE_RESIZE_SECONDARY_DIMENSION_ALT), Ctrl (M_MODE_RESIZE_SECONDARY_DIMENSION_CTRL), or Shift (M_MODE_RESIZE_SECONDARY_DIMENSION_SHIFT) key. You can use M_ORIENTED_RECT to define a square (which is special type of rectangle), by setting the InitFlag parameter to M_SQUARE_ASPECT_RATIO. You can also call MgraControlList() with M_SQUARE_ASPECT_RATIO as a combination value for M_MODE_RESIZE to specify this as the default behavior, or the behavior when pressing the Alt (M_MODE_RESIZE_ALT), Ctrl (M_MODE_RESIZE_CTRL), or Shift (M_MODE_RESIZE_SHIFT) key.															

The following GraphicType and corresponding CreationMode parameter settings change the 2D graphics list's state of interactivity to idle (M_STATE_IDLE) and will cancel or stop any user manipulations in progress.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For ending user interaction																	
GraphicType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	CreationMode																

M_CANCEL	Changes the 2D graphics list's state of interactivity to idle (M_STATE_IDLE) and cancels any user manipulations in progress. A cancel operation will stop any user manipulation in progress and undo any changes that occurred. If, for example, a user was rotating a graphic with the rotate handle, canceling will stop the user from rotating it further and return the graphic to the angle it was set to before the user started manipulating it. If a user is interactively defining a graphic to add to the 2D graphics list, canceling will stop the defining process and not add any graphic to the list.														
M_DEFAULT	Specifies that this parameter is not required.	c		e	f	g	a		i		j	k	l	m	n	b
M_STOP	Changes the 2D graphics list's state of interactivity to idle (M_STATE_IDLE) and stops any user manipulations in progress. A stop operation ends any user manipulations in progress and leaves any changes made to the graphic. If, for example, a user was rotating a graphic with the rotate handle, stopping will prevent the user from rotating it further and leave the graphic at that angle. If a user is interactively defining a graphic to add to the 2D graphics list, stopping will prevent the user from defining it further and adds the graphic to the list with the current dimensions.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to ignore this parameter is not required.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraLine

Synopsis

Draw a line in an image or add a line to a 2D graphics list.

Syntax

```
void MgraLine(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XStart,             //in
    AIL_DOUBLE YStart,             //in
    AIL_DOUBLE XEnd,               //in
    AIL_DOUBLE YEnd                //in
)
```

Description

This function draws a line destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of the line to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

The line is based on a start point (XStart, YStart) and an end point (XEnd, YEnd). The line inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the line falls outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

A line's coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraLine**() an uncalibrated image, the function will generate an error.

To create multiple lines, a polyline, or a polygon, use MgraLines().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraLine()** internally calls the *AIL_DOUBLE* version of this function (MgraLineDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraLineInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the line or the identifier of a valid 2D graphics list in which to add the line. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XStart

Specifies the X-coordinate of the start of the line in the input coordinate system.

YStart

Specifies the Y-coordinate of the start of the line in the input coordinate system.

XEnd

Specifies the X-coordinate of the end of the line in the input coordinate system.

YEnd

Specifies the Y-coordinate of the end of the line in the input coordinate system.

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraLineDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XStart, AIL_DOUBLE YStart,
AIL_DOUBLE XEnd, AIL_DOUBLE YEnd)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

XEnd

See XEnd of the main function for a description.

YEnd

See YEnd of the main function for a description.

```
void MgraLineInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XStart, AIL_INT64 YStart,
AIL_INT64 XEnd, AIL_INT64 YEnd)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

XEnd

See XEnd of the main function for a description.

YEnd

See YEnd of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraLines

Synopsis

Draw one or more lines, a polyline, or a polygon in an image, or add them to a 2D graphics list.

Syntax

```
void MgraLines(
    AIL_ID ContextGraId,             //in
    AIL_ID DstImageBufOrListGraId,   //out
    AIL_INT NumberOfLinesOrVertices, //in
    const AIL_DOUBLE *XPtr,          //in
    const AIL_DOUBLE *YPtr,          //in
    const AIL_DOUBLE *X2Ptr,         //in
    const AIL_DOUBLE *Y2Ptr,         //in
    AIL_INT64 ControlFlag            //in
)
```

Description

This function draws one or more lines, a polyline, or a polygon destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of the lines, polyline, or polygon to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

Lines, polylines, and polygons inherit all the relevant settings of the specified 2D graphics context, such as the foreground color. If part of a line, polyline, or polygon falls outside of the destination image (or, when drawn in a 2D graphics list, the associated display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(), respectively. To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList(), respectively.

The coordinates of the lines, polyline, or polygon are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraLines**() an uncalibrated image, the function will generate an error.

To create a single line without the option of creating multiple lines, a polyline, or a polygon, use MgraLine().

Note that prior to Aurora Imaging Library 9.0, this function only supported integer positions; the XPtr, YPtr, X2Ptr, and Y2Ptr parameters only accepted arrays of type *long*. As of Aurora Imaging Library 9.0, this function includes support for positions with floating-point precision. To implement this change and maintain backwards compatibility when using a C++ compiler (*.cpp), **MgraLines**() is available as an inline function which automatically calls MgraLinesDouble(), MgraLinesInt64(), and MgraLinesInt32(), depending on whether XPtr, YPtr, X2Ptr, and Y2Ptr receive arrays of type *AIL_DOUBLE*, *AIL_INT64*, or *AIL_INT32*, respectively. To maintain backwards compatibility when using a C compiler (*.c), **MgraLines**() maps to MgraLinesInt32() when working on a 32-bit system, or MgraLinesInt64() when working on a 64-bit system; you must explicitly call MgraLinesDouble() to pass XPtr, YPtr, X2Ptr, and Y2Ptr arrays of type *AIL_DOUBLE*, respectively. If you are an advanced user and want to retrieve a pointer to **MgraLines**(), you must use the Double, Int64, or Int32 version of this function, since **MgraLines**() is actually a macro or an overloaded function.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that presented above is an example prototype of this function. You can call MgraLines() with the same

> prototype as any of the type-specific functions below, without using the function's type-specific name; that is, use MgraLines() and not, for example, MgraLinesInt32(). The data is automatically cast into the appropriate type. This is true except when using a C compiler. When using a C compiler, you must always call this function with its type-specific name (you cannot use the standard name of this function).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies the identifier of the 2D graphics context, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the lines, polyline, or polygon or the identifier of a valid 2D graphics list in which to add the lines, polyline, or polygon. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

NumberOfLinesOrVertices

Specifies the number of lines to draw or add (M_LINE_LIST), or the number of vertices in the polyline or polygon to draw or add (M_POLYGON or M_POLYLINE), according to the array parameters (XPtr, YPtr, X2Ptr, and Y2Ptr).

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the XPtr, YPtr, X2Ptr, or Y2Ptr parameter.

XPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>]

Specifies the address of the array containing the X-coordinate(s) of the start of the lines, or containing the X-coordinates of the vertices in the polyline or polygon, in the input coordinate system. The number of elements in the array should be equal to the NumberOfLinesOrVertices parameter value.

See the Parameter associations section for possible values that can be specified.

YPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>]

Specifies the address of the array containing the Y-coordinate(s) of the start of the lines, or containing the Y-coordinates of the vertices in the polyline or polygon, in the input coordinate system. The number of elements in the array should be equal to the NumberOfLinesOrVertices parameter value.

See the Parameter associations section for possible values that can be specified.

X2Ptr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>]
- M_NULL

Specifies the address of the array containing the X-coordinate(s) of the end of the lines in the input coordinate system. The number of elements in the array should be equal to the NumberOfLinesOrVertices parameter value. If this information is not required (M_POLYGON and M_POLYLINE), set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

Y2Ptr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>]
- M_NULL

Specifies the address of the array containing the Y-coordinate(s) of the end of the lines in the input coordinate system. The number of elements in the array should be equal to the NumberOfLinesOrVertices parameter value. If this information is not required (M_POLYGON and M_POLYLINE), set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Specifies the type of line to draw in the image or to add to the 2D graphics list.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The table below lists possible values for the **XPtr**, **YPtr**, **X2Ptr**, **Y2Ptr**, and **ControlFlag** parameters.

- For specifying the type of line

Note that any unused parameters should be set to **M_NULL**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of line																	
ControlFlag		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	XPtr																
	YPtr																
	X2Ptr																
	Y2Ptr																

M_DEFAULT		Same as M_LINE_LIST.													
	XPtr		c		e	f	g	a		i		j	k	l	m	n	b
	YPtr		c		e	f	g	a		i		j	k	l	m	n	b
	X2Ptr		c		e	f	g	a		i		j	k	l	m	n	b
	Y2Ptr		c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE_LINES		Specifies a series of infinite lines, each with no start or end points. Each line is defined by two points ((XPtr, YPtr) and (X2Ptr, Y2Ptr)), but the line extends beyond these points.	c		e	f	g	a		i		j	k	l	m	n	b
	XPtr	Specifies the address of the array containing the X-coordinate(s) of the first point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
	YPtr	Specifies the address of the array containing the Y-coordinate(s) of the first point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
	X2Ptr	Specifies the address of the array containing the X-coordinate(s) of the second point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
	Y2Ptr	Specifies the address of the array containing the Y-coordinate(s) of the second point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_LIST		Specifies a series of lines of finite length. Each line is defined with two points ((XPtr, YPtr) and (X2Ptr, Y2Ptr)).	c		e	f	g	a		i		j	k	l	m	n	b
	XPtr	Specifies the address of the array containing the X-coordinate(s) of the first point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
	YPtr	Specifies the address of the array containing the Y-coordinate(s) of the first point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
	X2Ptr	Specifies the address of the array containing the X-coordinate(s) of the second point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
	Y2Ptr	Specifies the address of the array containing the Y-coordinate(s) of the second point used to define a line.	c		e	f	g	a		i		j	k	l	m	n	b
M_POLYGON +		Specifies a series of connected lines forming a closed polygon. The polygon is defined with two or more vertices (XPtr, YPtr). The final vertex is connected to the first.	c		e	f	g	a		i		j	k	l	m	n	b
	XPtr	Specifies the address of the array containing the X-coordinates of the vertices used to define the polygon.	c		e	f	g	a		i		j	k	l	m	n	b
	YPtr	Specifies the address of the array containing the Y-coordinates of the vertices used to define the polygon.	c		e	f	g	a		i		j	k	l	m	n	b

	X2Ptr	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Y2Ptr	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
M_POLYLINE		Specifies a series of connected lines forming an open polyline. The polyline is defined with two or more vertices (XPtr, YPtr). The final vertex is not connected to the first. Since the final vertex is not connected to the first, the number of line segments in the polyline is equal to NumberOfLinesOrVertices - 1.	c		e	f	g	a		i		j	k	l	m	n	b
	XPtr	Specifies the address of the array containing the X-coordinates of the vertices used to define the polyline.	c		e	f	g	a		i		j	k	l	m	n	b
	YPtr	Specifies the address of the array containing the Y-coordinates of the vertices used to define the polyline.	c		e	f	g	a		i		j	k	l	m	n	b
	X2Ptr	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Y2Ptr	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_POLYGON.

You can add the following value to the above-mentioned value to set whether the polygon is filled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether the polygon is filled																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILLED	Specifies that the polygon is filled with the foreground color of the 2D graphics context.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler

```
void MgraLinesDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumberOfLinesOrVertices, const AIL_DOUBLE *XPtr, const AIL_DOUBLE *YPtr, const AIL_DOUBLE *X2Ptr, const AIL_DOUBLE *Y2Ptr, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumberOfLinesOrVertices

See NumberOfLinesOrVertices of the main function for a description.

XPtr

See XPtr of the main function for a description.

YPtr

See YPtr of the main function for a description.

X2Ptr

See X2Ptr of the main function for a description.

Y2Ptr

See Y2Ptr of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraLinesInt32 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumberOfLinesOrVertices, const AIL_INT32 *XPtr, const AIL_INT32 *YPtr, const AIL_INT32 *X2Ptr, const AIL_INT32 *Y2Ptr, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumberOfLinesOrVertices

See NumberOfLinesOrVertices of the main function for a description.

XPtr

See XPtr of the main function for a description.

YPtr

See YPtr of the main function for a description.

X2Ptr

See X2Ptr of the main function for a description.

Y2Ptr

See Y2Ptr of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraLinesInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumberOfLinesOrVertices, const AIL_INT64 *XPtr, const AIL_INT64 *YPtr, const AIL_INT64 *X2Ptr, const AIL_INT64 *Y2Ptr, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumberOfLinesOrVertices

See NumberOfLinesOrVertices of the main function for a description.

XPtr

See XPtr of the main function for a description.

YPtr

See YPtr of the main function for a description.

X2Ptr

See X2Ptr of the main function for a description.

Y2Ptr

See Y2Ptr of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraRect

Synopsis

Draw a rectangle in an image without rotation or fill color, or add it to a 2D graphics list.

Syntax

```
void MgraRect(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XStart,             //in
    AIL_DOUBLE YStart,             //in
    AIL_DOUBLE XEnd,               //in
    AIL_DOUBLE YEnd                //in
)
```

Description

This function draws a rectangle destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of a rectangle to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

The rectangle is created from the top-left corner (XStart, YStart) to the bottom-right corner (XEnd, YEnd). The rectangle inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the rectangle falls outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

A rectangle's coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraRect**() an uncalibrated image, the function will generate an error.

To create a filled rectangle, use MgraRectFill(). To create a rectangle that can be optionally filled and rotated, use MgraRectAngle().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraRect()** internally calls the *AIL_DOUBLE* version of this function (MgraRectDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraRectInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the rectangle or the identifier of a valid 2D graphics list in which to add the rectangle. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XStart

Specifies the X-coordinate of the top-left corner of the rectangle in the input coordinate system.

YStart

Specifies the Y-coordinate of the top-left corner of the rectangle in the input coordinate system.

XEnd

Specifies the X-coordinate of the bottom-right corner of the rectangle in the input coordinate system.

YEnd

Specifies the Y-coordinate of the bottom-right corner of the rectangle in the input coordinate system.

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraRectDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XStart, AIL_DOUBLE YStart, AIL_DOUBLE XEnd, AIL_DOUBLE YEnd)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

XEnd

See XEnd of the main function for a description.

YEnd

See YEnd of the main function for a description.

```
void MgraRectInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XStart, AIL_INT64 YStart, AIL_INT64 XEnd, AIL_INT64 YEnd)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

XEnd

See XEnd of the main function for a description.

YEnd

See YEnd of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraRectAngle

Synopsis

Draw a rectangle in an image with optional rotation and fill color, or add it to a 2D graphics list.

Syntax

```
void MgraRectAngle(
    AIL_ID ContextGraId,              //in
    AIL_ID DestImageBufIdOrGraListId, //out
    AIL_DOUBLE XPos,                  //in
    AIL_DOUBLE YPos,                  //in
    AIL_DOUBLE Width,                 //in
    AIL_DOUBLE Height,                //in
    AIL_DOUBLE Angle,                 //in
    AIL_INT64 ControlFlag             //in
)
```

Description

This function draws a rectangle destructively (raster-based) in the specified image with an optional rotation and fill color. Alternatively, this function can add a vector-based version of a rectangle to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling.

The rectangle is created using a point (XPos, YPos), a width (Width), a height (Height), and an angle of rotation (Angle). The rectangle inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the rectangle falls outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

A rectangle's coordinates, dimensions, and angle are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup. Note that if you set your input coordinate system to M_WORLD and you pass **MgraRectAngle**() an uncalibrated image or the drawing target of the 2D graphics list is uncalibrated, the function will generate an error.

To create a filled rectangle that is not rotated, use MgraRectFill(). To create a rectangle that is not rotated or filled, use MgraRect().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DestImageBufIdOrGraListId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraRectAngle()** internally calls the *AIL_DOUBLE* version of this function (MgraRectAngleDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraRectAngleInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DestImageBufIdOrGraListId

Specifies the identifier of a valid image buffer in which to draw the rectangle or the identifier of a valid 2D graphics list in which to add the rectangle. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XPos

Specifies the X-coordinate of the rectangle's position in the input coordinate system.

YPos

Specifies the Y-coordinate of the rectangle's position in the input coordinate system.

Width INQ

Specifies the width of the rectangle, relative to the input coordinate system.

Height INQ

Specifies the height of the rectangle, relative to the input coordinate system.

Angle INQ

Specifies the angle with which to rotate the rectangle, in degrees, relative to the input coordinate system. The rectangle will either be rotated around its center (when using M_CENTER_AND_DIMENSION) or its top-left corner (when using M_CORNER_AND_DIMENSION).

ControlFlag INQ

Specifies how to draw the rectangle in the image or add the rectangle to the 2D graphics list.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to draw or add the rectangle																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_CORNER_AND_DIMENSION.	c		e	f	g	a		i		j	k	l	m	n	b

M_CENTER_AND_DIMENSION +	Specifies that the rectangle is drawn according to its center position, as defined by the specified coordinates. Coordinates are set with the XPos and YPos parameters.	c		e	f	g	a		i		j	k	l	m	n	b
M_CORNER_AND_DIMENSION +	Specifies that the rectangle is drawn according to the position of its top-left corner, as defined by the specified coordinates. Coordinates are set with the XPos and YPos parameters.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for any of the possible values of the ControlFlag parameter.

You can add the following value to the above-mentioned values to set whether the rectangle is filled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether the rectangle is filled																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILLED	Specifies that the rectangle is filled with specified foreground color (MgraControl() or MgraControlList() with M_COLOR).	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraRectAngleDouble (AIL_ID ContextGraId, AIL_ID DestImageBufIdOrGraListId, AIL_DOUBLE XPos, AIL_DOUBLE YPos, AIL_DOUBLE Width, AIL_DOUBLE Height, AIL_DOUBLE Angle, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DestImageBufIdOrGraListId

See DestImageBufIdOrGraListId of the main function for a description.

XPos

See XPos of the main function for a description.

YPos

See YPos of the main function for a description.

Width

See Width of the main function for a description.

Height

See Height of the main function for a description.

Angle

See Angle of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraRectAngleInt64 (AIL_ID ContextGraId, AIL_ID DestImageBufIdOrGraListId, AIL_INT64 XPos, AIL_INT64 YPos,
AIL_INT64 Width, AIL_INT64 Height, AIL_DOUBLE Angle, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DestImageBufIdOrGraListId

See DestImageBufIdOrGraListId of the main function for a description.

XPos

See XPos of the main function for a description.

YPos

See YPos of the main function for a description.

Width

See Width of the main function for a description.

Height

See Height of the main function for a description.

Angle

See Angle of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraRectFill

Synopsis

Draw a filled rectangle in an image without rotation, or add it to a 2D graphics list.

Syntax

```
void MgraRectFill(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XStart,             //in
    AIL_DOUBLE YStart,             //in
    AIL_DOUBLE XEnd,               //in
    AIL_DOUBLE YEnd                //in
)
```

Description

This function draws a filled rectangle destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of a filled rectangle to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling. Note that the rectangle is filled with the specified foreground color (MgraControl() or MgraControlList() with M_COLOR).

The filled rectangle is created from the top-left corner (XStart, YStart) to the bottom-right corner (XEnd, YEnd). The filled rectangle inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the filled rectangle falls outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

A filled rectangle's coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraRectFill**() an uncalibrated image, the function will generate an error.

To create a rectangle that is not filled, use MgraRect(). To create a rectangle that can be optionally filled and rotated, use MgraRectAngle().

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MgraRectFill()** internally calls the *AIL_DOUBLE* version of this function (MgraRectFillDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MgraRectFillInt64()).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the filled rectangle or the identifier of a valid 2D graphics list in which to add the filled rectangle. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XStart

Specifies the X-coordinate of the top-left corner of the filled rectangle in the input coordinate system.

YStart

Specifies the Y-coordinate of the top-left corner of the filled rectangle in the input coordinate system.

XEnd

Specifies the X-coordinate of the bottom-right corner of the filled rectangle in the input coordinate system.

YEnd

Specifies the Y-coordinate of the bottom-right corner of the filled rectangle in the input coordinate system.

Type-specific versions of the function when using a C compiler under 64-bit

```
void MgraRectFillDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_DOUBLE XStart, AIL_DOUBLE YStart, AIL_DOUBLE XEnd, AIL_DOUBLE YEnd)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

XEnd

See XEnd of the main function for a description.

YEnd

See YEnd of the main function for a description.

```
void MgraRectFillInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT64 XStart, AIL_INT64 YStart, AIL_INT64 XEnd, AIL_INT64 YEnd)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

XStart

See XStart of the main function for a description.

YStart

See YStart of the main function for a description.

XEnd

See XEnd of the main function for a description.

YEnd

See YEnd of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraText

Synopsis

Draw text in an image or add text to a 2D graphics list.

Syntax

```
void MgraText(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_DOUBLE XStart,             //in
    AIL_DOUBLE YStart,             //in
    AIL_CONST_TEXT_PTR StringPtr   //in
)
```

Description

This function draws text destructively (raster-based) in the specified image. Alternatively, this function can add a vector-based version of the text to the specified 2D graphics list.

Note that while the location of text can sometimes change (for example, by applying a camera calibration or zooming the display), the text's font will always remain the same, and the text itself will never be magnified or distorted.

Text is defined by a string (StringPtr) and a starting position (XStart, YStart). Text inherits all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the text falls outside of the specified area (image or display), that part is clipped off. By default, a text's background is filled; to make it transparent, use MgraControl() with M_BACKGROUND_MODE.

To modify or inquire graphic context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

A text's coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraText**() an uncalibrated image, the function will generate an error.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the graphic or the identifier of a valid 2D graphics list in which to add the graphic. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

XStart INQ

Specifies the X-coordinate at which to start drawing the top-left corner of the first character in the input coordinate system.

YStart INQ

Specifies the Y-coordinate at which to start drawing the top-left corner of the first character in the input coordinate system.

StringPtr

Specifies the address of the string containing the text to draw in the destination buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the string																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
AIL_TEXT("*String*") 1	Specifies the address of the null-terminated (\0) ASCII string that must be drawn in the destination buffer (DstImageBufOrListGraId). There is no restriction on the length of the string.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraVectors

Synopsis

Draw vectors at arbitrary positions in an image, or add them to a 2D graphics list.

Syntax

```
void MgraVectors(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_INT NumVectors,            //in
    const AIL_DOUBLE *XArrayPtr,   //in
    const AIL_DOUBLE *YArrayPtr,   //in
    const AIL_DOUBLE *UArrayPtr,   //in
    const AIL_DOUBLE *VArrayPtr,   //in
    AIL_INT64 ScaleMode,           //in
    AIL_DOUBLE ScaleValue,         //in
    AIL_INT64 ControlFlag          //in
)
```

Description

This function draws vectors destructively (raster-based) in the specified image at arbitrary positions. Alternatively, this function can add a vector-based version of the vectors to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling. To draw vectors at equally spaced positions, use MgraVectorsGrid().

The vectors are based on the specified starting positions (XArrayPtr and YArrayPtr), and the specified horizontal and vertical displacements (UArrayPtr and VArrayPtr). For example, the values at a given index in the specified arrays define a vector that goes from (x[*index*], y[*index*]) to (x[*index*] + u[*index*], y[*index*] + v[*index*]).

The vectors inherit all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the vectors fall outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

The coordinates of the vectors are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraVectors**() an uncalibrated image, the function will generate an error.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

> Note that presented above is an example prototype of this function. You can call MgraVectors() with the same prototype as any of the type-specific functions below, without using the function's type-specific name; that is, use MgraVectors() and not, for example, MgraVectorsFloat(). The data is automatically cast into the appropriate type. This is true except when using a C compiler. When using a C compiler, you must always call this function with its type-specific name (you cannot use the standard name of this function).

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the identifier of the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies the identifier of the 2D graphics context, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the vectors or the identifier of a valid 2D graphics list in which to add the vectors. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

NumVectors

Specifies the number of vectors to draw.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the XArrayPtr, YArrayPtr, UArrayPtr, or VArrayPtr parameter.

XArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumVectors)

Specifies the address of the array containing the X-coordinates of the starting points of the vectors to draw. The number of elements in the array should be equal to the value specified for the NumVectors parameter.

YArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumVectors)

Specifies the address of the array containing the Y-coordinates of the starting points of the vectors to draw. The number of elements in the array should be equal to the value specified for the NumVectors parameter.

UArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)

Required array size:(NumVectors)

Specifies the address of the array containing the displacements, in the X-direction, of the vectors to draw. The number of elements in the array should be equal to the value specified for the NumVectors parameter.

VArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumVectors)

Specifies the address of the array containing the displacements, in the Y-direction, of the vectors to draw. The number of elements in the array should be equal to the value specified for the NumVectors parameter.

ScaleMode

Specifies how to use the scale value (ScaleValue) to scale the vectors. By default, vectors are not scaled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to use the scale value to scale the vectors																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ABSOLUTE.	c		e	f	g	a		i		j	k	l	m	n	b
M_ABSOLUTE	Specifies to use the scale value (ScaleValue) as a scaling factor that directly multiplies the vectors' horizontal and vertical displacements (UArrayPtr and VArrayPtr).	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies that Aurora Imaging Library determines the scale mode. To do this, Aurora Imaging Library makes internal assumptions to establish an ideal scaling factor that optimizes the space taken by the vectors in the destination (DstImageBufOrListGrald) while reducing the chance of overlapping vectors. When using M_AUTO, Aurora Imaging Library multiplies the vectors' horizontal and vertical displacements (UArrayPtr and VArrayPtr) by both the scale value (ScaleValue) and the internally established scale factor.	c		e	f	g	a		i		j	k	l	m	n	b

ScaleValue

Specifies the value by which to scale the vectors, according to the specified mode (ScaleMode). By default, vectors are not scaled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the value by which to scale the vectors, according to the specified mode																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the scale value.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Specifies how to configure the arrowheads of the vectors, and how to handle null vectors.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to configure the arrowheads of the vectors																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_FIXED_LENGTH_ARROWHEADS + M_SKIP_NULL_VECTORS.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIXED_LENGTH_ARROWHEADS +	Specifies that arrowheads are drawn identically for all vectors, regardless of each vector's length.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROPORTIONAL_ARROWHEADS +	Specifies that arrowheads are drawn proportionally to each vector's length.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FIXED_LENGTH_ARROWHEADS; M_PROPORTIONAL_ARROWHEADS.

You can add one of the following values to the above-mentioned values to determine how to handle null vectors.

Null vectors are vectors with a length of 0. This occurs when a vector's displacement values (UArrayPtr and VArrayPtr) are both 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining how to handle null vectors																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_NULL_VECTORS	Specifies that null vectors are drawn as a dot in the destination (DstImageBufOrListGrald).	c		e	f	g	a		i		j	k	l	m	n	b
M_SKIP_NULL_VECTORS	Specifies that null vectors are not drawn in the destination (DstImageBufOrListGrald).	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler

```
void MgraVectorsDouble (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumVectors, const AIL_DOUBLE *XArrayPtr, const AIL_DOUBLE *YArrayPtr, const AIL_DOUBLE *UArrayPtr, const AIL_DOUBLE *VArrayPtr, AIL_INT64 ScaleMode, AIL_DOUBLE ScaleValue, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumVectors

See NumVectors of the main function for a description.

XArrayPtr

See XArrayPtr of the main function for a description.

YArrayPtr

See YArrayPtr of the main function for a description.

UArrayPtr

See UArrayPtr of the main function for a description.

VArrayPtr

See VArrayPtr of the main function for a description.

ScaleMode

See ScaleMode of the main function for a description.

ScaleValue

See ScaleValue of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraVectorsFloat (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumVectors, const AIL_FLOAT *XArrayPtr, const AIL_FLOAT *YArrayPtr, const AIL_FLOAT *UArrayPtr, const AIL_FLOAT *VArrayPtr, AIL_INT64 ScaleMode, AIL_DOUBLE ScaleValue, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumVectors

See NumVectors of the main function for a description.

XArrayPtr

See XArrayPtr of the main function for a description.

YArrayPtr

See YArrayPtr of the main function for a description.

UArrayPtr

See UArrayPtr of the main function for a description.

VArrayPtr

See VArrayPtr of the main function for a description.

ScaleMode

See ScaleMode of the main function for a description.

ScaleValue

See ScaleValue of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraVectorsInt32 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumVectors, const AIL_INT32 *XArrayPtr, const AIL_INT32 *YArrayPtr, const AIL_INT32 *UArrayPtr, const AIL_INT32 *VArrayPtr, AIL_INT64 ScaleMode, AIL_DOUBLE ScaleValue, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumVectors

See NumVectors of the main function for a description.

XArrayPtr

See XArrayPtr of the main function for a description.

YArrayPtr

See YArrayPtr of the main function for a description.

UArrayPtr

See UArrayPtr of the main function for a description.

VArrayPtr

See VArrayPtr of the main function for a description.

ScaleMode

See ScaleMode of the main function for a description.

ScaleValue

See ScaleValue of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

```
void MgraVectorsInt64 (AIL_ID ContextGraId, AIL_ID DstImageBufOrListGraId, AIL_INT NumVectors, const AIL_INT64 *XArrayPtr, const AIL_INT64 *YArrayPtr, const AIL_INT64 *UArrayPtr, const AIL_INT64 *VArrayPtr, AIL_INT64 ScaleMode, AIL_DOUBLE ScaleValue, AIL_INT64 ControlFlag)
```

Details

Parameters

ContextGraId

See ContextGraId of the main function for a description.

DstImageBufOrListGraId

See DstImageBufOrListGraId of the main function for a description.

NumVectors

See NumVectors of the main function for a description.

XArrayPtr

See XArrayPtr of the main function for a description.

YArrayPtr

See YArrayPtr of the main function for a description.

UArrayPtr

See UArrayPtr of the main function for a description.

VArrayPtr

See VArrayPtr of the main function for a description.

ScaleMode

See ScaleMode of the main function for a description.

ScaleValue

See ScaleValue of the main function for a description.

ControlFlag

See ControlFlag of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MgraVectorsGrid

Synopsis

Draw vectors at equally spaced positions in an image, or add them to a 2D graphics list.

Syntax

```
void MgraVectorsGrid(
    AIL_ID ContextGraId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_ID UImageBufId,            //in
    AIL_ID VImageBufId,            //in
    AIL_INT Stride,                //in
    AIL_INT64 ScaleMode,           //in
    AIL_DOUBLE ScaleValue,         //in
    AIL_INT64 ControlFlag          //in
)
```

Description

This function draws vectors destructively (raster-based) in the specified image at equally spaced positions. Alternatively, this function can add a vector-based version of the vectors to the specified 2D graphics list, allowing you to, for example, non-destructively annotate a display without pixelation effects upon scaling. To draw vectors at arbitrary positions, use MgraVectors().

The vectors are based on the specified image buffers (UImageBufId and VImageBufId) containing values that correspond to the horizontal and vertical displacements of the vectors. Aurora Imaging Library uses the displacements at a given location to calculate the starting and ending positions of the vectors. For example, the values located at (x, y) in the specified buffers (UImageBufId and VImageBufId) define a vector that goes from (x, y) to ($x + u[x][y]$, $y + v[x][y]$). The number of pixels in the buffers (*SizeX* x *SizeY*) determine the number of vectors to draw.

Typically, the image buffers from which the vectors' positions and lengths are inferred are the result of a previous image processing operation, such as an edge detection (for example, MimConvolve()).

The vectors inherit all the relevant settings of the specified 2D graphics context, such as the foreground color (see MgraAlloc() for default context settings). If part of the vectors fall outside of the specified area (image or display), that part is clipped off.

To modify or inquire 2D graphics context settings, use MgraControl() or MgraInquire(). To modify or inquire 2D graphics list settings, use MgraControlList() or MgraInquireList().

The starting coordinates of the first vector drawn are (0,0). Coordinates are interpreted with respect to the input coordinate system, specified using MgraControl() or MgraControlList() with M_INPUT_UNITS. Note that if you set your input coordinate system to M_WORLD and you pass **MgraVectorsGrid**() uncalibrated images, the function will generate an error.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call this function concurrently from multiple threads on the same Aurora Imaging Library 2D graphics list (DstImageBufOrListGraId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the identifier of the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies the identifier of the 2D graphics context, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DstImageBufOrListGraId

Specifies the identifier of a valid image buffer in which to draw the vectors or the identifier of a valid 2D graphics list in which to add the vectors. You must have allocated the image buffer or the 2D graphics list using MbufAlloc...() or MgraAllocList(), respectively.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

UImageBufId

Specifies the identifier of a valid image buffer containing the values corresponding to the displacements, in the X-direction, of the vectors to draw. You must have allocated the image buffer using MbufAlloc...().

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error. The image buffer must be a single-band buffer.

VImageBufId

Specifies the identifier of a valid image buffer containing the values corresponding to the displacements, in the Y-direction, of the vectors to draw. You must have allocated the image buffer using MbufAlloc...().

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error. The image buffer must be a single-band buffer.

Stride

Specifies a value that establishes the positions for which vectors are drawn. Aurora Imaging Library always draws a vector for the first position (0,0), and then draws subsequent vectors that are a multiple of the specified stride, in both the X- and Y-direction. For example, if you specify a stride of 2, a vector is drawn for position (0,0), and for every second position after that, in the X-direction and in the Y-direction. Specifying a stride of 1 draws a vector for every position. Increasing the stride increases the space between the vectors drawn.

ScaleMode

Specifies how to use the scale value (ScaleValue) to scale the vectors. By default, vectors are not scaled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to use the scale value to scale the vectors																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ABSOLUTE.	c		e	f	g	a		i		j	k	l	m	n	b

M_ABSOLUTE	Specifies to use the scale value (ScaleValue) as a scaling factor that directly multiplies the vectors' horizontal and vertical displacements (UImageBufId and VImageBufId).	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies that Aurora Imaging Library determines the scale mode. To do this, Aurora Imaging Library makes internal assumptions to establish an ideal scaling factor that optimizes the space taken by the vectors in the destination (DstImageBufOrListGrafId) while reducing the chance of overlapping vectors. When using M_AUTO, Aurora Imaging Library multiplies the vectors' horizontal and vertical displacements (UImageBufId and VImageBufId) by both the scale value (ScaleValue) and the internally established scale factor.	c		e	f	g	a		i		j	k	l	m	n	b

ScaleValue

Specifies the value by which to scale the vectors, according to the specified mode (ScaleMode). By default, vectors are not scaled.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the value by which to scale the vectors, according to the specified mode																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the scale value.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Specifies how to configure the arrowheads of the vectors, and how to handle null vectors.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying how to configure the arrowheads of the vectors																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_FIXED_LENGTH_ARROWHEADS + M_SKIP_NULL_VECTORS.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIXED_LENGTH_ARROWHEADS +	Specifies that arrowheads are drawn identically for all vectors, regardless of each vector's length.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROPORTIONAL_ARROWHEADS +	Specifies that arrowheads are drawn proportionally to each vector's length.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FIXED_LENGTH_ARROWHEADS; M_PROPORTIONAL_ARROWHEADS.

You can add one of the following values to the above-mentioned values to determine how to handle null vectors.

Null vectors are vectors with a length of 0. This occurs when a vector's displacement values (UImageBufId and VImageBufId) are both 0.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining how to handle null vectors																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_NULL_VECTORS	Specifies that null vectors are drawn as a dot in the destination (DstImageBufOrListGraId).	c		e	f	g	a		i		j	k	l	m	n	b
M_SKIP_NULL_VECTORS	Specifies that null vectors are not drawn in the destination (DstImageBufOrListGraId).	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.